Puberty and emotions study supplementary model outputs

H1 Traits Model output

```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.3911244 0.05164203 0.2806487 0.4820849
```

Summary of Posterior Distribution

Parameter	Mean	Median	CI	pd
b_Intercept	4.89e-03	4.86e-03	[-0.14, 0.14]	52.75%
b_pds_total_s	0.26	0.26	[0.07, 0.45]	99.58%
b_swan_total_s	0.55	0.55	[0.41, 0.69]	100%
b_age_cont_s	-0.02	-0.02	[-0.21, 0.18]	57.24%
b_sed_percentile_s	0.05	0.05	[-0.10, 0.19]	73.62%
b_pds_total_s:swan_total_s	-0.06	-0.06	[-0.20, 0.07]	83.16%

Parameter	% in ROPE	meaningful
b_Intercept	83.42%	FALSE
b_pds_total_s	5.02%	TRUE
b_swan_total_s	0%	TRUE
b_age_cont_s	68.41%	FALSE
b_sed_percentile_s	74.69%	FALSE
b_pds_total_s:swan_total_s	69.20%	FALSE

H1 Traits Model diagnostics and PP checks

```
=== Rhat and ESS ===
```

```
Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

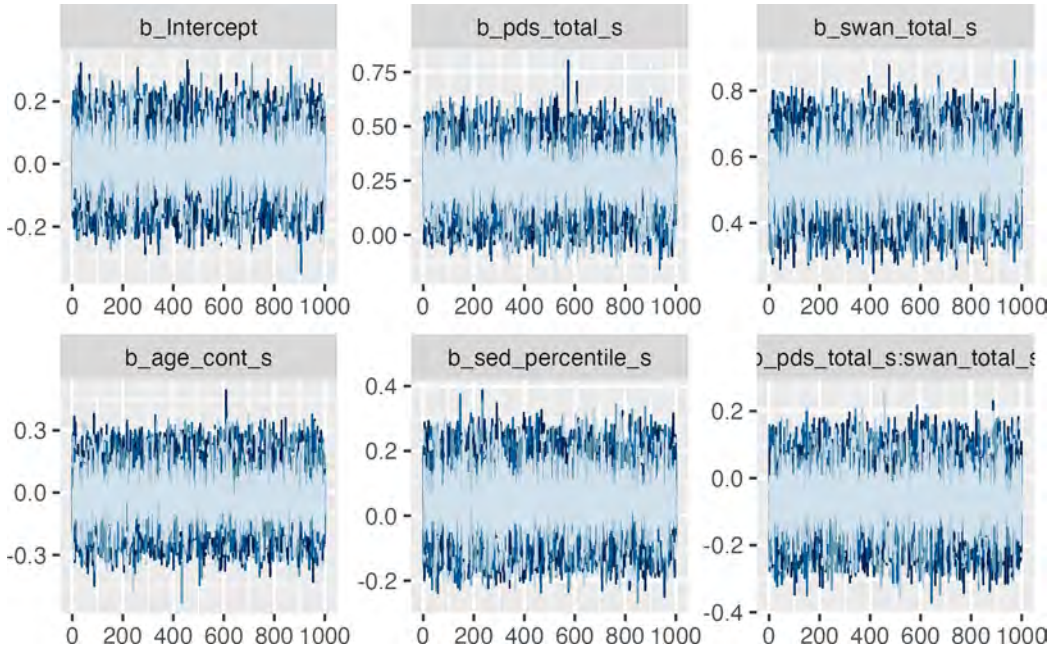
# A tibble: 10 x 4
   variable                    rhat ess_bulk ess_tail
   <chr>                      <dbl>    <dbl>    <dbl>
 1 b_Intercept                 1.00  189104.   92281.
 2 b_pds_total_s               1.00  114275.   95797.
 3 b_swan_total_s              1.00   58908.   88782.
 4 b_age_cont_s                1.00   51116.   94956.
 5 b_sed_percentile_s          1.01    5970.   62828.
 6 b_pds_total_s:swan_total_s  1.00  113038.   91764.
 7 sigma                       1.00  145922.   90941.
 8 Intercept                   1.00  189099.   93533.
 9 lprior                      1.00  144129.   94232.
10 lp__                        1.06    1192.    1767.


=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```



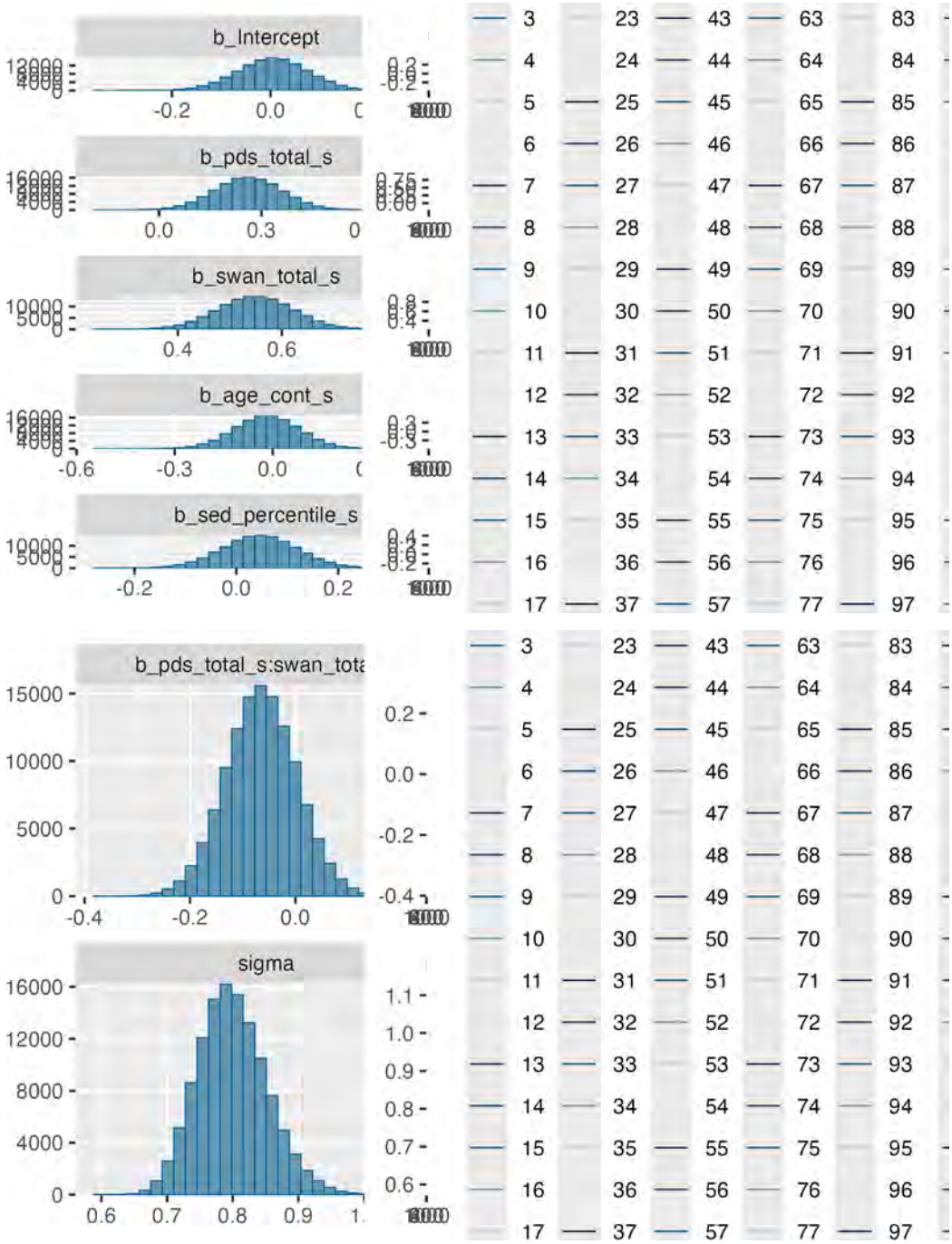

```
=== Prior Summary ===

Priors specified for the model.

          prior     class                     coef group resp dpar nlpar lb ub
   normal(0, 1)         b
   normal(0, 1)         b               age_cont_s
   normal(0, 1)         b              pds_total_s
   normal(0, 1)         b pds_total_s:swan_total_s
   normal(0, 1)         b         sed_percentile_s
   normal(0, 1)         b             swan_total_s
   normal(0, 1) Intercept
 exponential(1)     sigma                                                  0
 tag       source
             user
     (vectorized)
     (vectorized)
     (vectorized)
     (vectorized)
     (vectorized)
             user
             user
```

=== Posterior Predictive Checks ===

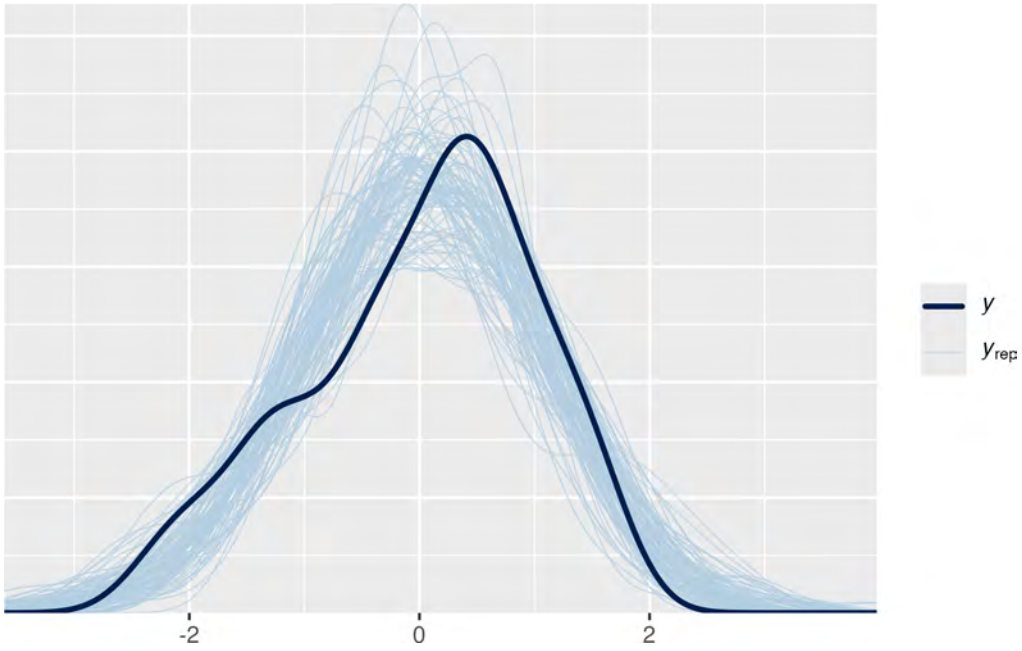
Density plots for each parameter.



```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```



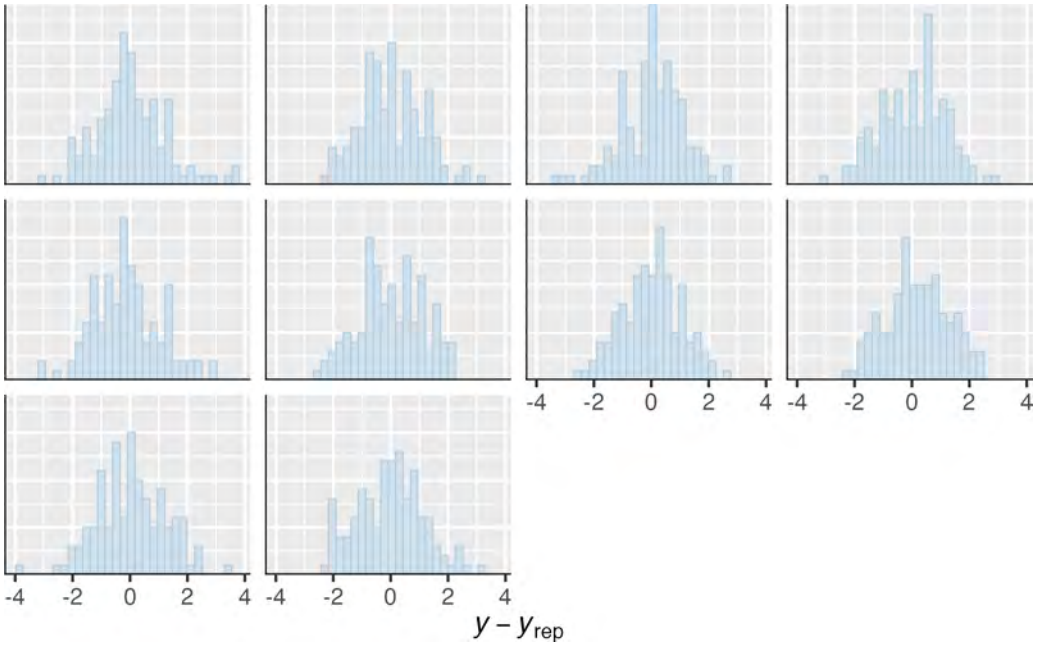

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

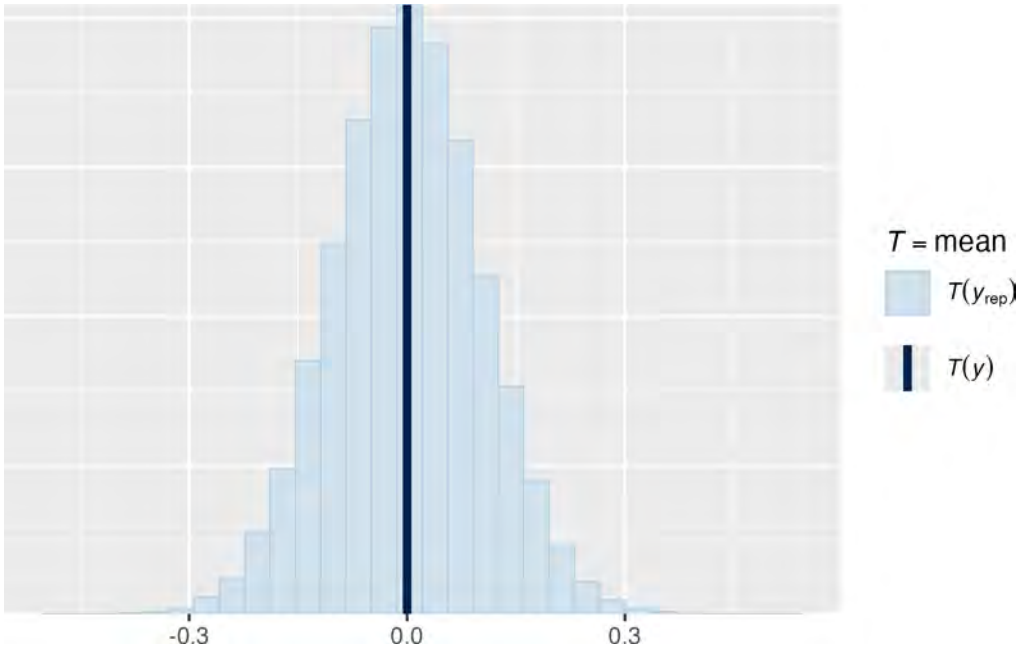
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



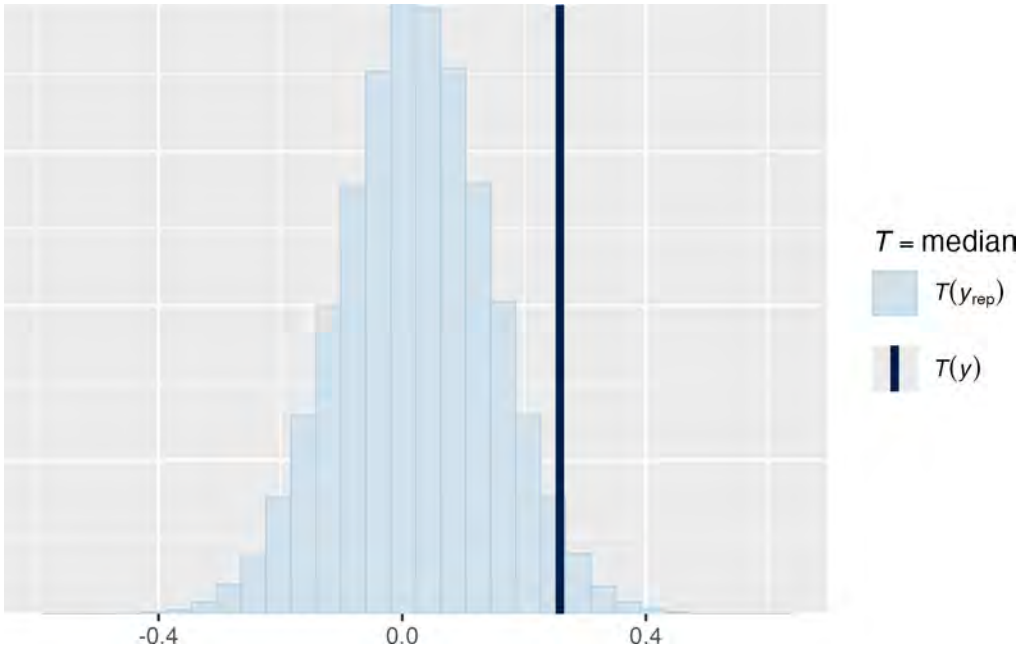

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



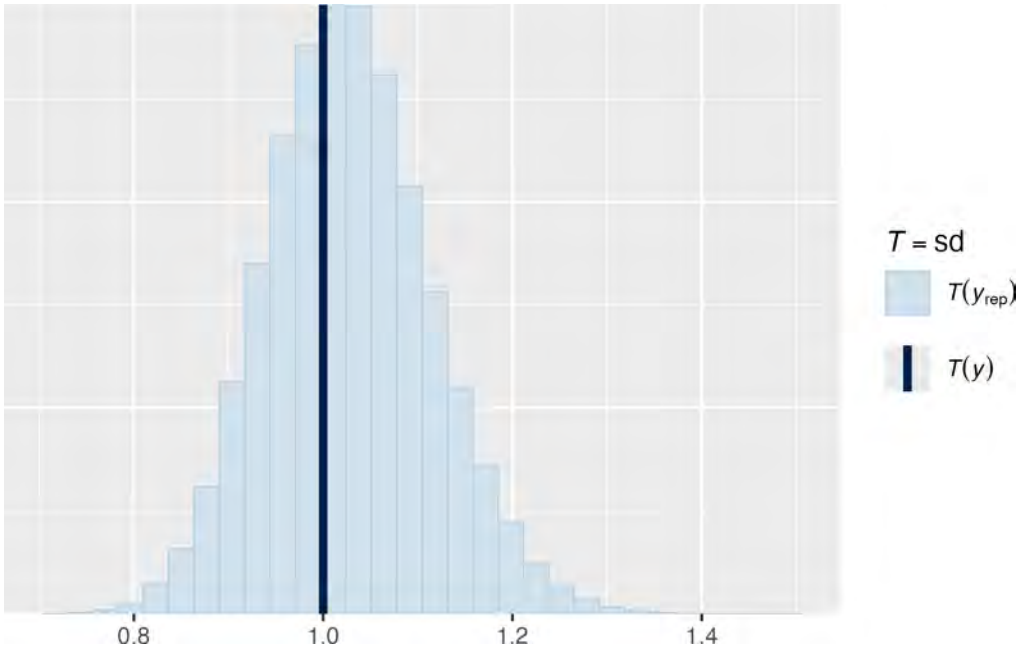

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

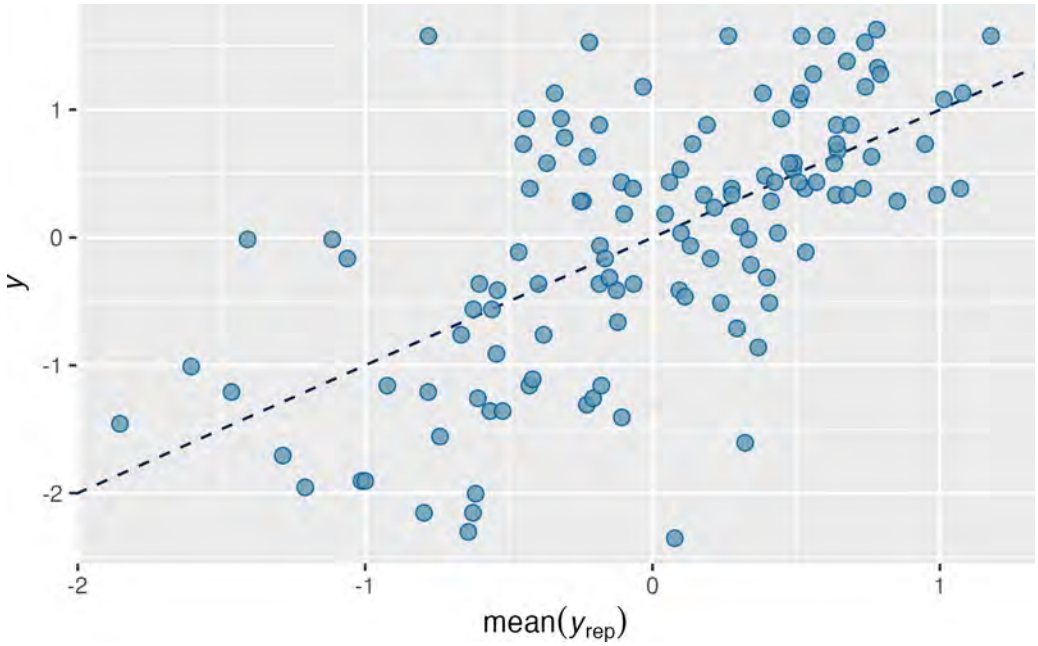



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

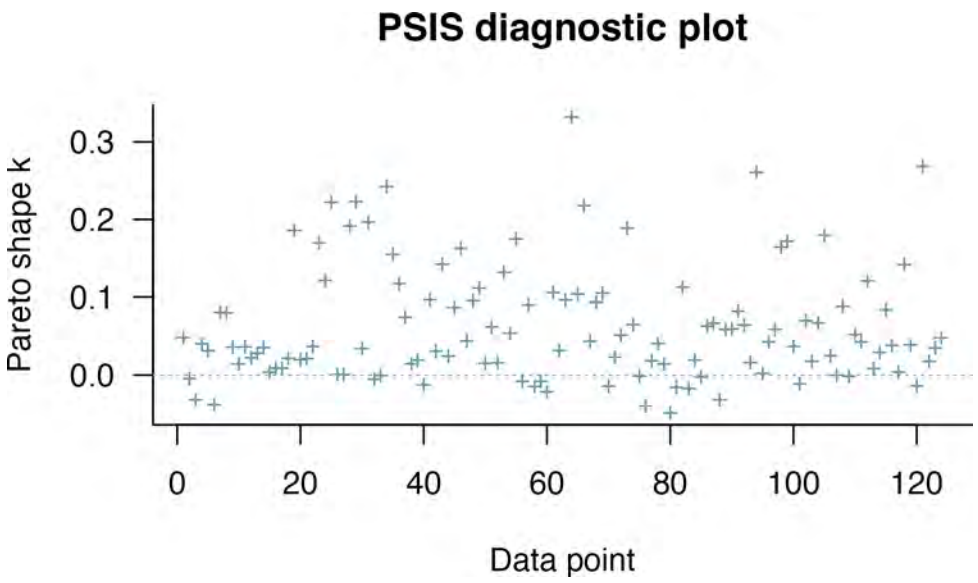
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H1 Traits Model no interaction output

```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.3853823 0.05221773 0.2742917 0.4774747
```

```
Summary of Posterior Distribution

Parameter          |      Mean |    Median |            CI |     pd | % in ROPE | meaningful
-------------------------------------------------------------------------------------------
b_Intercept        | -1.07e-04 | -2.52e-04 | [-0.14, 0.14] | 50.15% |    83.50% |      FALSE
b_pds_total_s      |      0.26 |      0.26 | [ 0.07, 0.46] | 99.59% |     4.81% |       TRUE
b_swan_total_s     |      0.55 |      0.55 | [ 0.40, 0.69] |   100% |        0% |       TRUE
b_age_cont_s       |     -0.02 |     -0.02 | [-0.21, 0.17] | 57.52% |    68.50% |      FALSE
b_sed_percentile_s |      0.04 |      0.04 | [-0.10, 0.19] | 72.21% |    75.46% |      FALSE
```

```
=== Pubertal status and ADHD traits effects in raw outcome units ===
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

effect	mean	lower	upper
pds_raw_effect	1.82	0.50	3.16
swan_raw_effect	0.58	0.43	0.73

H1 Traits Model no interaction diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

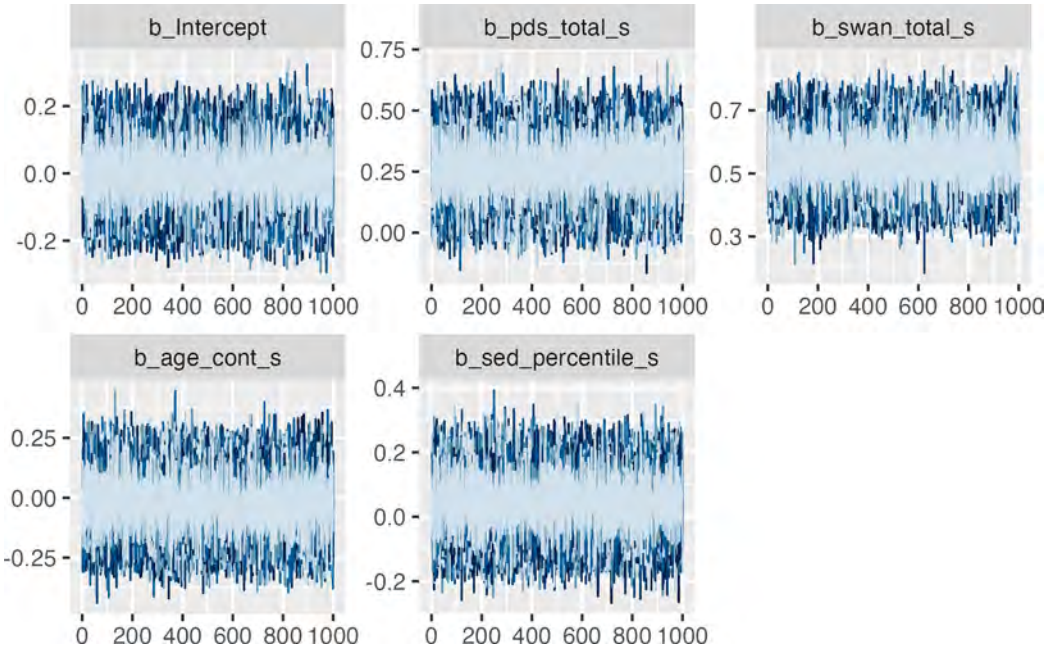
# A tibble: 9 x 4
  variable            rhat ess_bulk ess_tail
  <chr>              <dbl>    <dbl>    <dbl>
1 b_Intercept         1.00  168857.   90425.
2 b_pds_total_s       1.00  109348.   96513.
3 b_swan_total_s      1.00   97111.   89934.
4 b_age_cont_s        1.00   37941.   93770.
5 b_sed_percentile_s  1.01    6807.   54985.
6 sigma               1.00  140181.   90781.
7 Intercept           1.00  168857.   90425.
8 lprior              1.00  133723.   91000.
9 lp__                1.07     966.    1262.


=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```



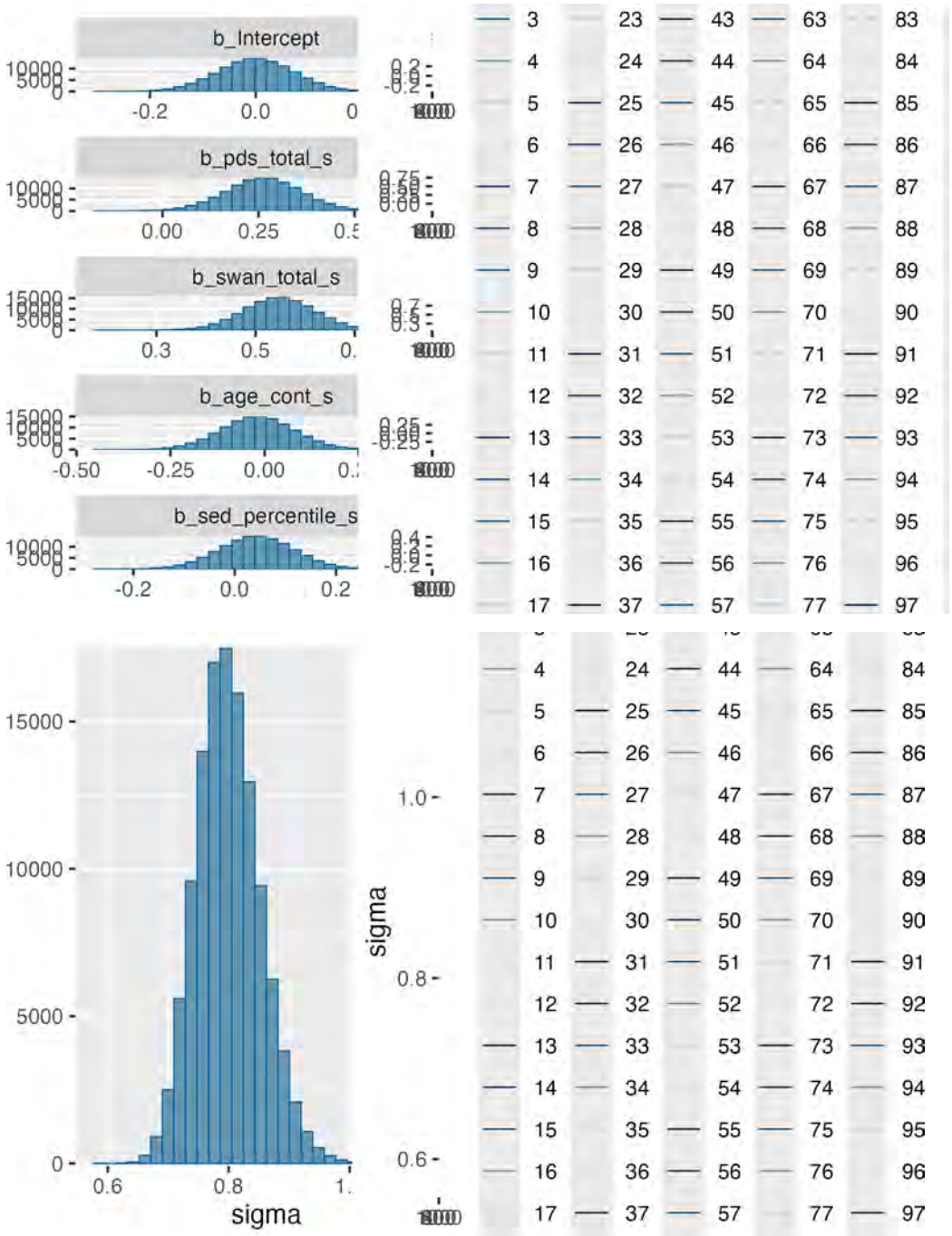

```
=== Prior Summary ===

Priors specified for the model.

          prior     class             coef group resp dpar nlpar lb ub tag
   normal(0, 1)         b
   normal(0, 1)         b       age_cont_s
   normal(0, 1)         b      pds_total_s
   normal(0, 1)         b sed_percentile_s
   normal(0, 1)         b     swan_total_s
   normal(0, 1) Intercept
 exponential(1)     sigma                                          0
       source
         user
 (vectorized)
 (vectorized)
 (vectorized)
 (vectorized)
         user
         user


=== Posterior Predictive Checks ===
```

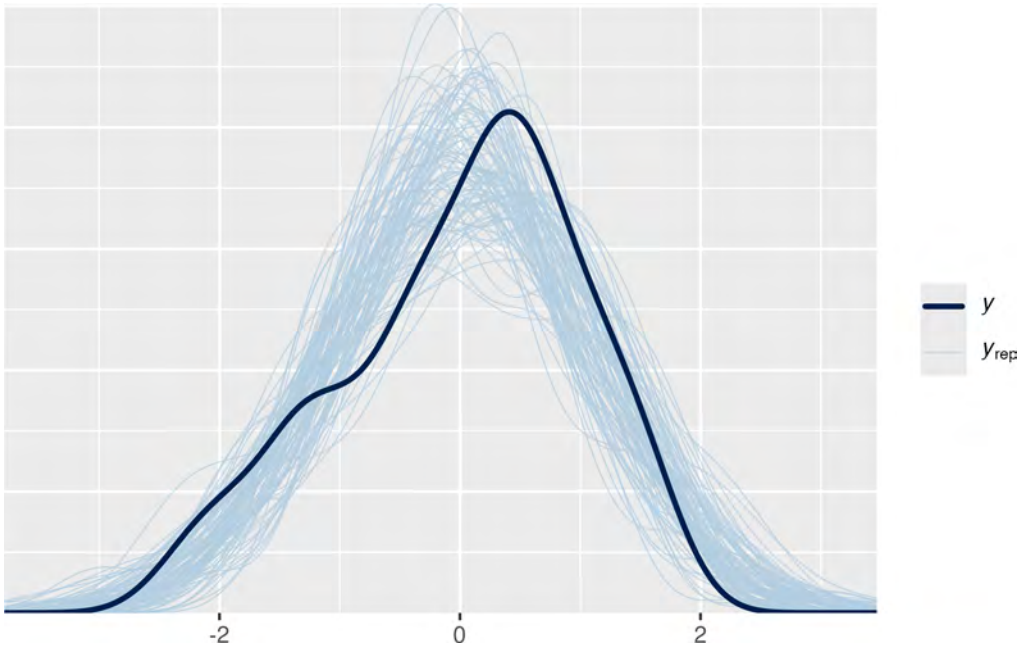
Density plots for each parameter.



```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```



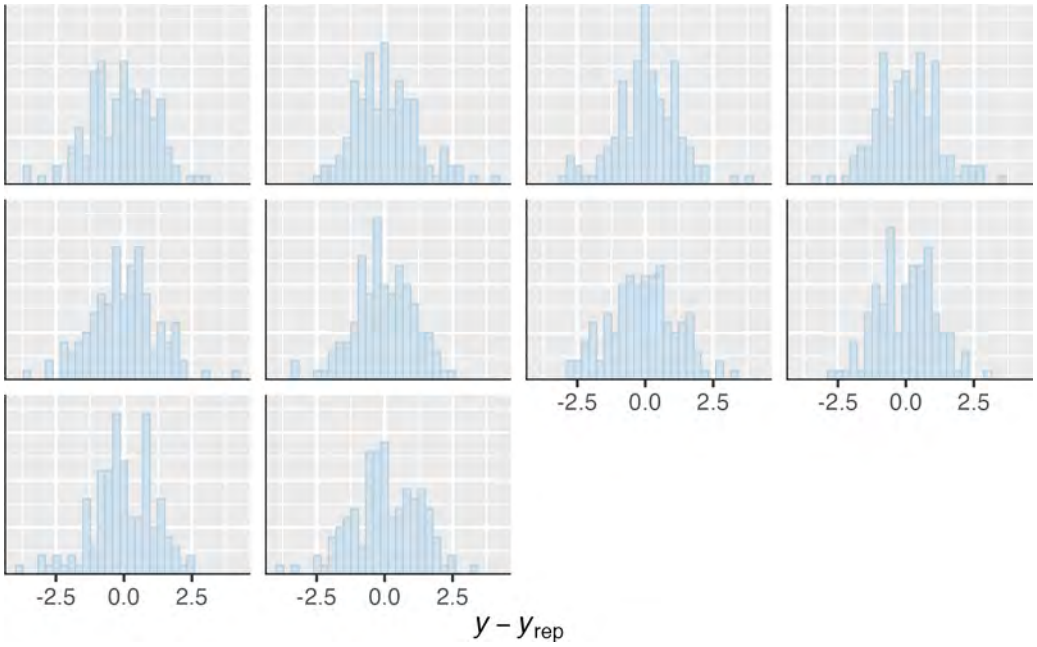

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

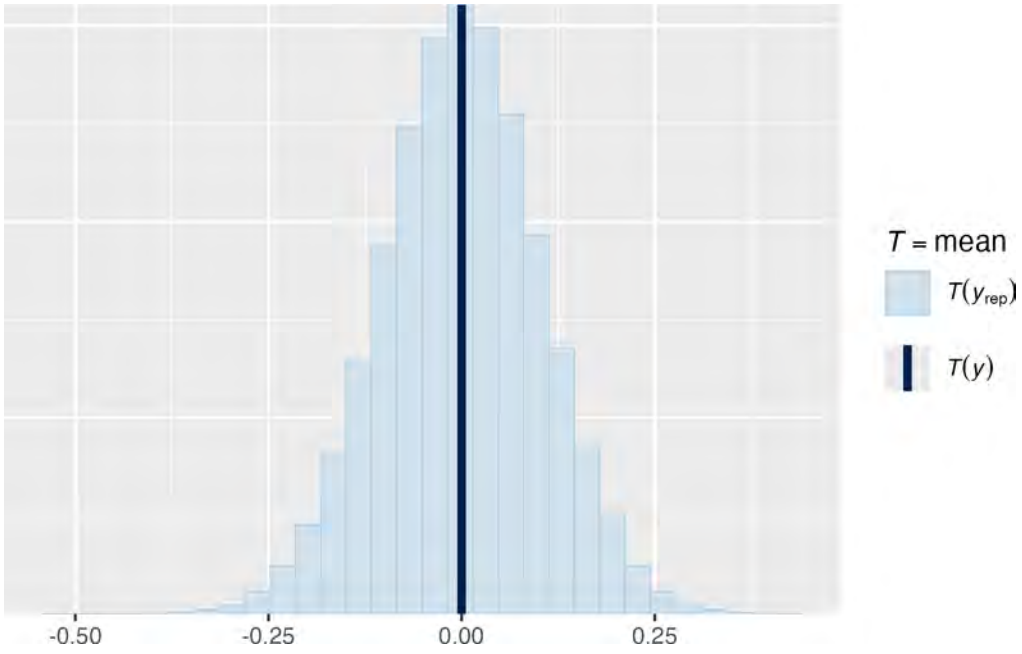
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



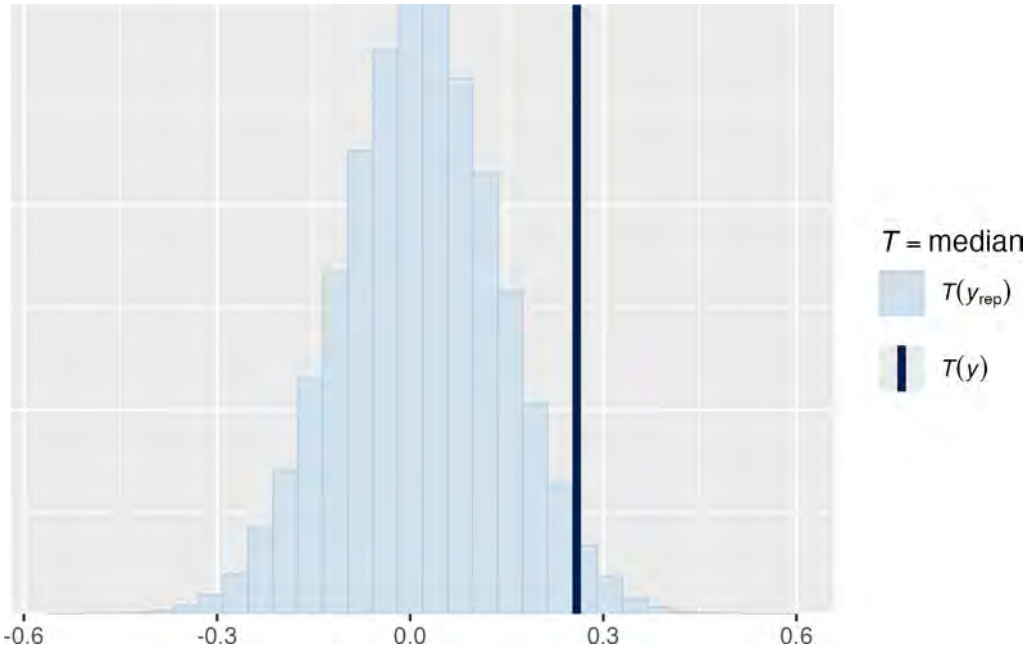

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



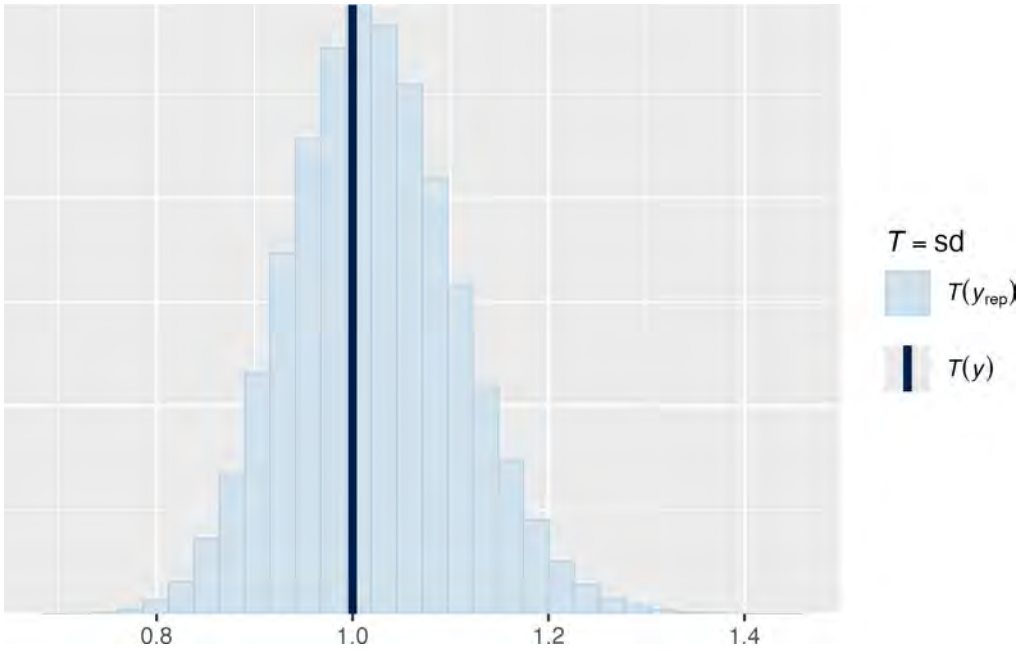

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

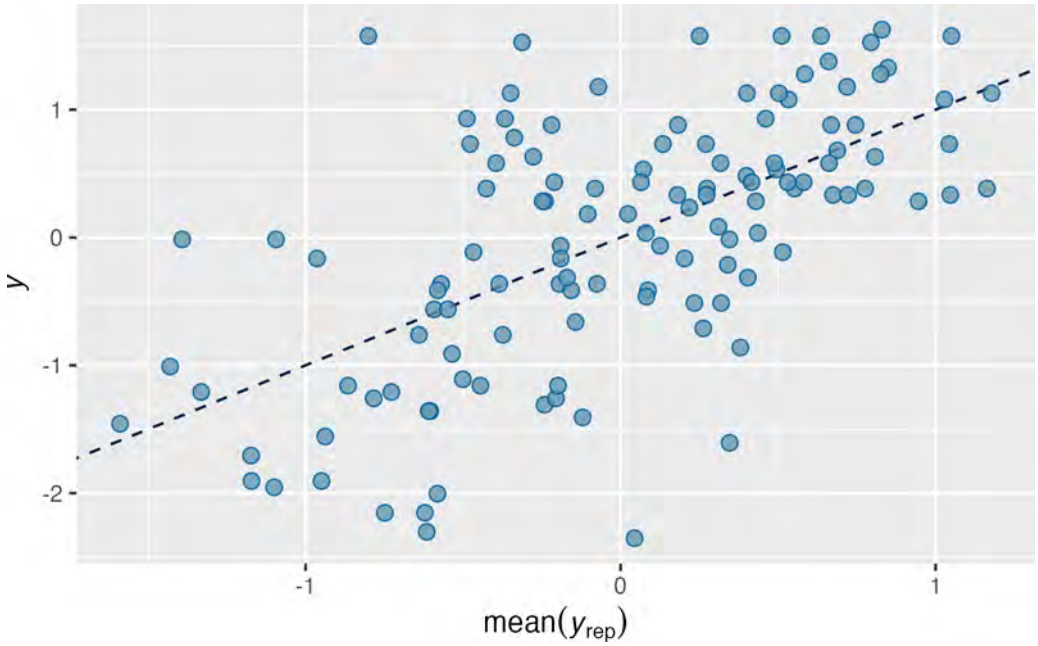



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

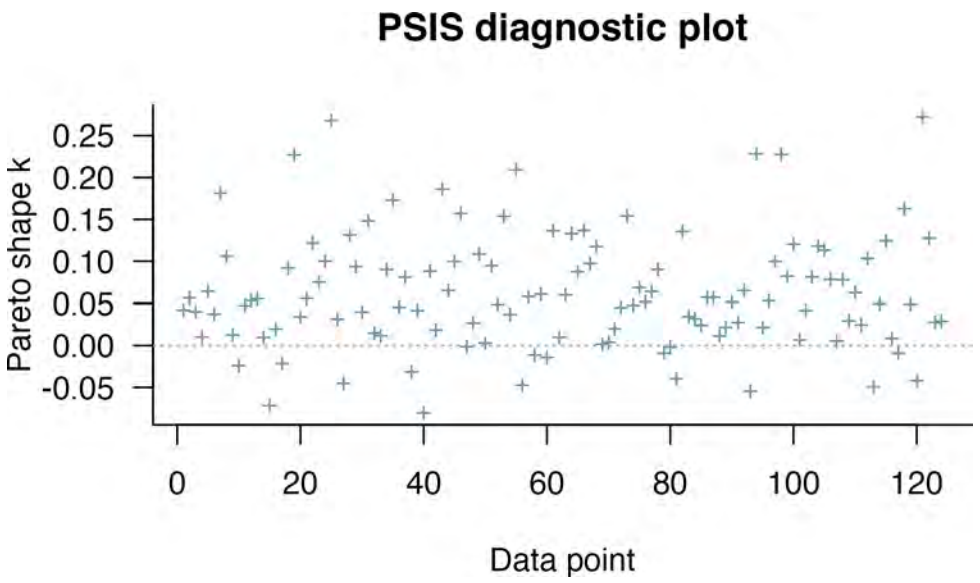
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H2 Traits Model output

```
    Estimate Est.Error      Q2.5     Q97.5
R2 0.3665063 0.0536121 0.2526493 0.4622324
```

Summary of Posterior Distribution

Parameter	Mean	Median	CI	pd
b_Intercept	1.31e-03	1.29e-03	[-0.14, 0.14]	50.70%
b_pubertal_timing_s	0.19	0.19	[0.05, 0.34]	99.54%
b_swan_total_s	0.55	0.55	[0.41, 0.70]	100%
b_sed_percentile_s	0.06	0.06	[-0.08, 0.21]	80.10%
b_pubertal_timing_s:swan_total_s	-0.06	-0.06	[-0.21, 0.08]	80.80%

Parameter	% in ROPE	meaningful
b_Intercept	83.28%	FALSE
b_pubertal_timing_s	10.32%	FALSE
b_swan_total_s	0%	TRUE
b_sed_percentile_s	67.87%	FALSE
b_pubertal_timing_s:swan_total_s	67.81%	FALSE

H2 Traits Model diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

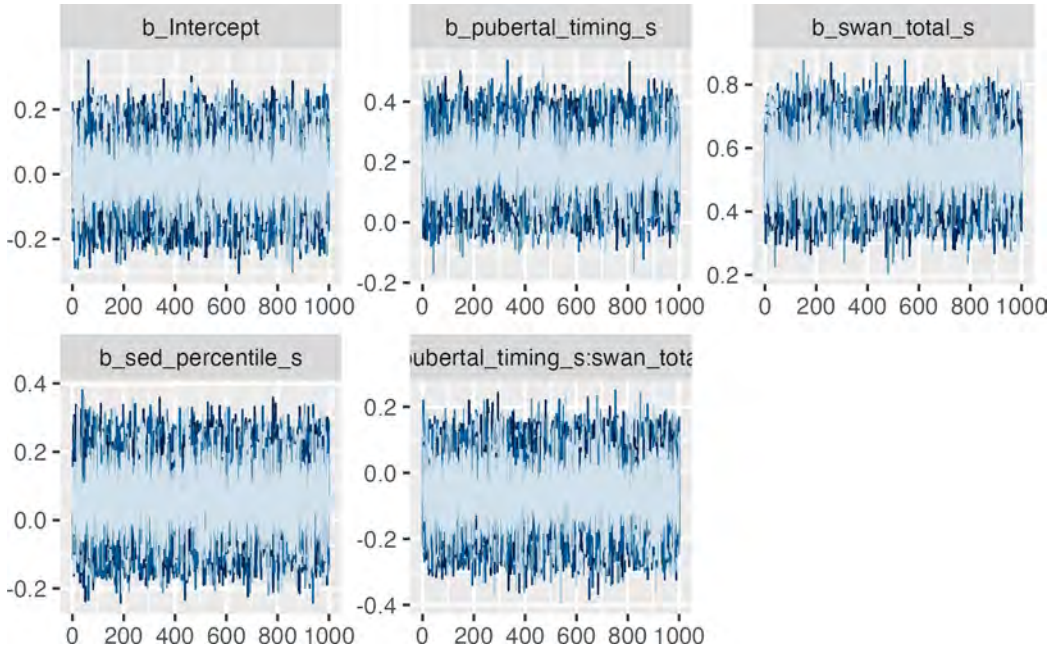
# A tibble: 9 x 4
  variable                         rhat ess_bulk ess_tail
  <chr>                           <dbl>    <dbl>    <dbl>
1 b_Intercept                      1.00  170944.   92972.
2 b_pubertal_timing_s              1.00  161774.   94776.
3 b_swan_total_s                   1.00   39443.   94736.
4 b_sed_percentile_s               1.01    7304.   67816.
5 b_pubertal_timing_s:swan_total_s 1.00  105535.   91836.
6 sigma                            1.00  132547.   93128.
7 Intercept                        1.00  171212.   93129.
8 lprior                           1.00  147837.   93832.
9 lp__                             1.09     814.    1115.


=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```



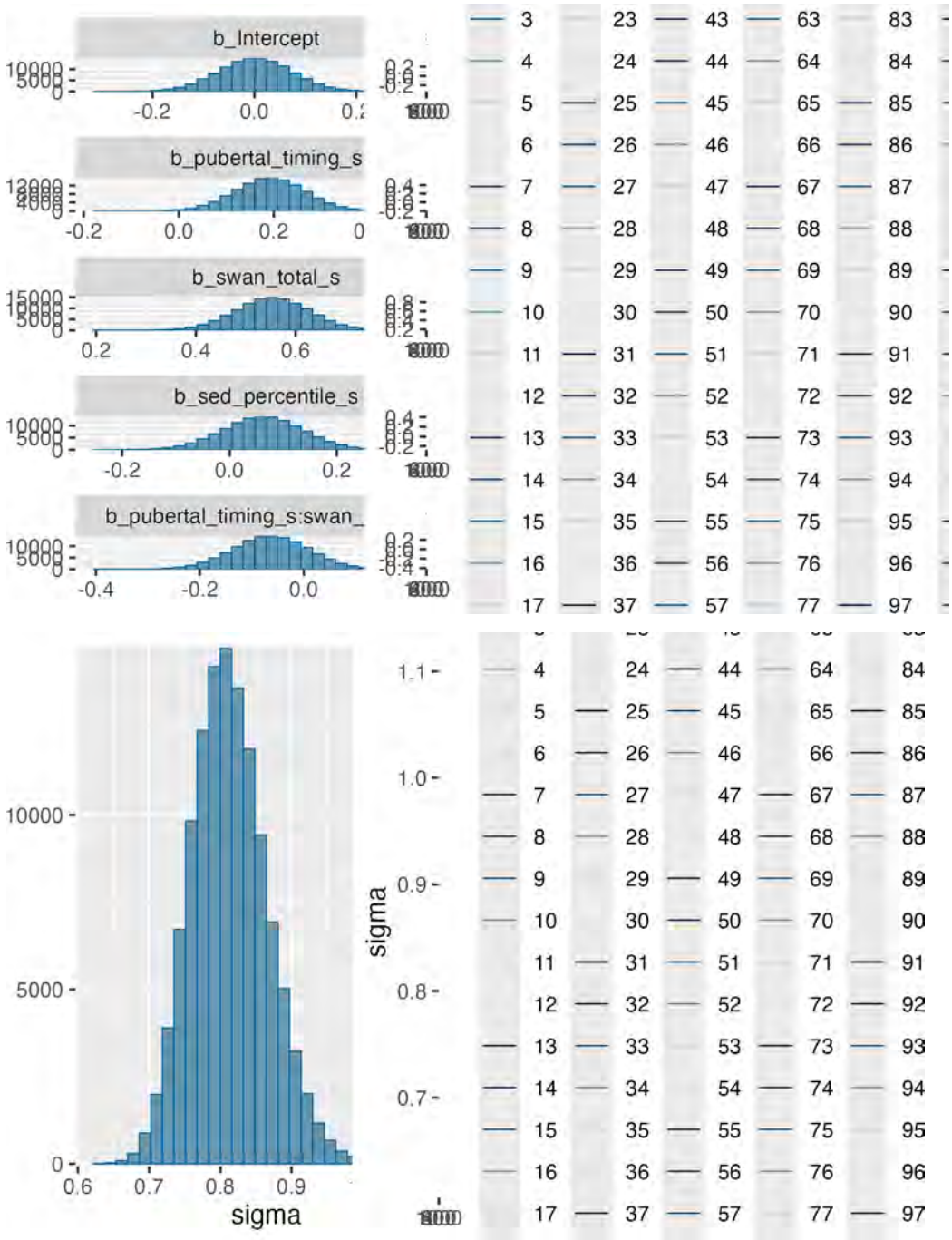

```
=== Prior Summary ===

Priors specified for the model.

          prior     class                         coef group resp dpar nlpar
   normal(0, 1)         b
   normal(0, 1)         b              pubertal_timing_s
   normal(0, 1)         b pubertal_timing_s:swan_total_s
   normal(0, 1)         b               sed_percentile_s
   normal(0, 1)         b                   swan_total_s
   normal(0, 1) Intercept
 exponential(1)     sigma
 lb ub tag       source
                   user
           (vectorized)
           (vectorized)
           (vectorized)
           (vectorized)
                   user
  0                user


=== Posterior Predictive Checks ===
```

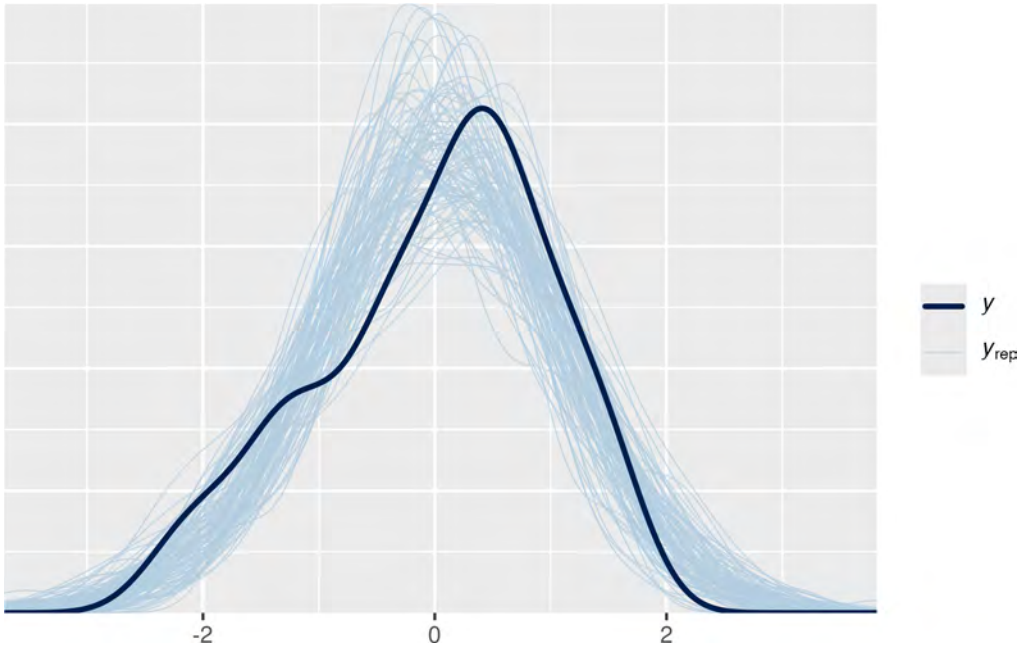
Density plots for each parameter.



```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

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```



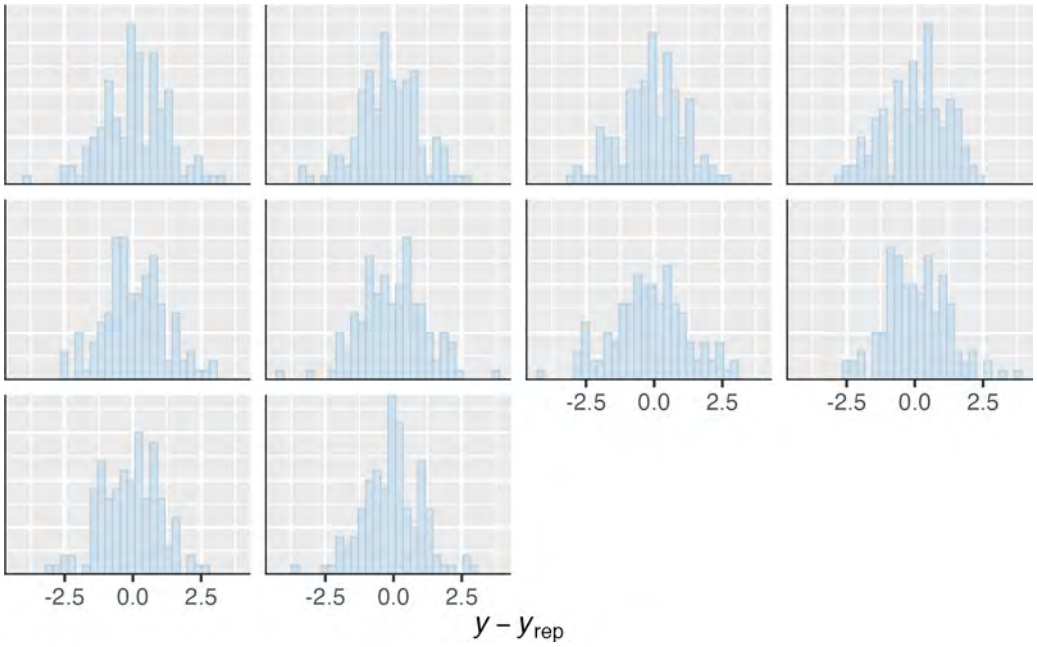

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

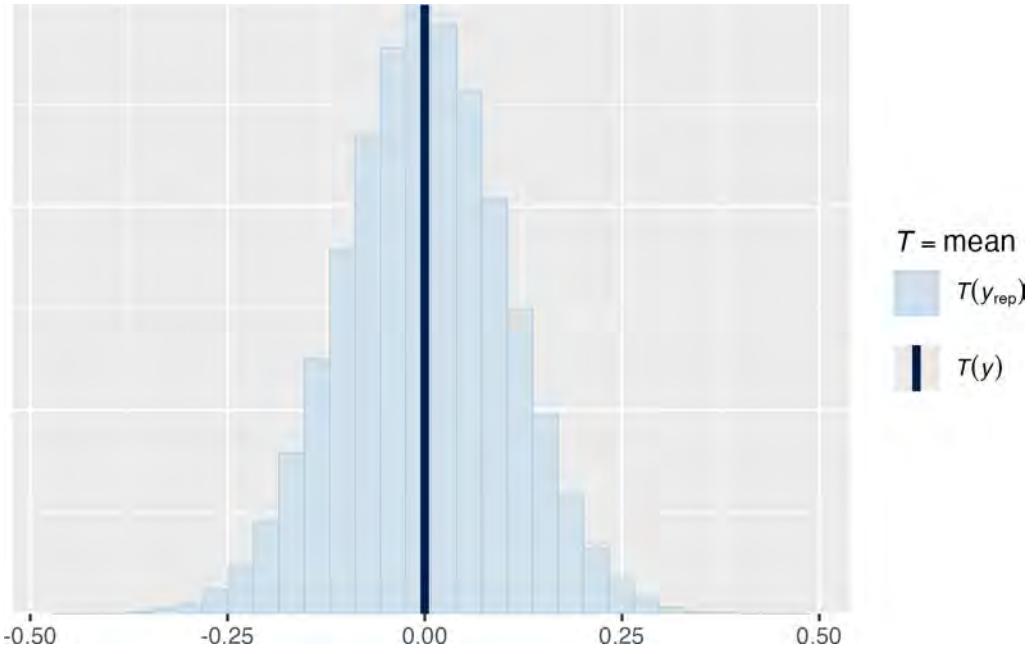
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



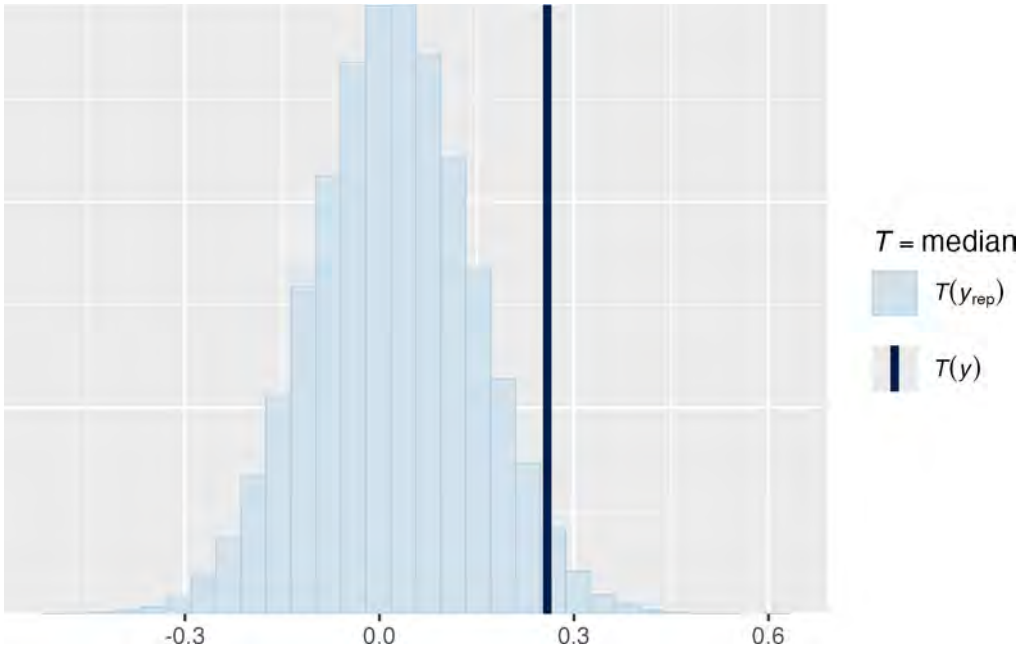

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



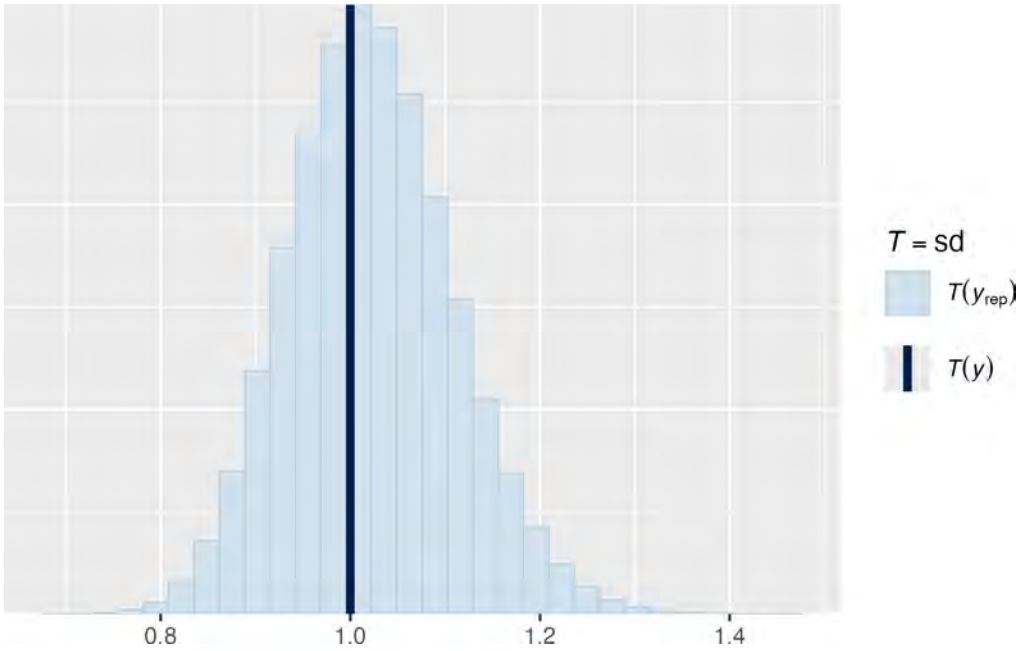

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

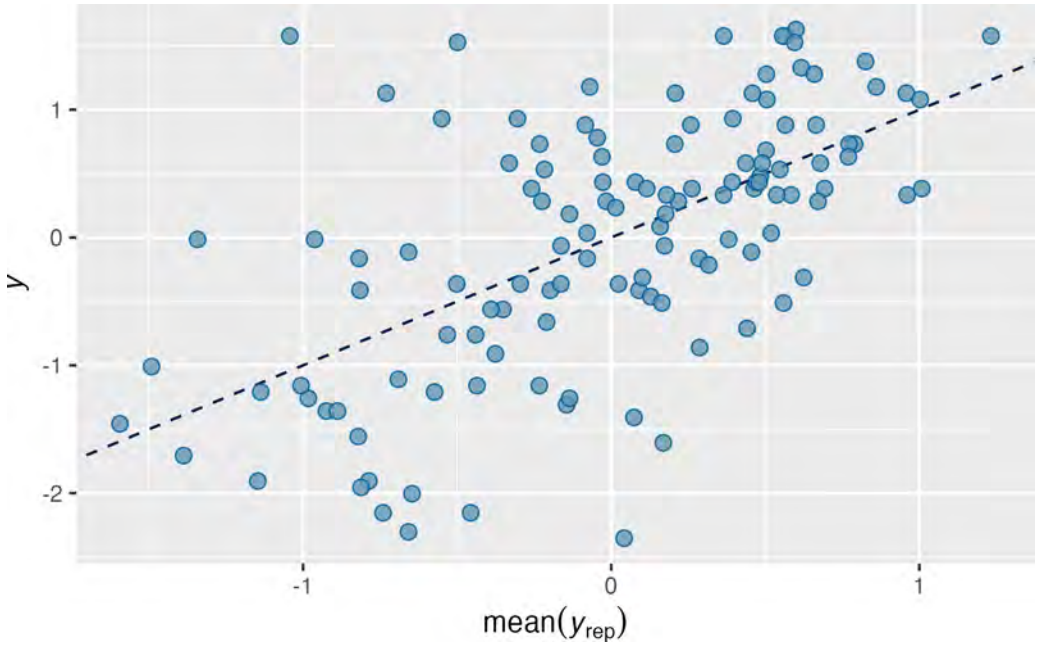



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

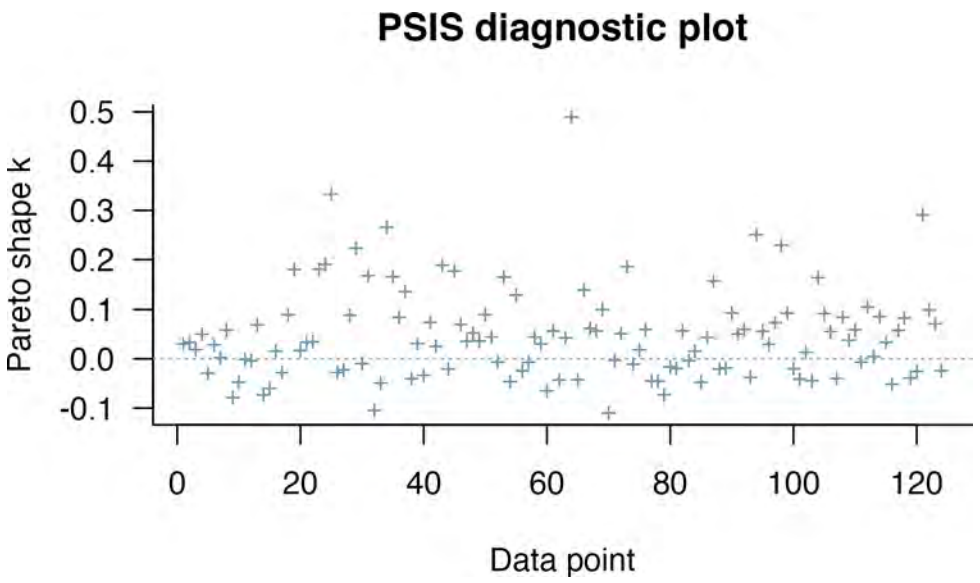
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H2 Traits Model no interaction output

```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.3607555 0.05432147 0.2459129 0.4578504
```

```
Summary of Posterior Distribution

Parameter           |      Mean |    Median |            CI |     pd
------------------------------------------------------------------------
b_Intercept         | -4.06e-04 | -2.65e-04 | [-0.15, 0.14] | 50.13%
b_pubertal_timing_s |      0.19 |      0.19 | [ 0.05, 0.34] | 99.58%
b_swan_total_s      |      0.56 |      0.56 | [ 0.42, 0.71] |   100%
b_sed_percentile_s  |      0.06 |      0.06 | [-0.09, 0.20] | 77.44%

Parameter           | % in ROPE | meaningful
-------------------------------------------
b_Intercept         |    82.87% |      FALSE
b_pubertal_timing_s |    10.12% |      FALSE
b_swan_total_s      |        0% |       TRUE
b_sed_percentile_s  |    70.97% |      FALSE

=== ADHD traits and pubertal timing effects in raw ERS units ===

Warning: Dropping 'draws_df' class as required metadata was removed.
```

effect	mean	lower	upper
swan_raw_effect	0.60	0.44	0.75
timing_raw_effect	1.82	0.45	3.16

H2 Traits Model no interaction diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

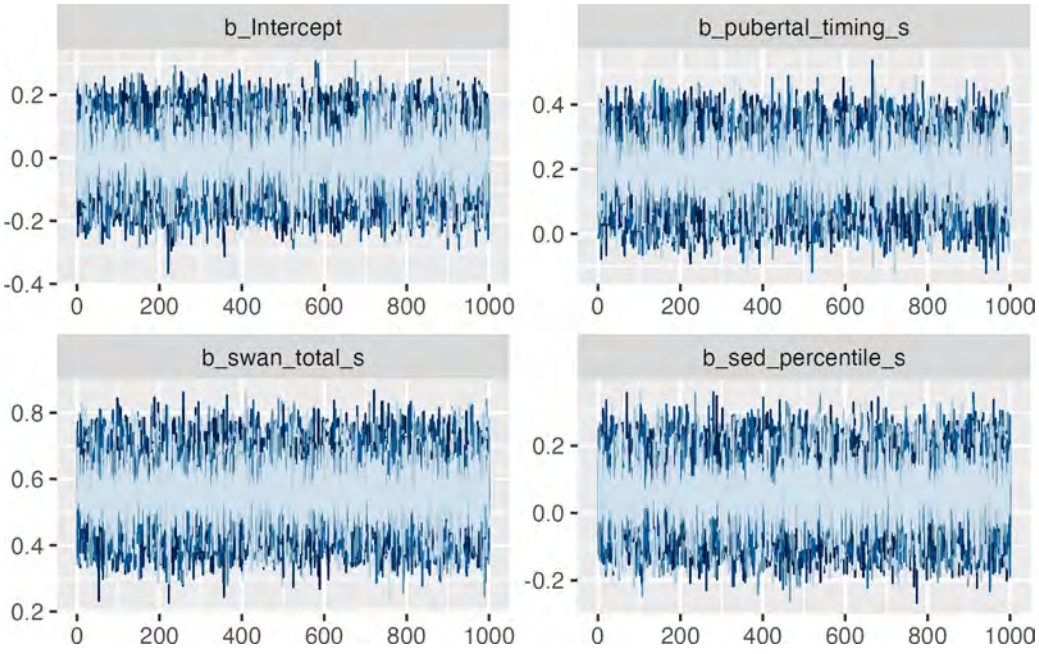
# A tibble: 8 x 4
  variable             rhat ess_bulk ess_tail
  <chr>               <dbl>    <dbl>    <dbl>
1 b_Intercept          1.00  152104.   94410.
2 b_pubertal_timing_s  1.00  145193.   92539.
3 b_swan_total_s       1.00   47185.   92365.
4 b_sed_percentile_s   1.01   10521.   89679.
5 sigma                1.00   41707.   92299.
6 Intercept            1.00  152104.   94410.
7 lprior               1.00  137541.   92601.
8 lp__                 1.11     651.     814.


=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```



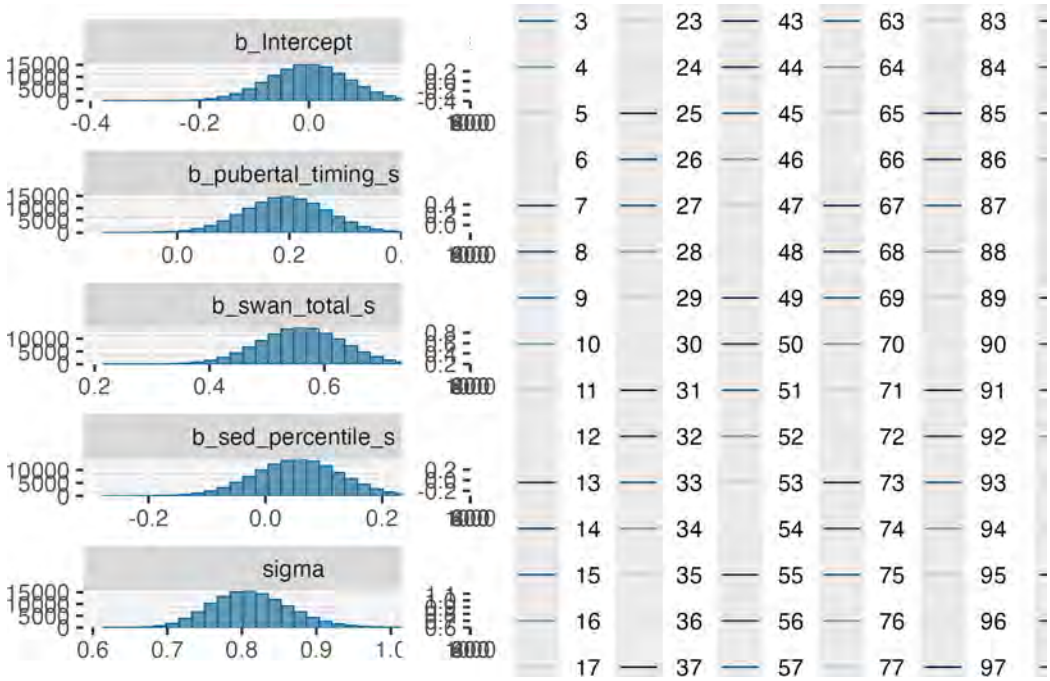

```
=== Prior Summary ===

Priors specified for the model.

          prior     class              coef group resp dpar nlpar lb ub tag
   normal(0, 1)         b
   normal(0, 1)         b pubertal_timing_s
   normal(0, 1)         b  sed_percentile_s
   normal(0, 1)         b      swan_total_s
   normal(0, 1) Intercept
 exponential(1)     sigma                                                0
       source
         user
 (vectorized)
 (vectorized)
 (vectorized)
         user
         user


=== Posterior Predictive Checks ===
```

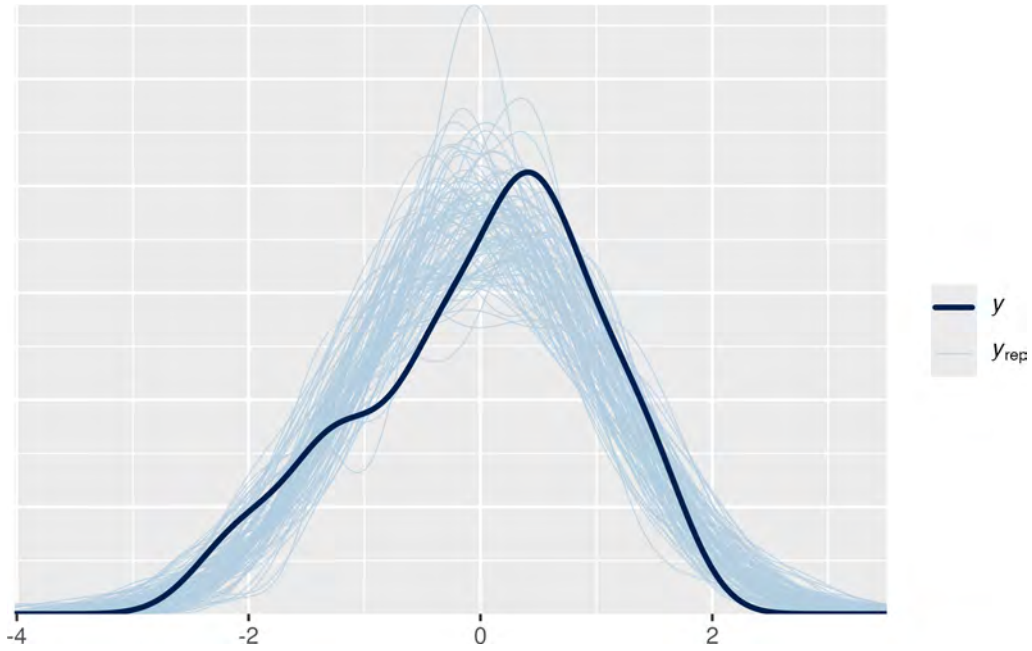
Density plots for each parameter.



=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results with caution until a more principled approach has been implemented.



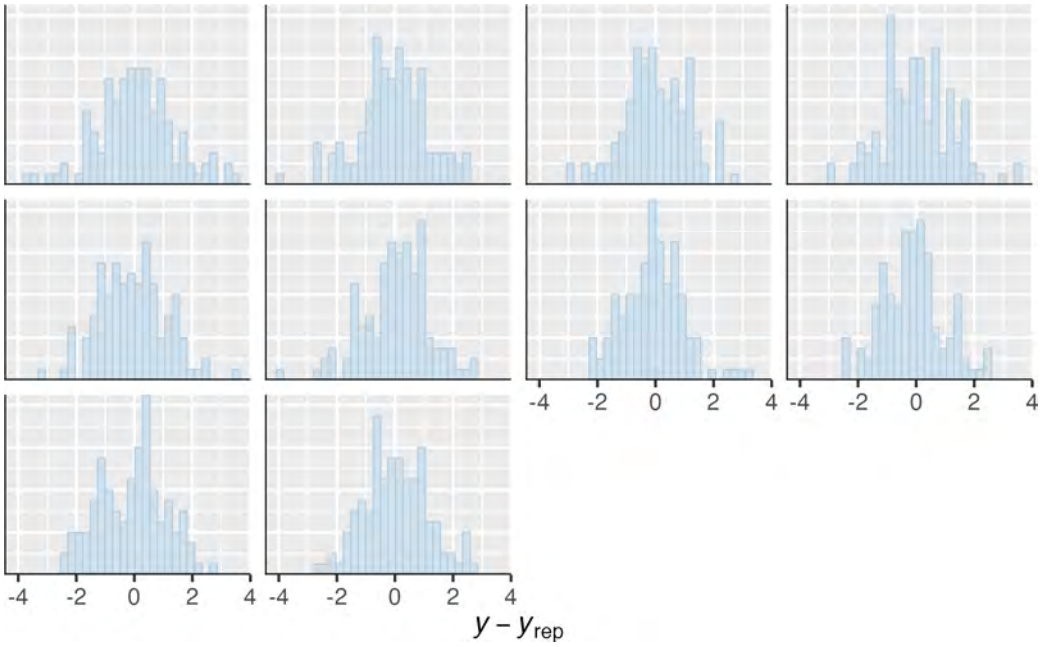

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

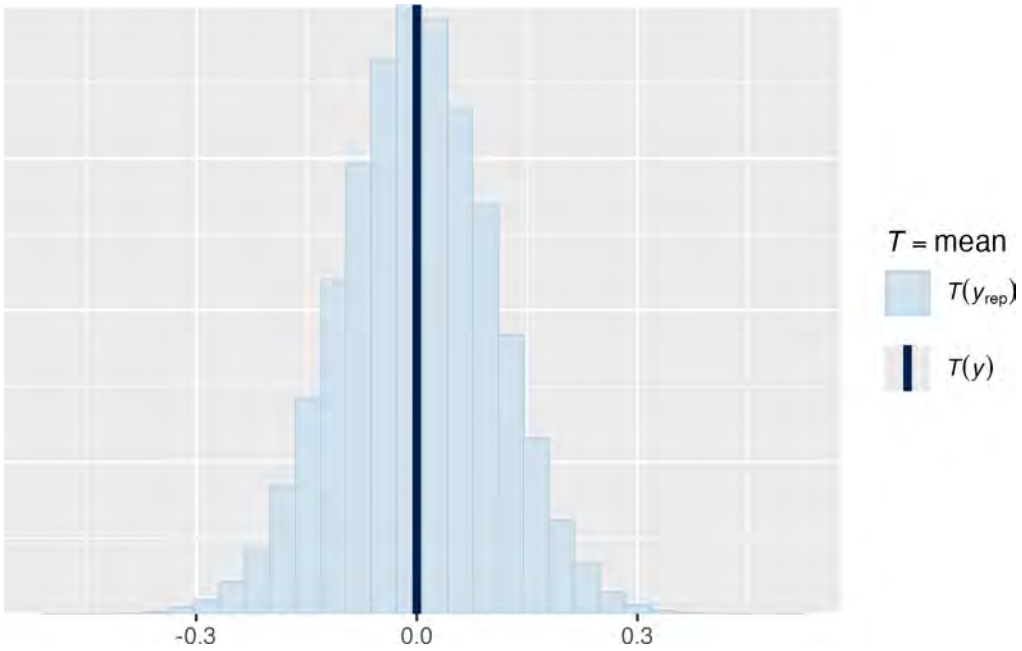
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



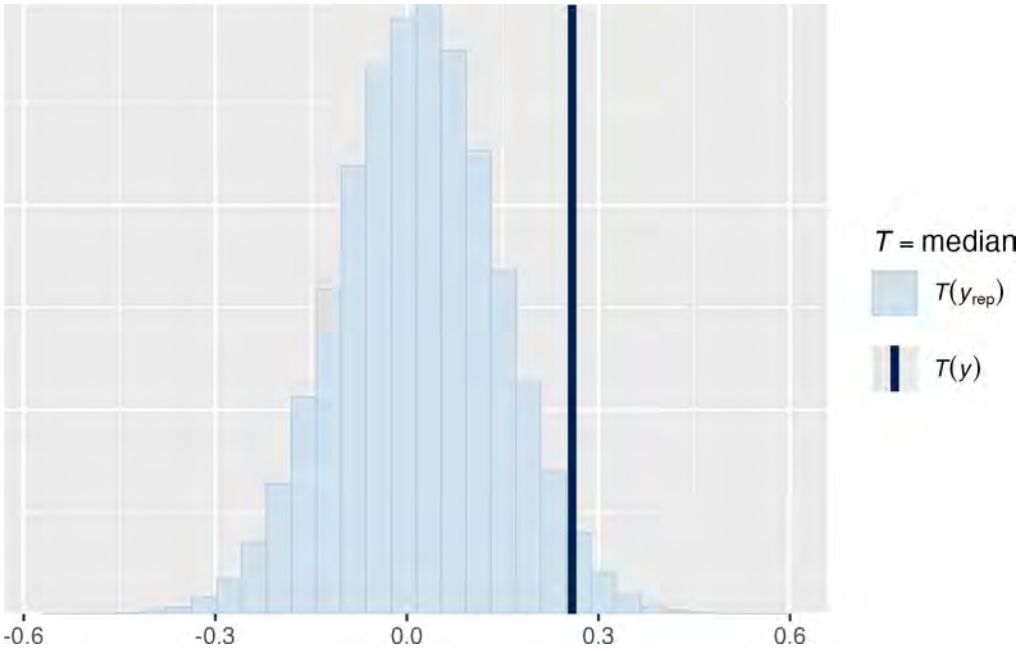

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



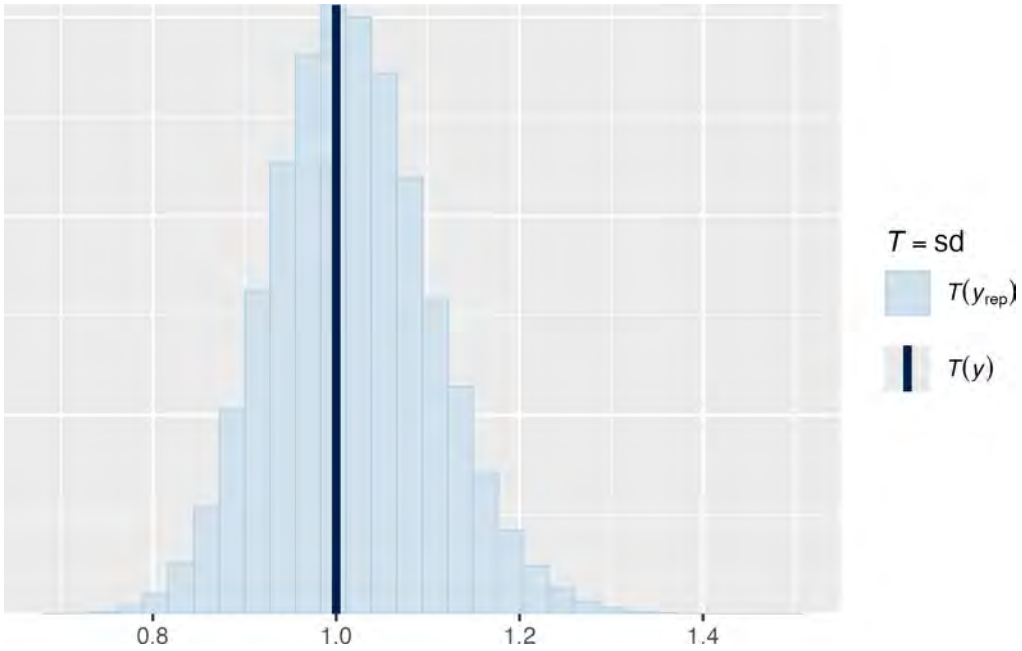

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

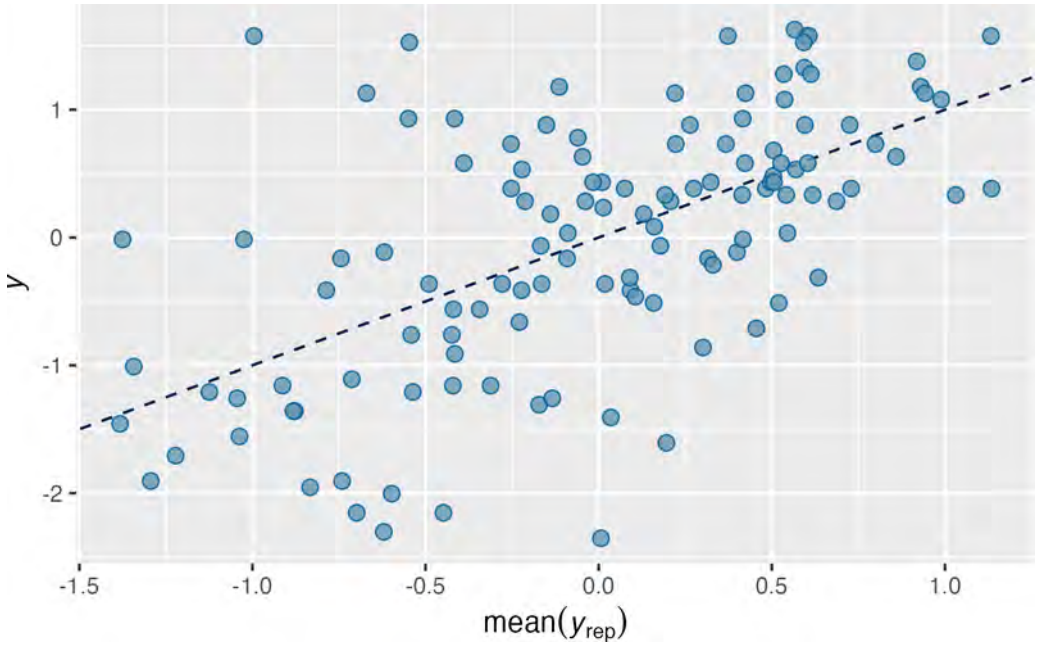



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

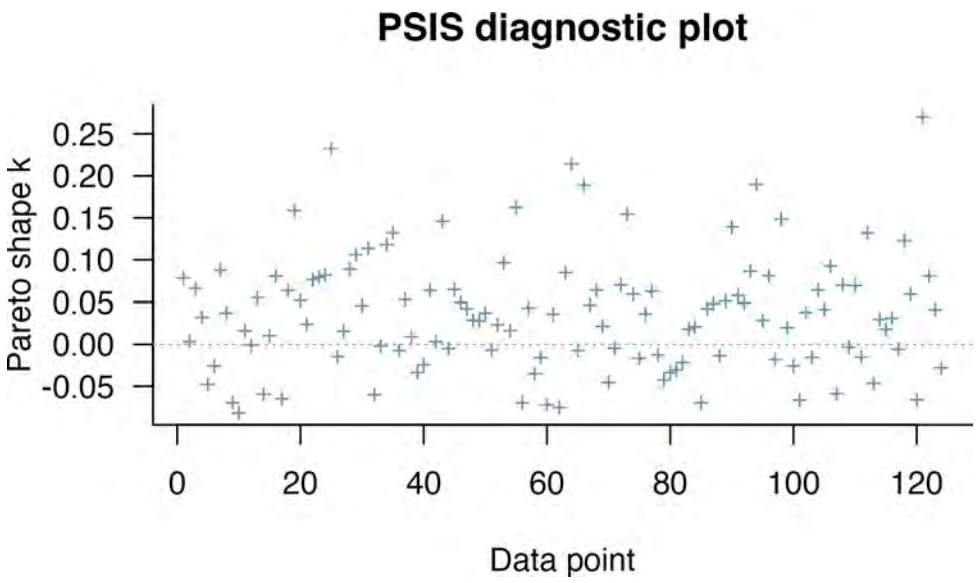
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H1 Menarche Status Model output

```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.3158884 0.05557063 0.2006375 0.4166143
```

```
Summary of Posterior Distribution

Parameter                          |  Mean | Median |             CI |     pd
-------------------------------------------------------------------------------
b_Intercept                        | -1.04 |  -1.04 | [-1.48, -0.59] |   100%
b_menarche_statusY                 |  0.53 |   0.53 | [ 0.00,  1.06] | 97.47%
b_adhd_statusY                     |  0.98 |   0.98 | [ 0.34,  1.61] | 99.86%
b_age_cont_s                       |  0.01 |   0.01 | [-0.17,  0.19] | 55.42%
b_sed_percentile_s                 |  0.10 |   0.10 | [-0.06,  0.25] | 89.14%
b_menarche_statusY:adhd_statusY    | -0.04 |  -0.04 | [-0.74,  0.68] | 53.98%

Parameter                          | % in ROPE | meaningful
-----------------------------------------------------------
b_Intercept                        | 3.33e-03% |       TRUE
b_menarche_statusY                 |     4.50% |       TRUE
b_adhd_statusY                     |     0.30% |       TRUE
b_age_cont_s                       |    72.00% |      FALSE
b_sed_percentile_s                 |    50.98% |      FALSE
b_menarche_statusY:adhd_statusY    |    21.59% |      FALSE
```

H1 Menarche Status Model diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

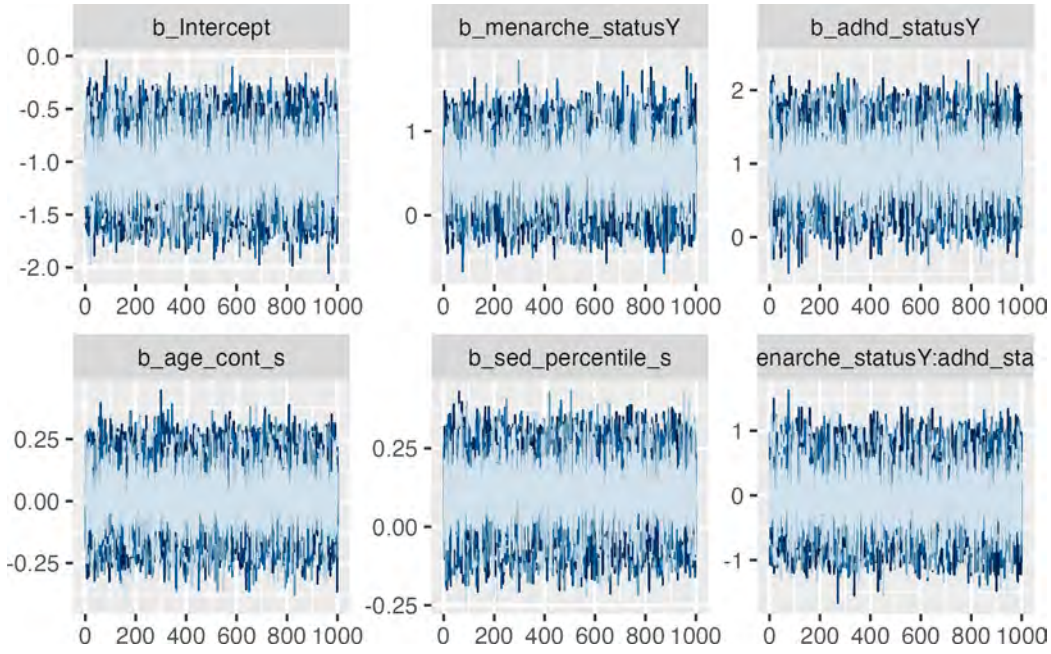
# A tibble: 10 x 4
   variable                         rhat ess_bulk ess_tail
   <chr>                           <dbl>    <dbl>    <dbl>
 1 b_Intercept                      1.00  102389.   98473.
 2 b_menarche_statusY               1.00  100593.   90512.
 3 b_adhd_statusY                   1.00   88454.   84467.
 4 b_age_cont_s                     1.00  119295.   86794.
 5 b_sed_percentile_s               1.01    5677.   42133.
 6 b_menarche_statusY:adhd_statusY  1.00   83269.   80242.
 7 sigma                            1.00  114844.   85588.
 8 Intercept                        1.00  126352.   84179.
 9 lprior                           1.00   84564.   79427.
10 lp__                             1.03    2821.    4521.


=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```



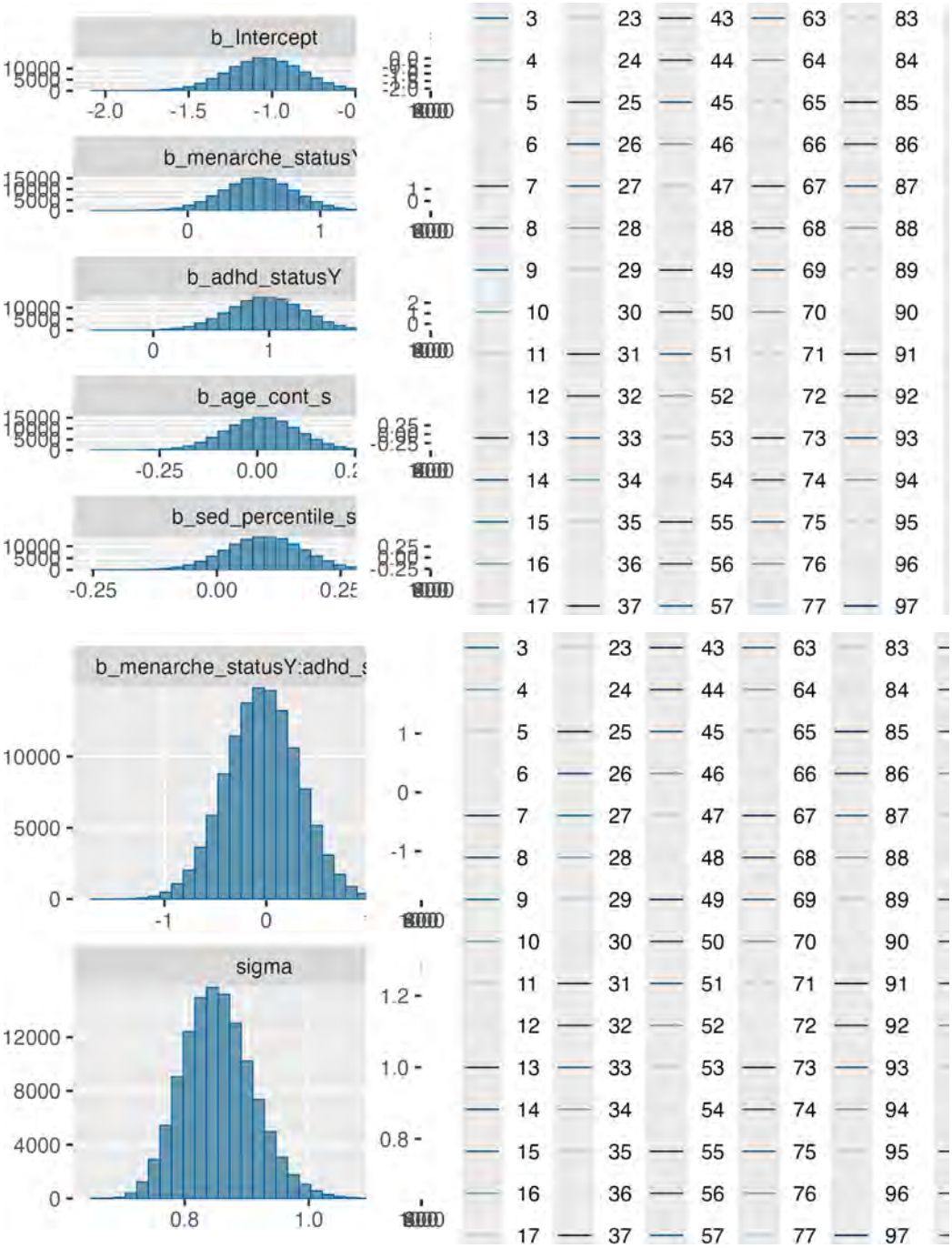

```
=== Prior Summary ===

Priors specified for the model.

          prior     class                         coef group resp dpar nlpar
   normal(0, 1)         b
   normal(0, 1)         b                 adhd_statusY
   normal(0, 1)         b                   age_cont_s
   normal(0, 1)         b              menarche_statusY
   normal(0, 1)         b menarche_statusY:adhd_statusY
   normal(0, 1)         b              sed_percentile_s
   normal(0, 1) Intercept
 exponential(1)     sigma
 lb ub tag       source
                   user
           (vectorized)
           (vectorized)
           (vectorized)
           (vectorized)
           (vectorized)
                   user
  0                user
```

```
=== Posterior Predictive Checks ===

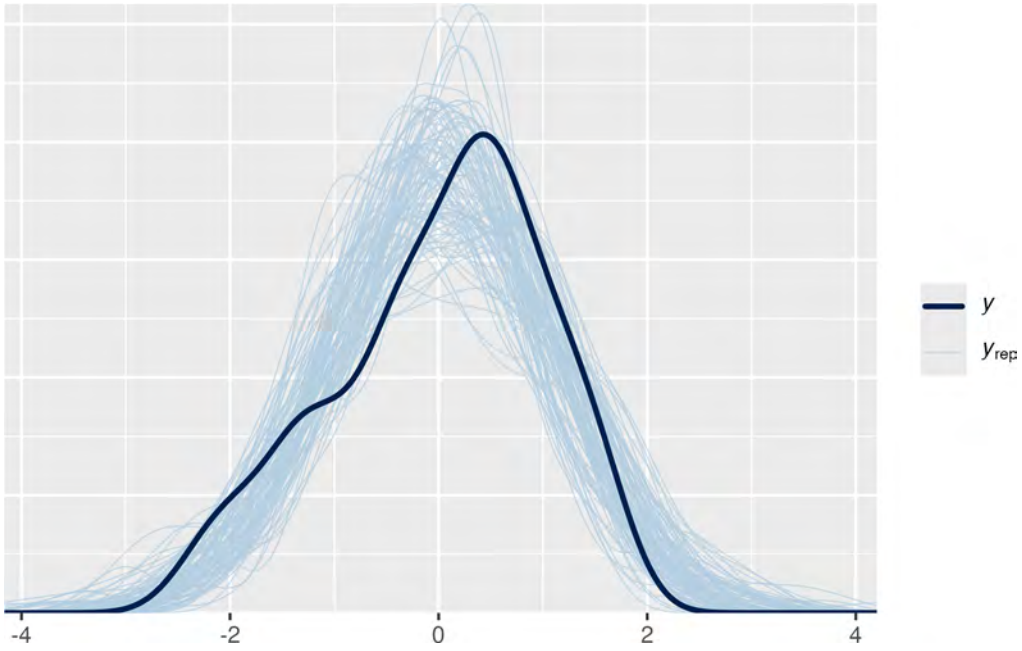
Density plots for each parameter.
```



```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```



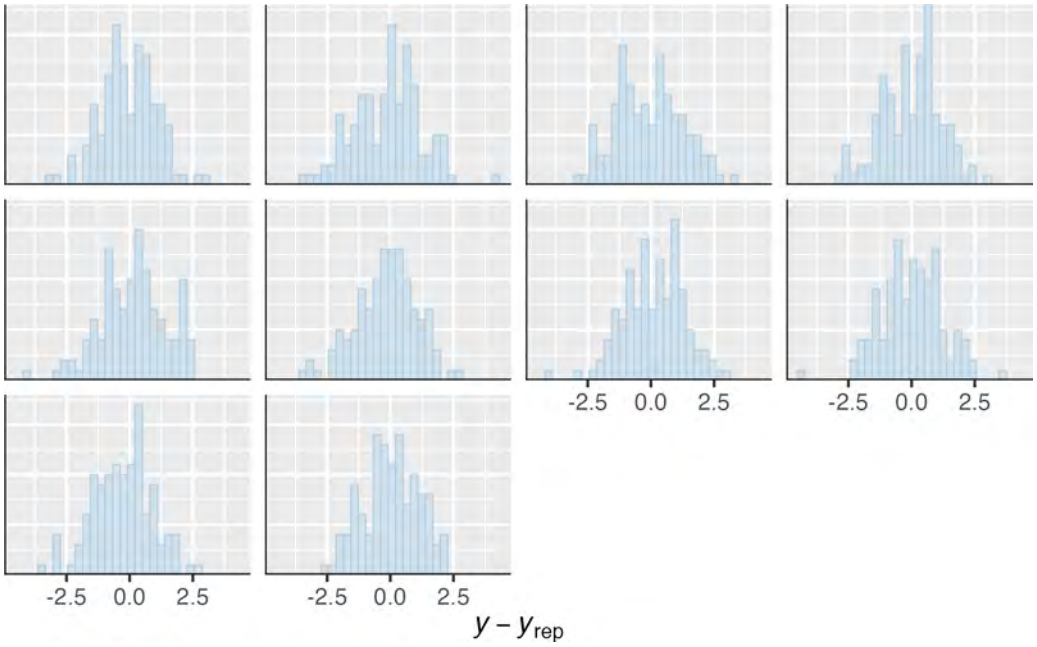

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

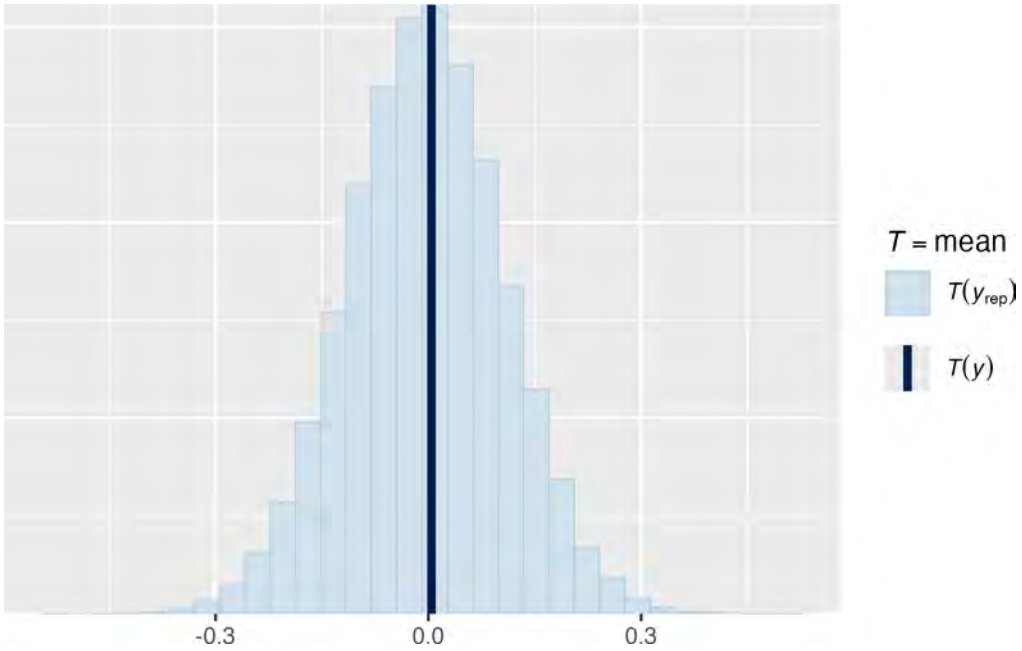
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



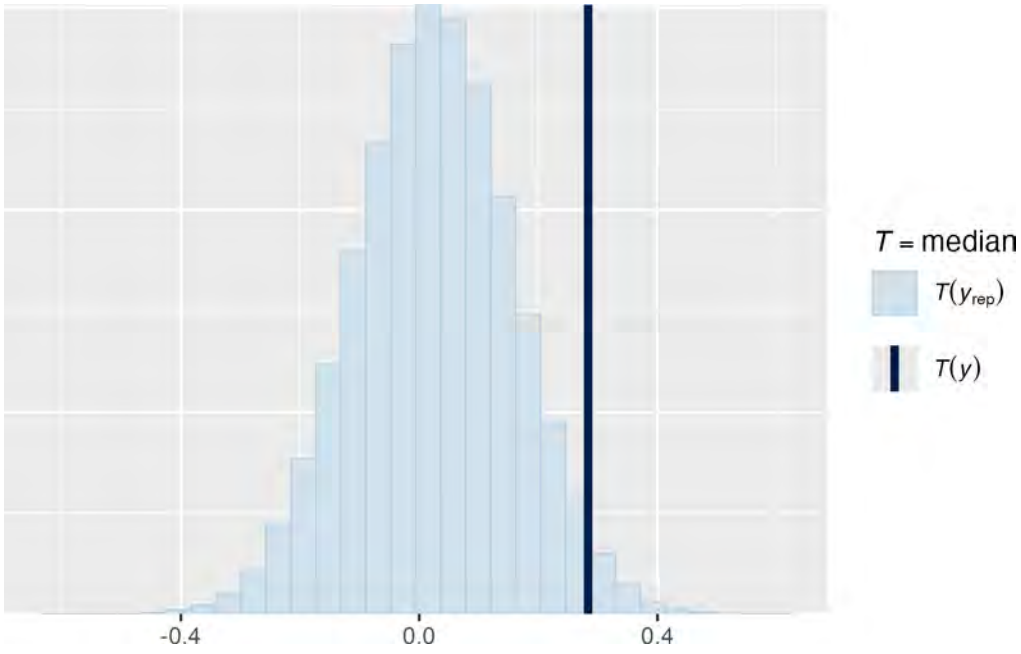

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



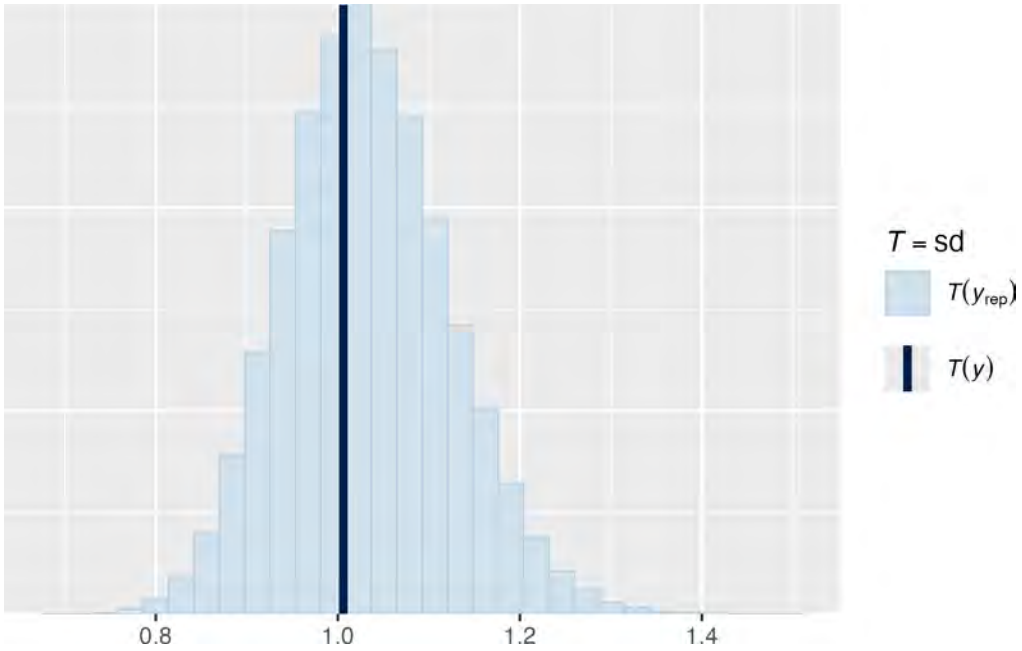

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

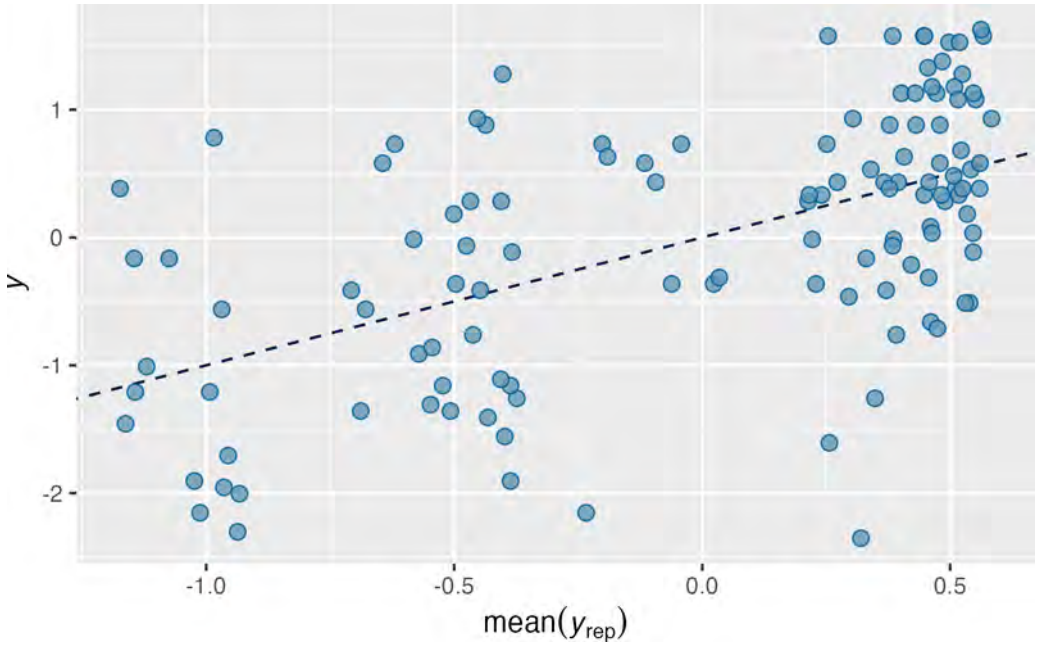



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

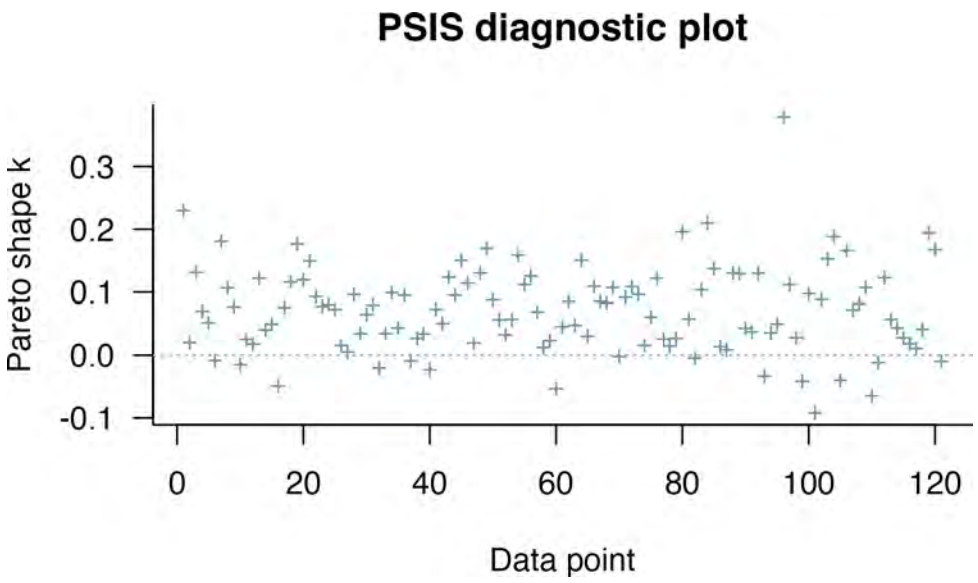
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H1 Menarche Status Model no interaction output

```
    Estimate  Est.Error      Q2.5   Q97.5
R2 0.3142051 0.05582554 0.1983157 0.41635
```

```
Summary of Posterior Distribution

Parameter          |  Mean | Median |             CI |     pd | % in ROPE | meaningful
-------------------------------------------------------------------------------------
b_Intercept        | -1.03 |  -1.03 | [-1.44, -0.63] |   100% |        0% |       TRUE
b_menarche_statusY |  0.52 |   0.52 | [ 0.08,  0.97] | 98.81% |     2.96% |       TRUE
b_adhd_statusY     |  0.95 |   0.95 | [ 0.62,  1.28] |   100% |        0% |       TRUE
b_age_cont_s       |  0.01 |   0.01 | [-0.17,  0.19] | 55.16% |    72.52% |      FALSE
b_sed_percentile_s |  0.10 |   0.10 | [-0.06,  0.25] | 88.94% |    51.18% |      FALSE
```

```
=== ADHD and menarche differences in raw outcome units ===
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

effect	mean	lower	upper
adhd_raw_effect	19.11	12.52	25.79
menarche_raw_effect	10.39	1.57	19.46

H1 Menarche Status Model no interaction diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

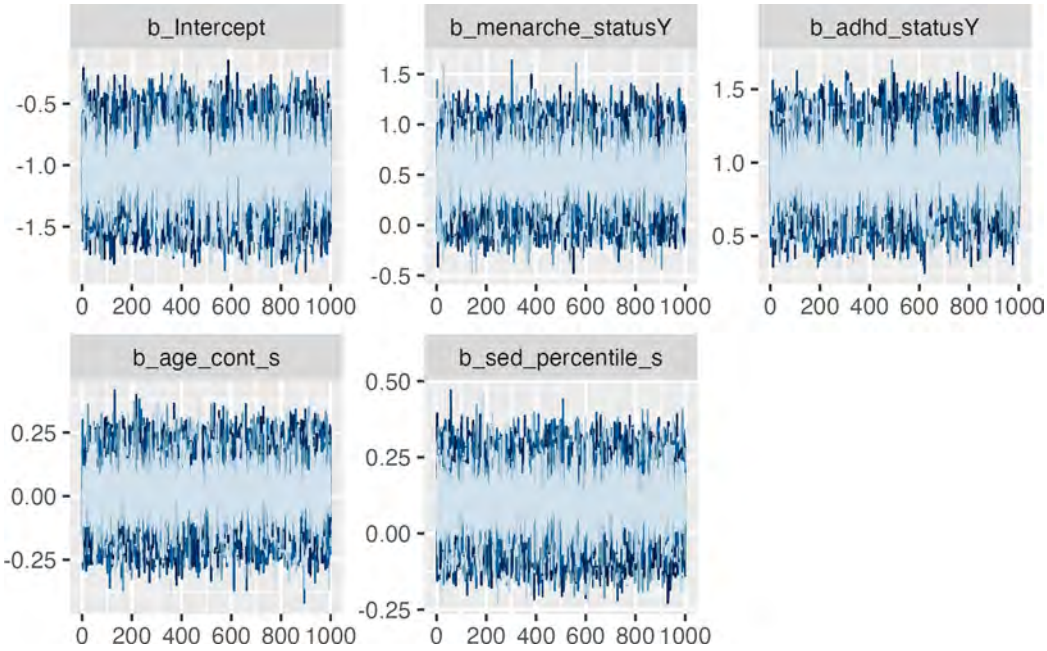
# A tibble: 9 x 4
  variable            rhat ess_bulk ess_tail
  <chr>              <dbl>    <dbl>    <dbl>
1 b_Intercept         1.00  149197.  100747.
2 b_menarche_statusY  1.00  136612.   97059.
3 b_adhd_statusY      1.00  158360.   92462.
4 b_age_cont_s        1.00  137926.   97343.
5 b_sed_percentile_s  1.01    6077.   76445.
6 sigma               1.00  149421.   92781.
7 Intercept           1.00  163442.   90587.
8 lprior              1.00  146786.   96100.
9 lp__                1.03    2385.    3401.


=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```



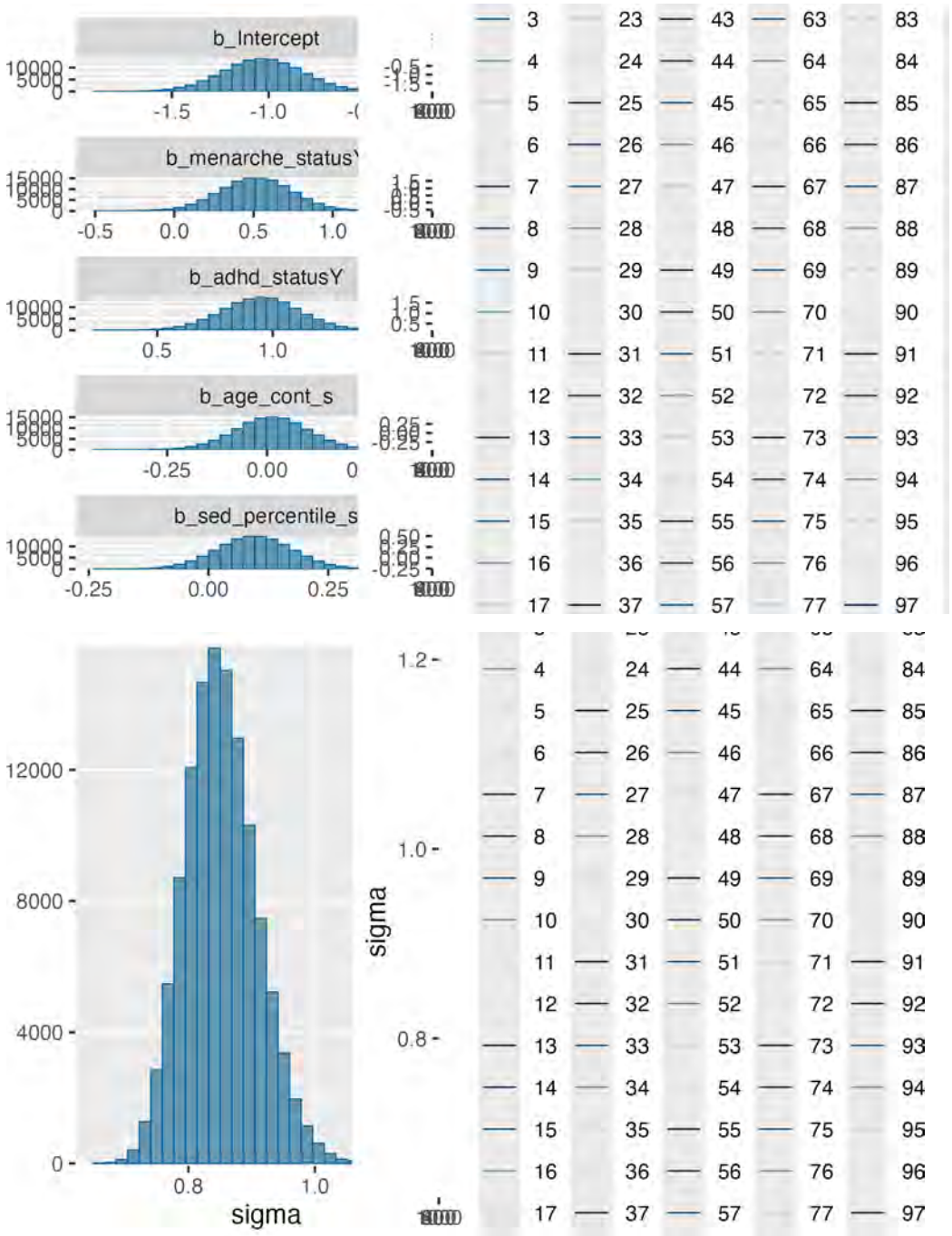

```
=== Prior Summary ===

Priors specified for the model.

          prior     class             coef group resp dpar nlpar lb ub tag
   normal(0, 1)         b
   normal(0, 1)         b     adhd_statusY
   normal(0, 1)         b       age_cont_s
   normal(0, 1)         b menarche_statusY
   normal(0, 1)         b sed_percentile_s
   normal(0, 1) Intercept
 exponential(1)     sigma                                         0
       source
         user
 (vectorized)
 (vectorized)
 (vectorized)
 (vectorized)
         user
         user


=== Posterior Predictive Checks ===
```

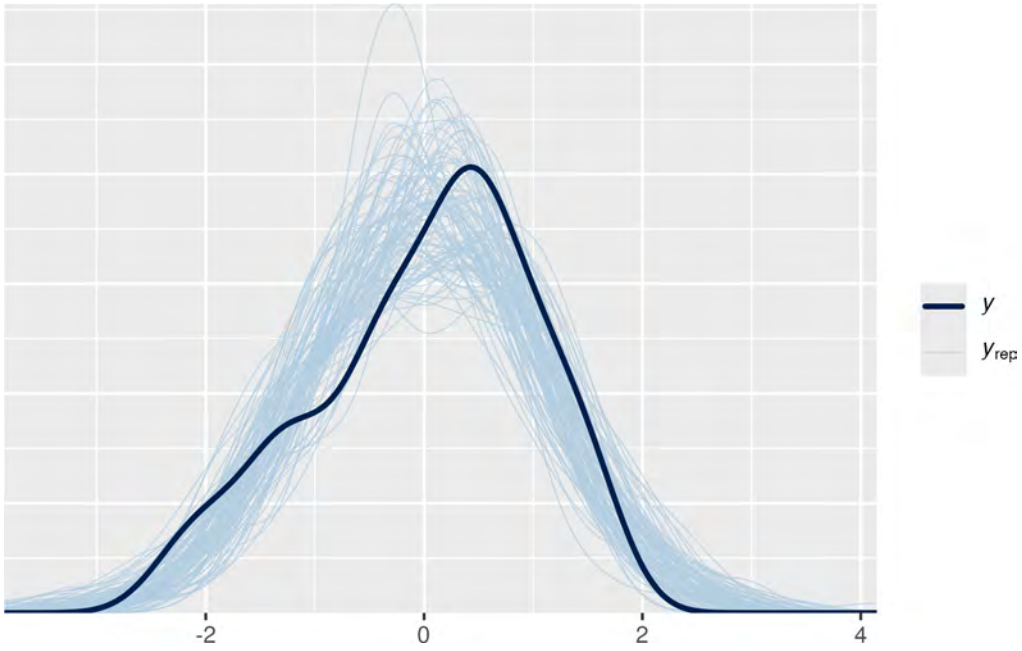
Density plots for each parameter.



```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```



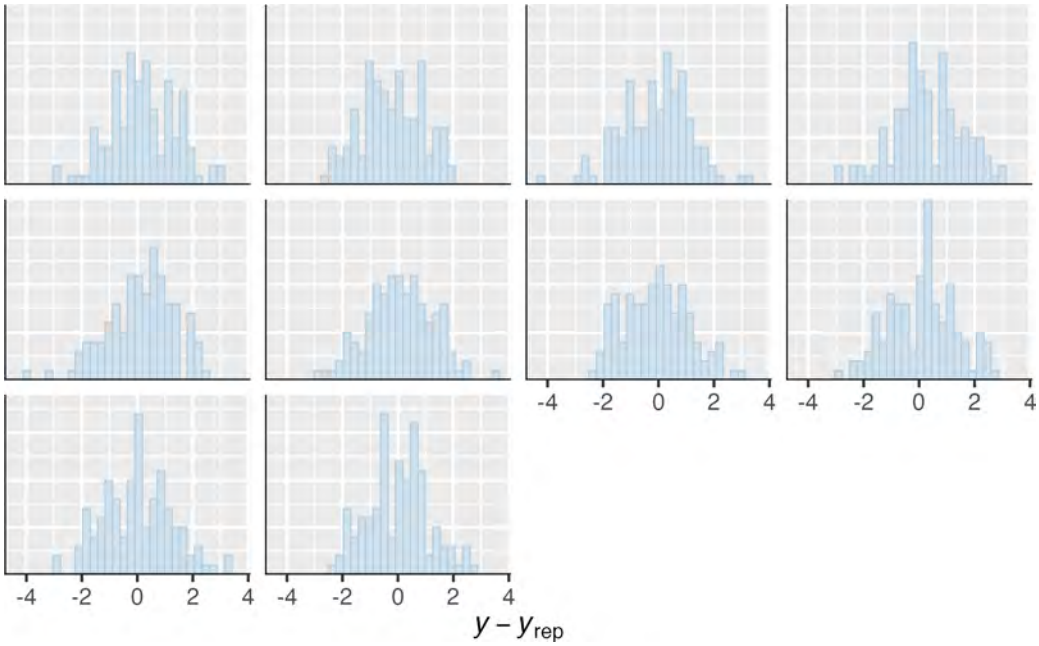

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

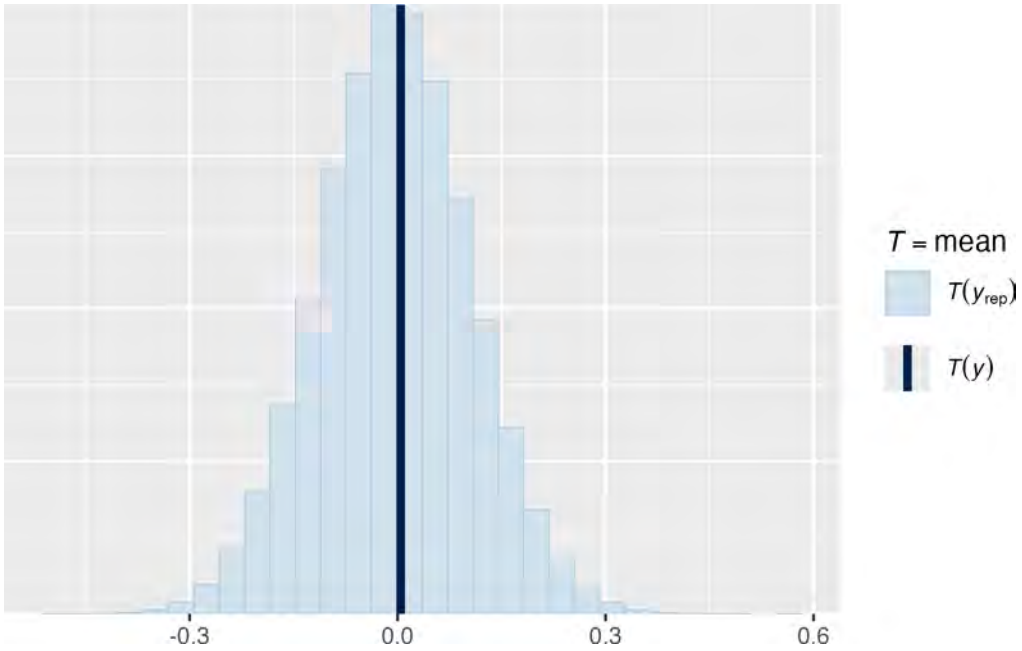
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



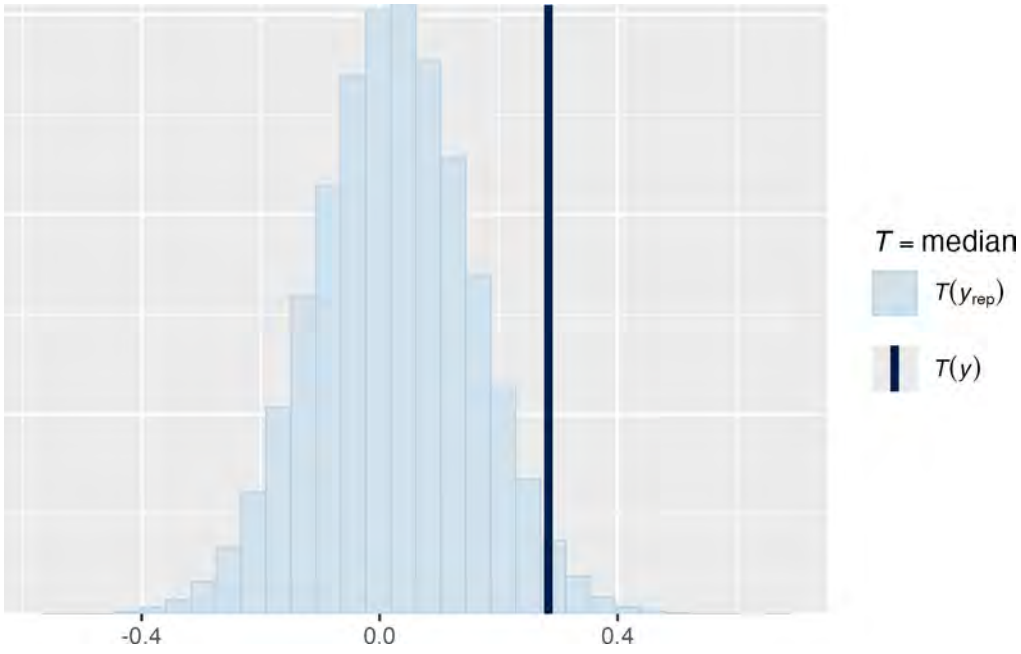

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



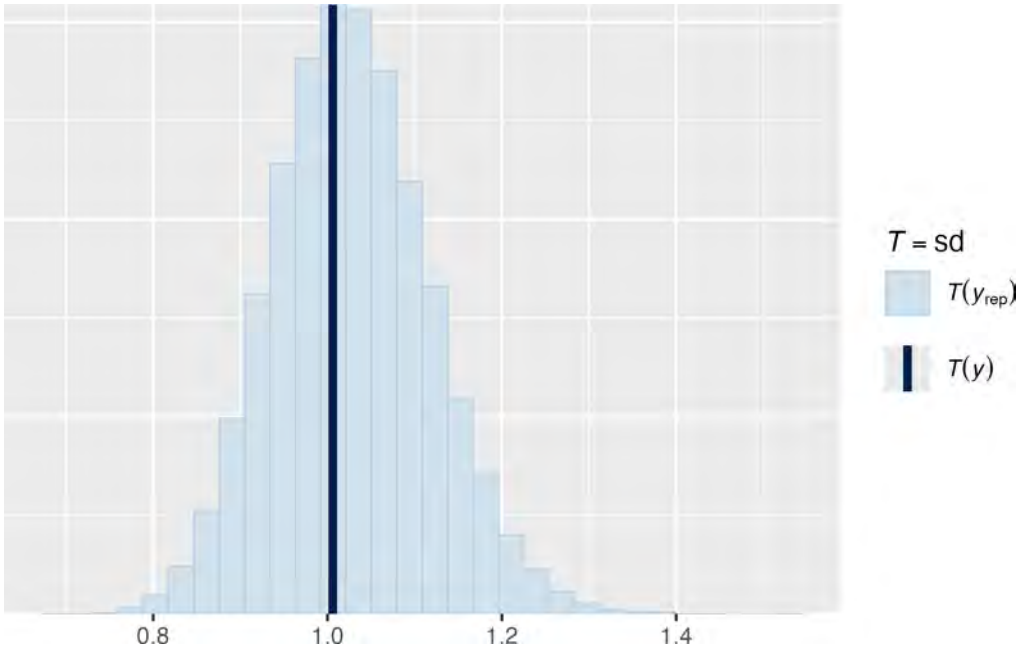

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

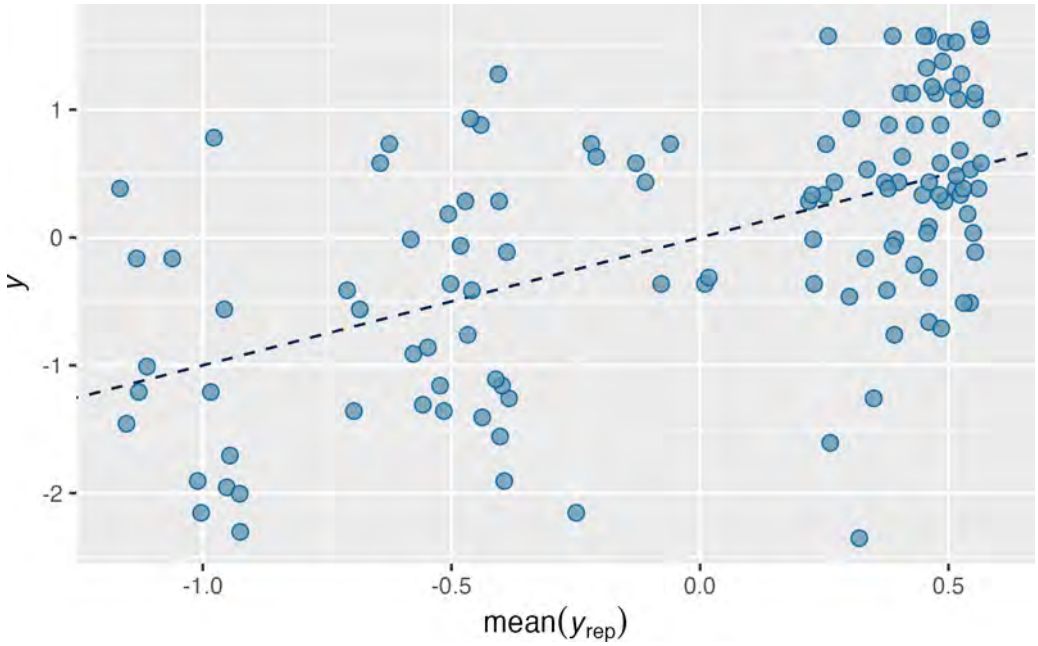



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

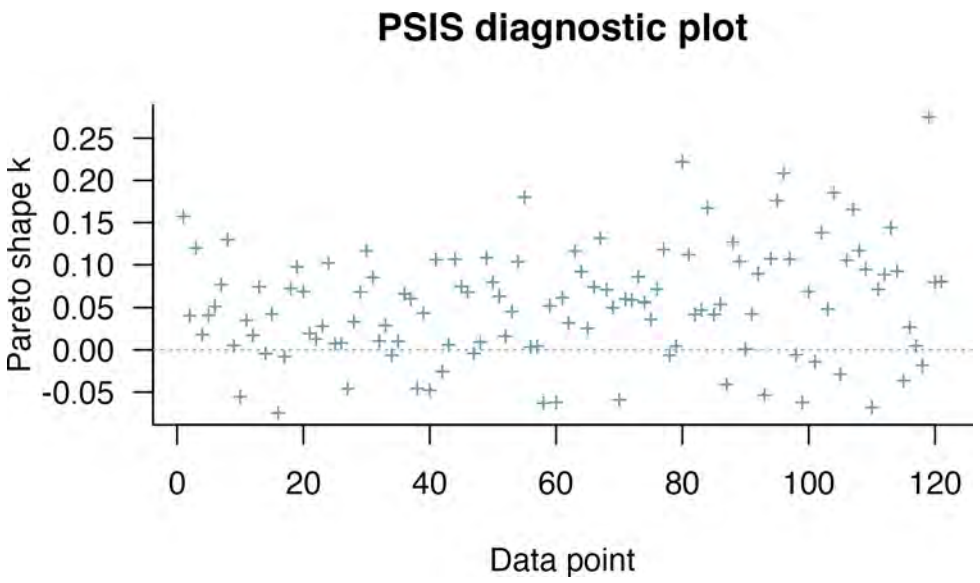
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H2 Menarche Timing Model output

```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.2406736 0.06242984 0.1160537 0.3587444
```

```
Summary of Posterior Distribution

Parameter                          |  Mean | Median |             CI |      pd
-------------------------------------------------------------------------------
b_Intercept                        | -0.47 |  -0.47 | [-0.78, -0.16] |  99.82%
b_menarche_timing_s                |  0.06 |   0.06 | [-0.26,  0.38] |  64.72%
b_adhd_statusY                     |  0.90 |   0.90 | [ 0.54,  1.28] | 100.00%
b_sed_percentile_s                 |  0.17 |   0.17 | [ 0.00,  0.33] |  97.42%
b_menarche_timing_s:adhd_statusY   | -0.13 |  -0.13 | [-0.50,  0.24] |  75.97%

Parameter                          | % in ROPE | meaningful
-------------------------------------------------------------
b_Intercept                        |     1.04% |       TRUE
b_menarche_timing_s                |    43.61% |      FALSE
b_adhd_statusY                     | 3.33e-03% |       TRUE
b_sed_percentile_s                 |    21.87% |       TRUE
b_menarche_timing_s:adhd_statusY   |    32.12% |      FALSE
```

H2 Menarche Timing Model diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

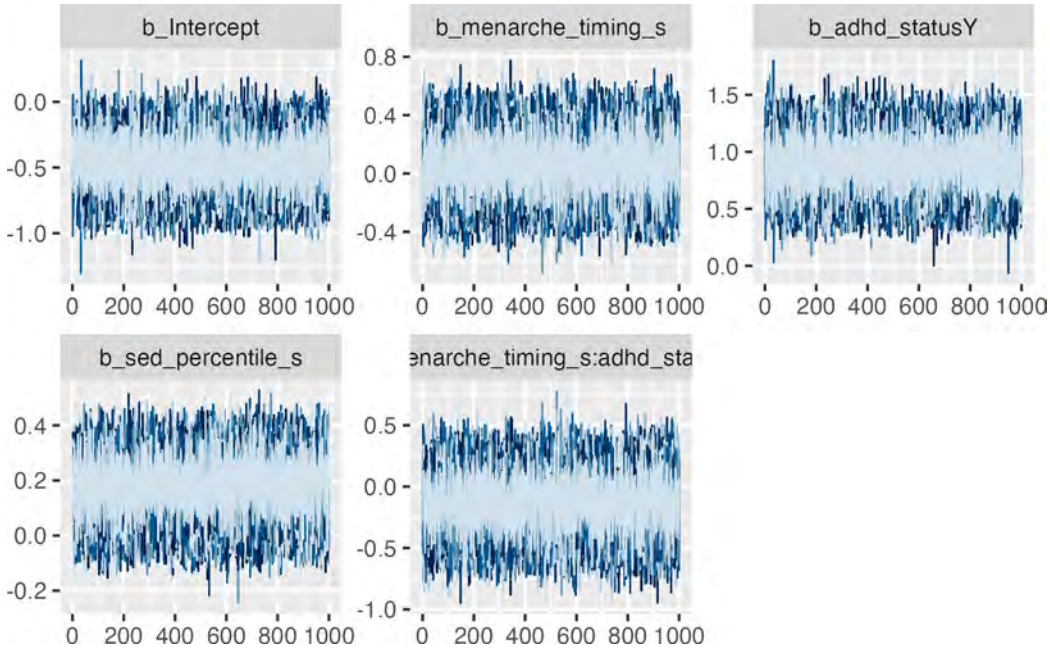
# A tibble: 9 x 4
  variable                         rhat ess_bulk ess_tail
  <chr>                           <dbl>    <dbl>    <dbl>
1 b_Intercept                      1.00  119577.   87293.
2 b_menarche_timing_s              1.00   77485.   79656.
3 b_adhd_statusY                   1.00  119869.   84177.
4 b_sed_percentile_s               1.02    4981.   48080.
5 b_menarche_timing_s:adhd_statusY 1.00   77634.   80656.
6 sigma                            1.00  109439.   86769.
7 Intercept                        1.00  122058.   87180.
8 lprior                           1.00  108284.   84612.
9 lp__                             1.03    2557.    3374.


=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```



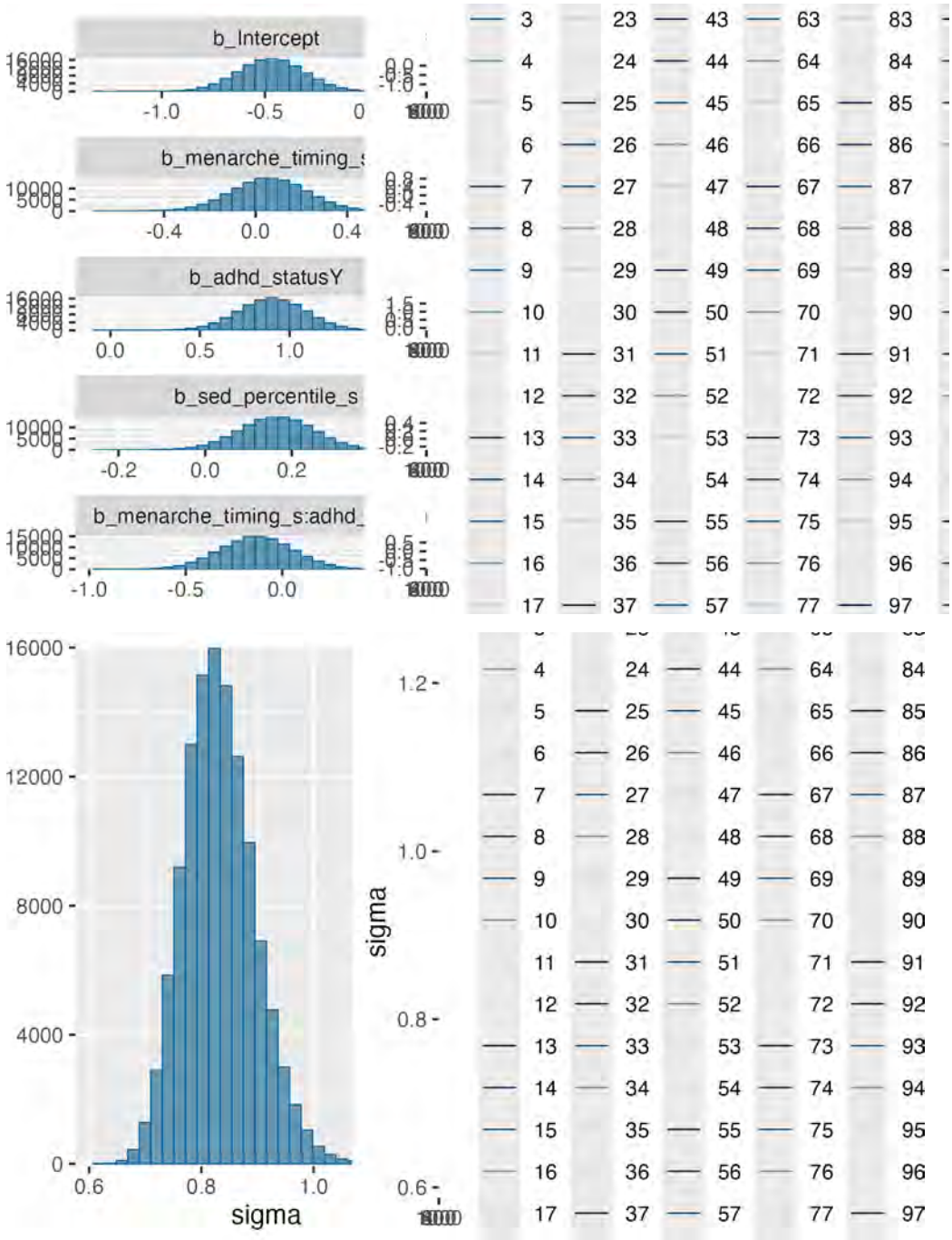

```
=== Prior Summary ===

Priors specified for the model.

          prior     class                           coef group resp dpar nlpar
   normal(0, 1)         b
   normal(0, 1)         b                   adhd_statusY
   normal(0, 1)         b              menarche_timing_s
   normal(0, 1)         b menarche_timing_s:adhd_statusY
   normal(0, 1)         b               sed_percentile_s
   normal(0, 1) Intercept
 exponential(1)     sigma
 lb ub tag       source
                   user
           (vectorized)
           (vectorized)
           (vectorized)
           (vectorized)
                   user
  0                user


=== Posterior Predictive Checks ===
```

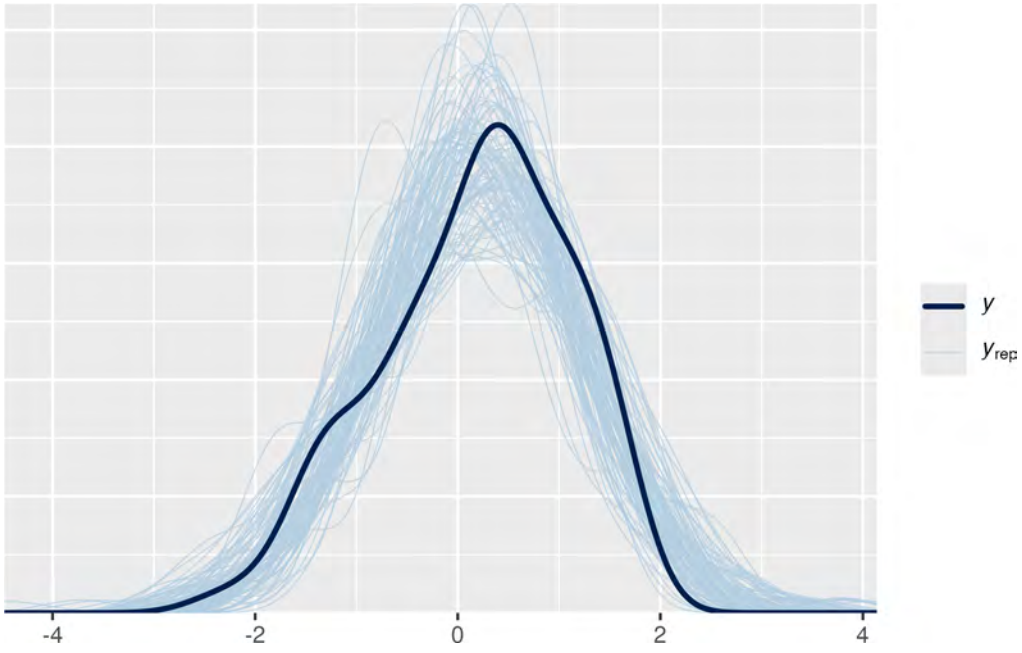
Density plots for each parameter.



```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```



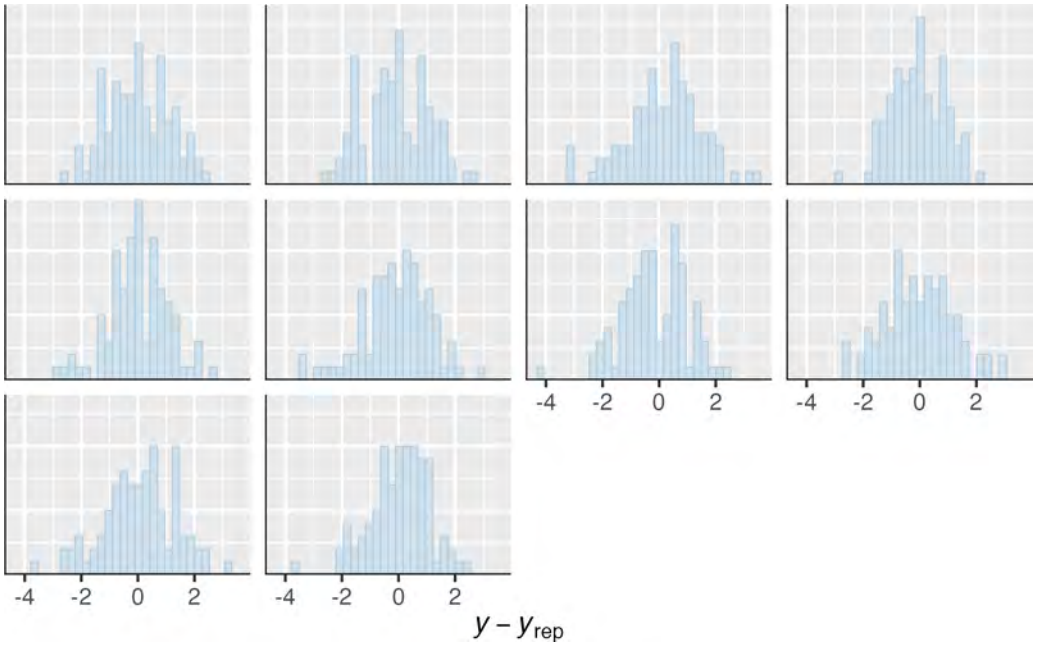

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

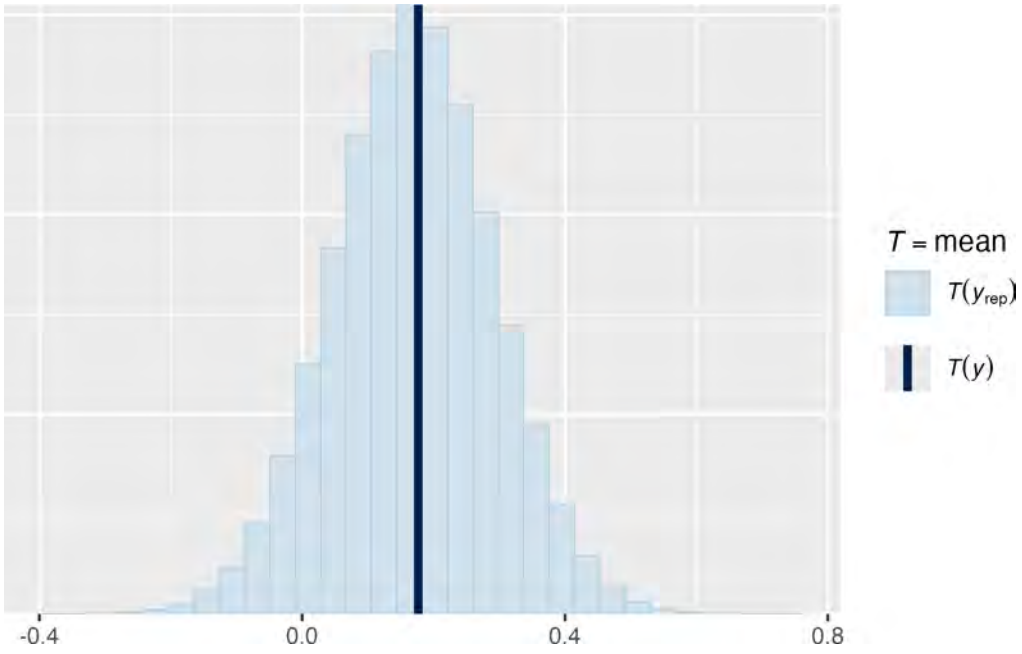
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



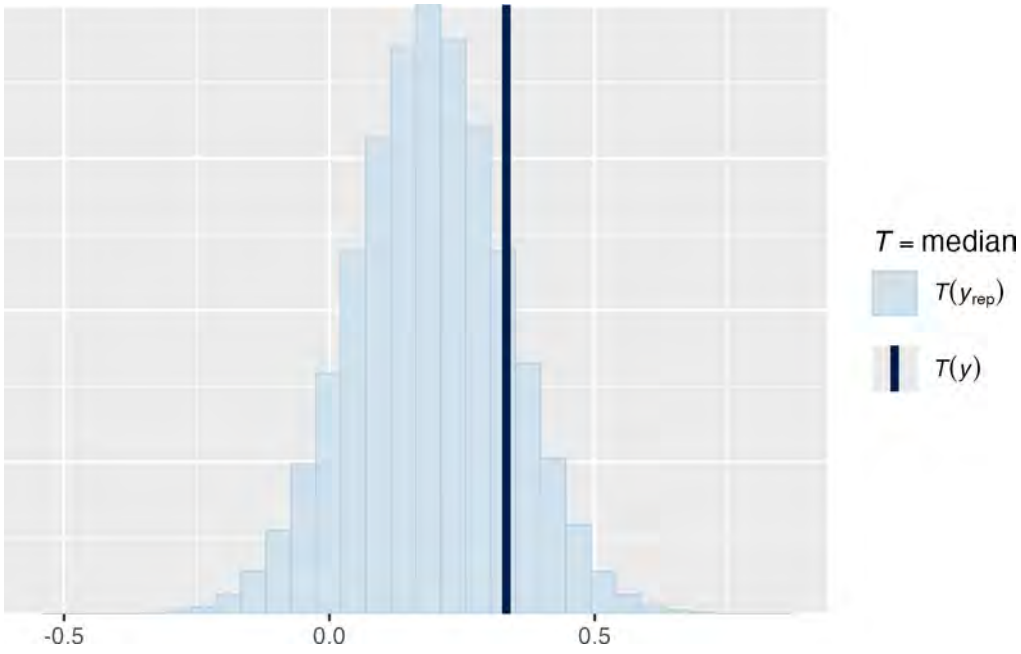

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



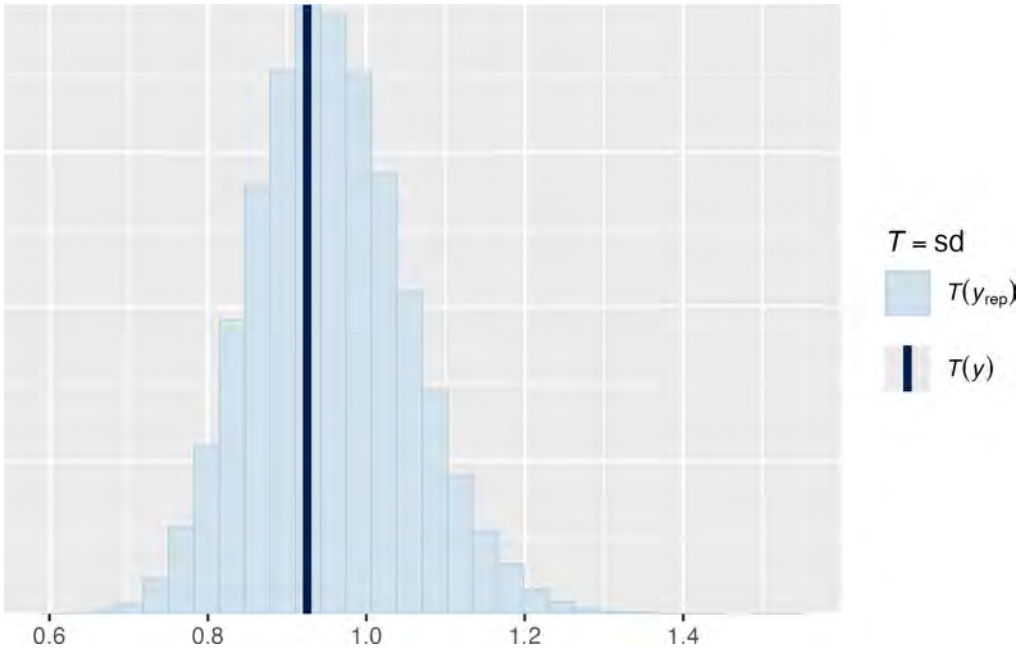

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

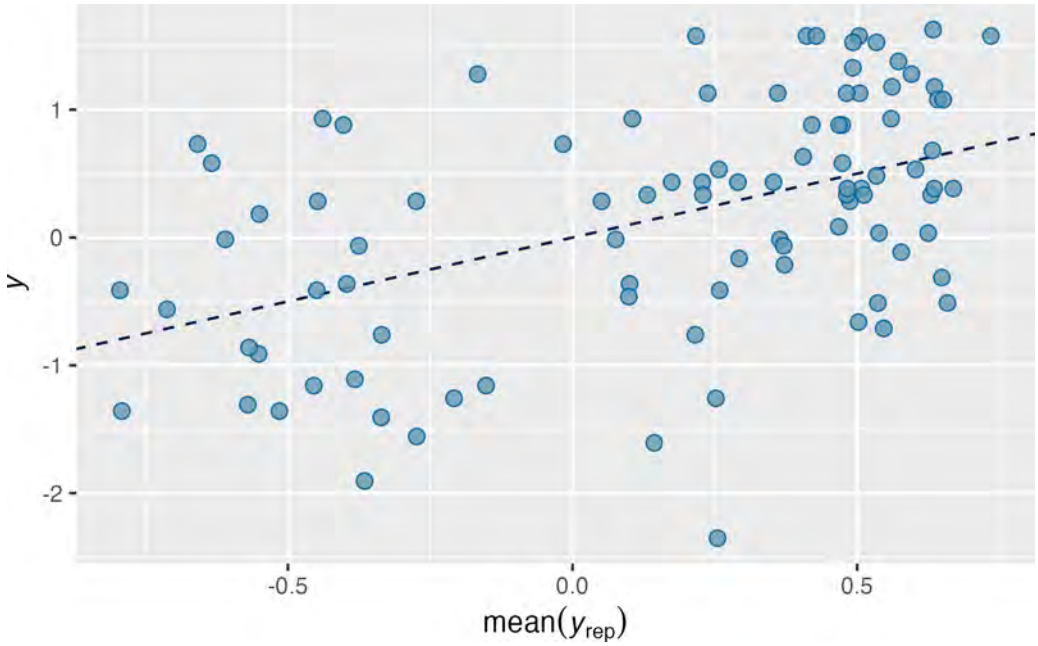



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

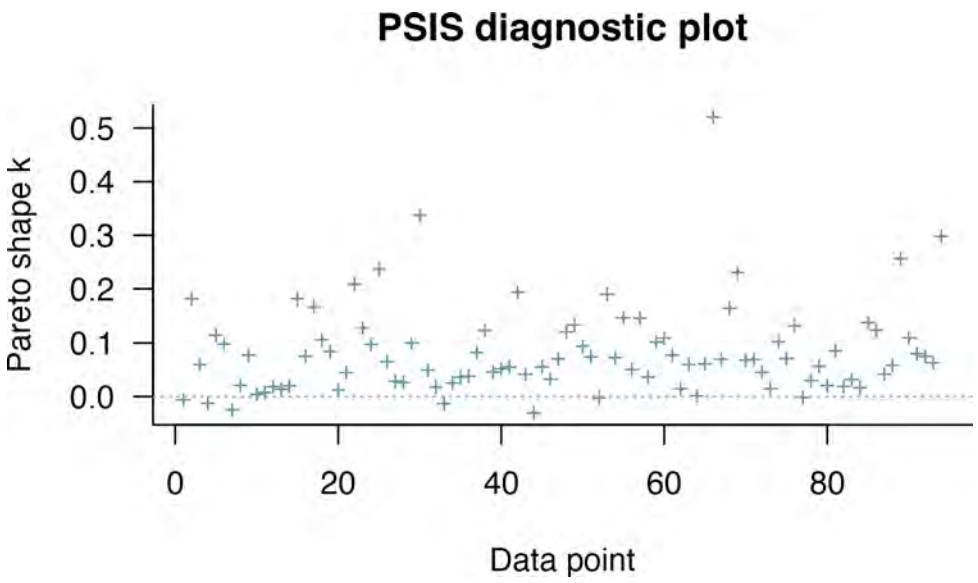
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H2 Menarche Timing Model no interaction output

```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.2326067 0.06317975 0.1065182 0.3523025
```

```
Summary of Posterior Distribution

Parameter           |  Mean | Median |             CI |      pd | % in ROPE | meaningful
-------------------------------------------------------------------------------------------
b_Intercept         | -0.47 |  -0.47 | [-0.78, -0.16] |  99.79% |     1.04% |       TRUE
b_menarche_timing_s | -0.03 |  -0.03 | [-0.21,  0.13] |  65.73% |    71.82% |      FALSE
b_adhd_statusY      |  0.90 |   0.90 | [ 0.53,  1.27] | 100.00% | 5.00e-
03% |       TRUE
b_sed_percentile_s  |  0.16 |   0.16 | [ 0.00,  0.33] |  97.20% |    22.50% |      FALSE

=== ADHD difference in raw outcome units ===

Warning: Dropping 'draws_df' class as required metadata was removed.
```

adhd_raw_effect_mean	adhd_raw_effect_lower	adhd_raw_effect_upper
18.16	10.65	25.54

H2 Menarche Timing Model no interaction diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

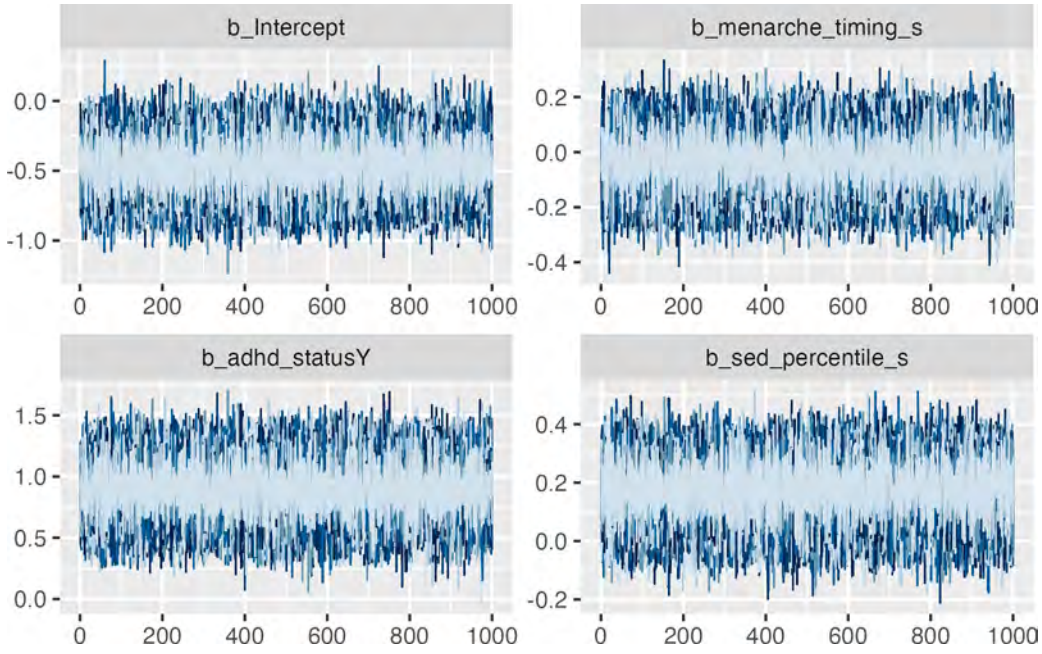
# A tibble: 8 x 4
  variable             rhat ess_bulk ess_tail
  <chr>               <dbl>    <dbl>    <dbl>
1 b_Intercept          1.00  148927.   93523.
2 b_menarche_timing_s  1.00  149165.   92829.
3 b_adhd_statusY       1.00  146109.   92654.
4 b_sed_percentile_s   1.02    4499.   47326.
5 sigma                1.00  133229.   90443.
6 Intercept            1.00  138075.   91982.
7 lprior               1.00  145468.   90925.
8 lp__                 1.04    1735.    2307.


=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```



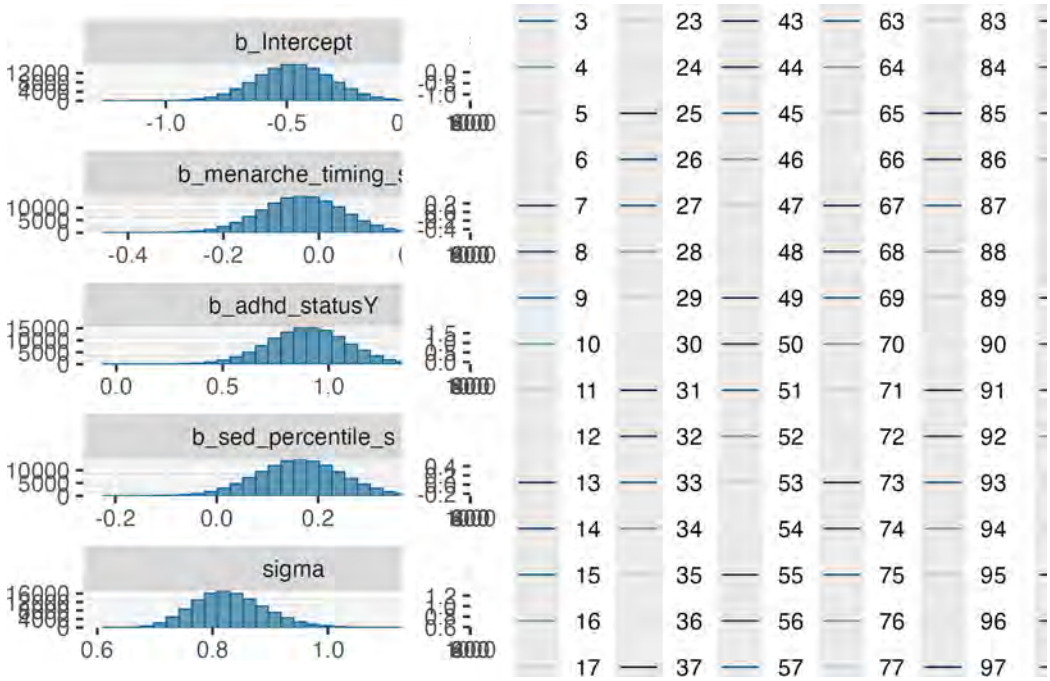

```
=== Prior Summary ===

Priors specified for the model.

          prior     class              coef group resp dpar nlpar lb ub tag
   normal(0, 1)         b
   normal(0, 1)         b      adhd_statusY
   normal(0, 1)         b menarche_timing_s
   normal(0, 1)         b  sed_percentile_s
   normal(0, 1) Intercept
 exponential(1)     sigma                                                0
       source
         user
 (vectorized)
 (vectorized)
 (vectorized)
         user
         user


=== Posterior Predictive Checks ===
```

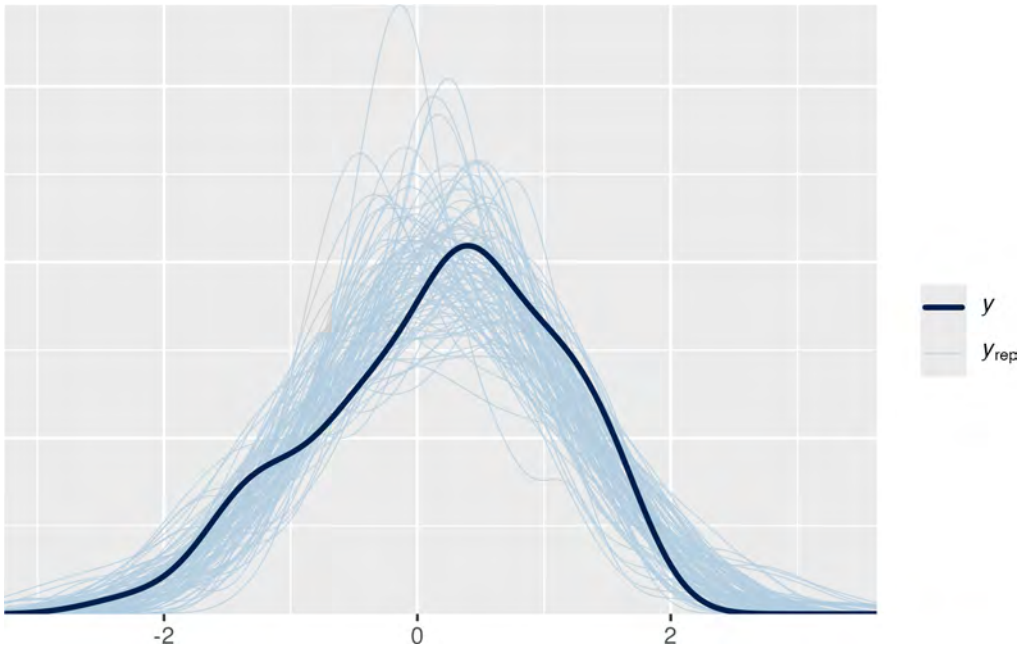
Density plots for each parameter.



=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results with caution until a more principled approach has been implemented.



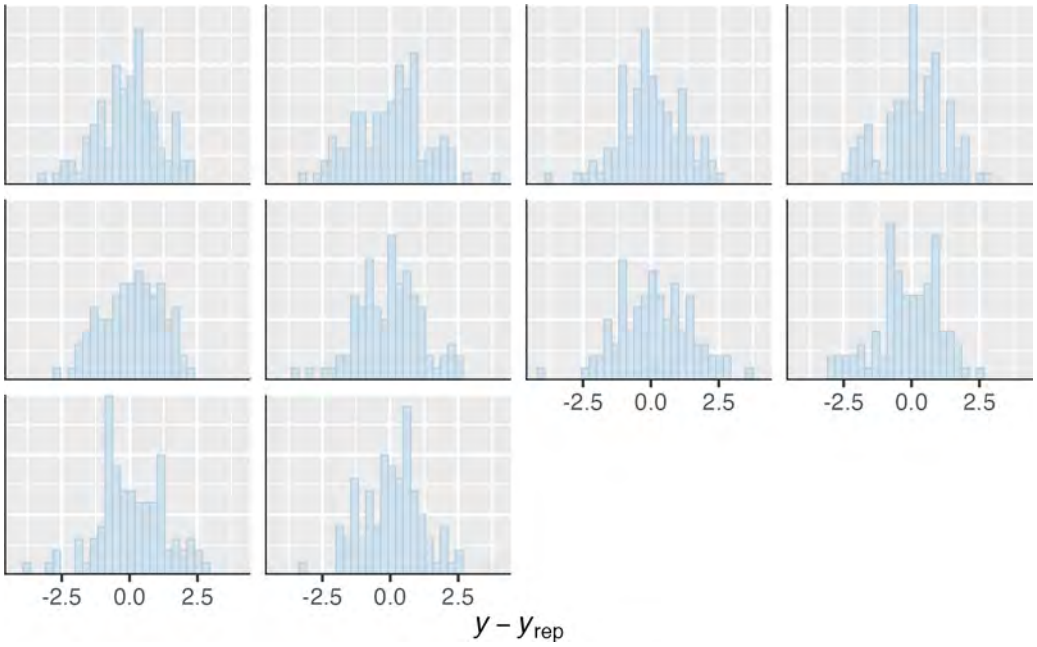

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

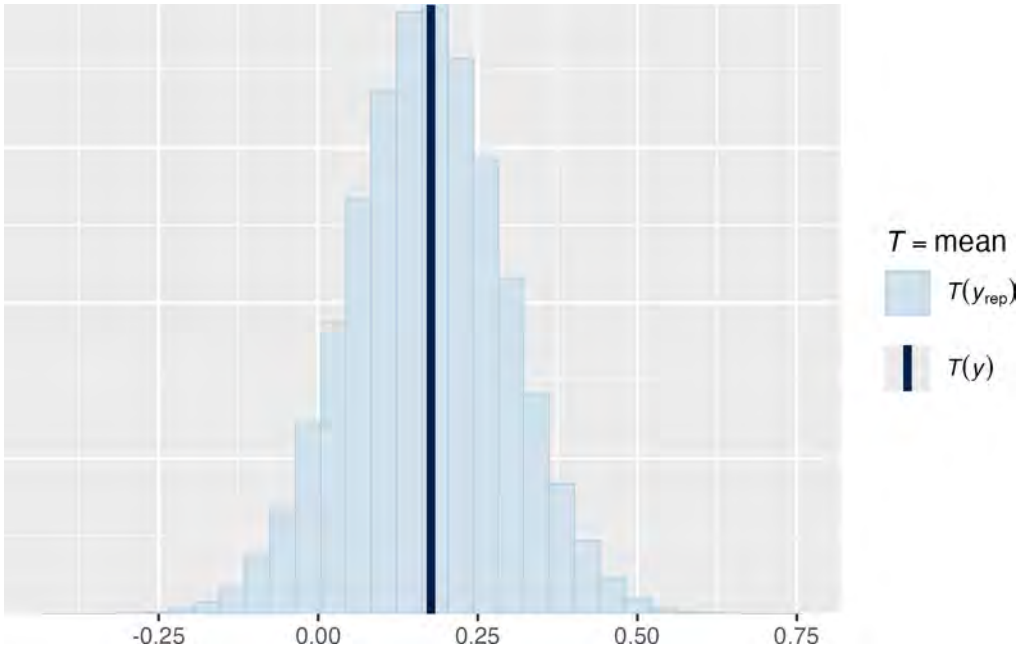
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



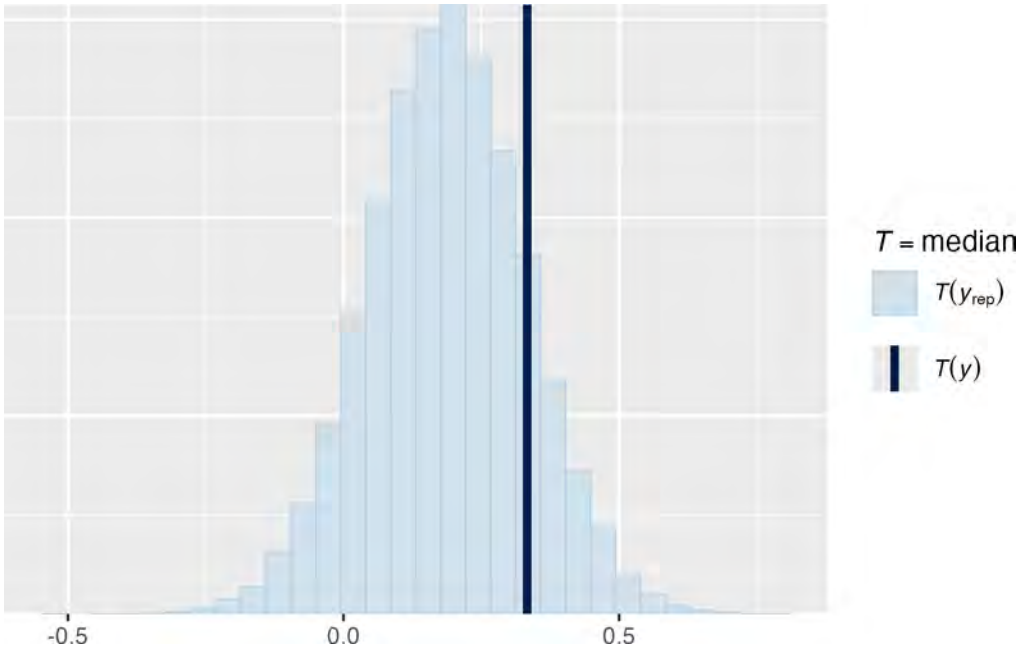

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



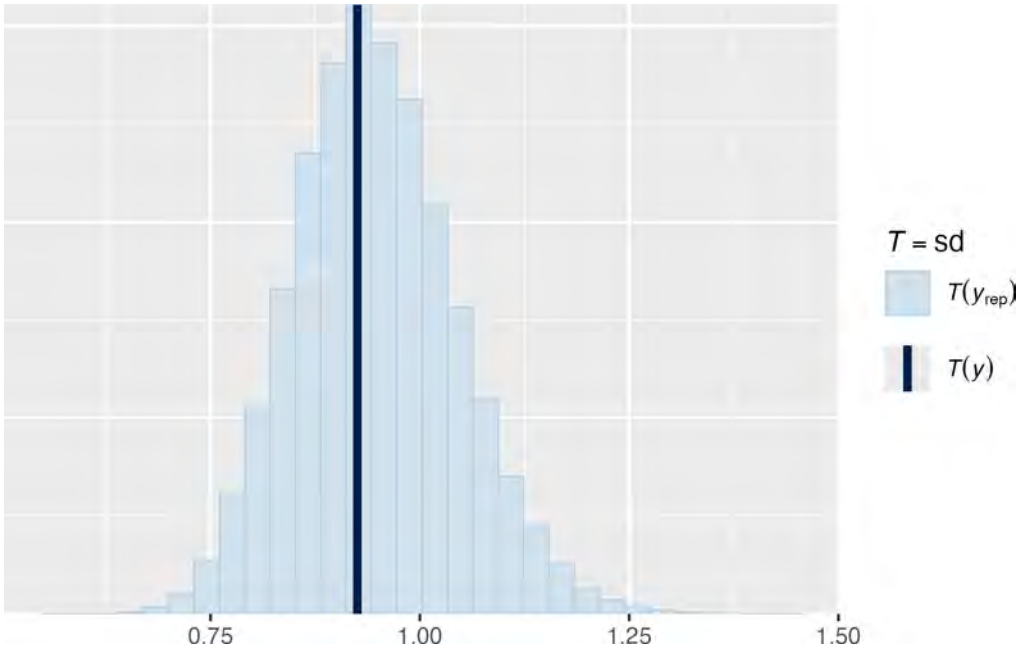

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

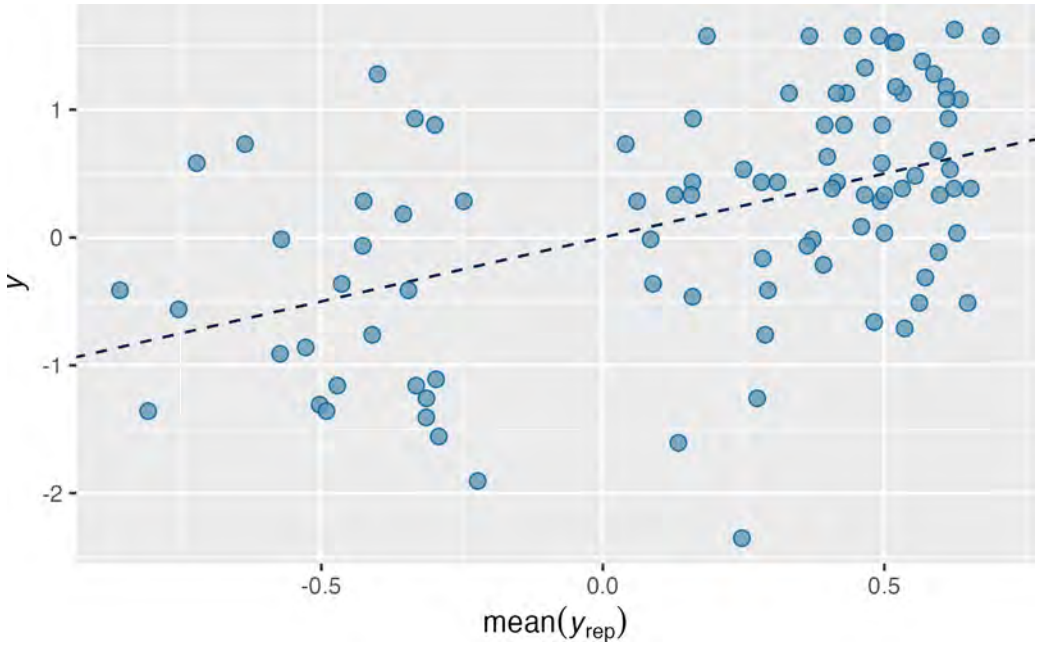



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

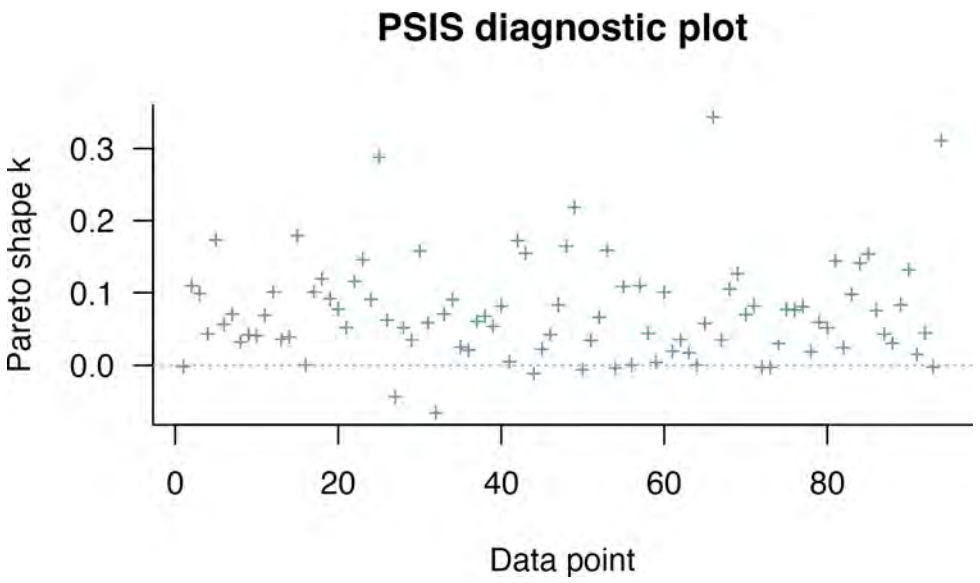
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H1 Young Person Model output

```
    Estimate Est.Error      Q2.5     Q97.5
R2 0.3005776 0.0722977 0.1514729 0.4316031
```

```
Summary of Posterior Distribution
```

Parameter	Mean	Median	CI	pd	% in ROPE	meaningful
b_Intercept	-0.51	-0.51	[-0.84, -0.18]	99.85%	0%	TRUE
b_pds_total_y_s	0.30	0.30	[-0.04, 0.64]	95.91%	9.99%	TRUE
b_adhd_yY	0.81	0.81	[0.39, 1.23]	99.98%	0%	TRUE
b_age_cont_y_s	-0.16	-0.16	[-0.44, 0.11]	87.96%	31.31%	FALSE
b_sed_percentile_s	0.09	0.09	[-0.14, 0.32]	77.39%	50.22%	FALSE
b_pds_total_y_s:adhd_yY	0.07	0.07	[-0.33, 0.49]	63.98%	37.25%	FALSE

H1 Young Person Model diagnostics and PP checks

```
=== Rhat and ESS ===
```

```
Rhat values close to 1 indicate convergence. ESS should ideally be > 400.
```

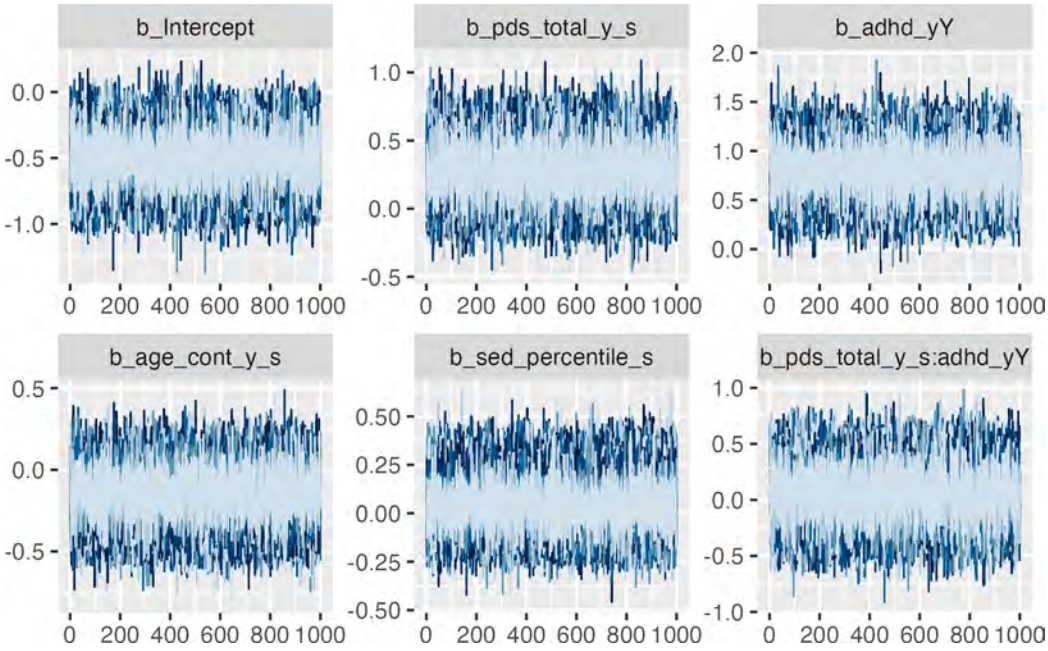
```
# A tibble: 10 x 4
   variable                 rhat ess_bulk ess_tail
   <chr>                   <dbl>    <dbl>    <dbl>
 1 b_Intercept              1.01   20241.   92248.
 2 b_pds_total_y_s          1.01   13832.   82556.
 3 b_adhd_yY                1.01    8121.   86162.
 4 b_age_cont_y_s           1.02    3196.   18929.
 5 b_sed_percentile_s       1.15     496.    2132.
 6 b_pds_total_y_s:adhd_yY  1.01    8012.   78749.
 7 sigma                    1.01    8258.   84380.
 8 Intercept                1.00  141107.   92710.
 9 lprior                   1.01   16110.   89236.
10 lp__                     1.12     606.     981.
```

```
=== Divergent Transitions ===
```

```
Ideally 0. Large numbers indicate sampler problems.
```

```
Number of divergent transitions: 0
```

```
=== Trace Plots ===
```



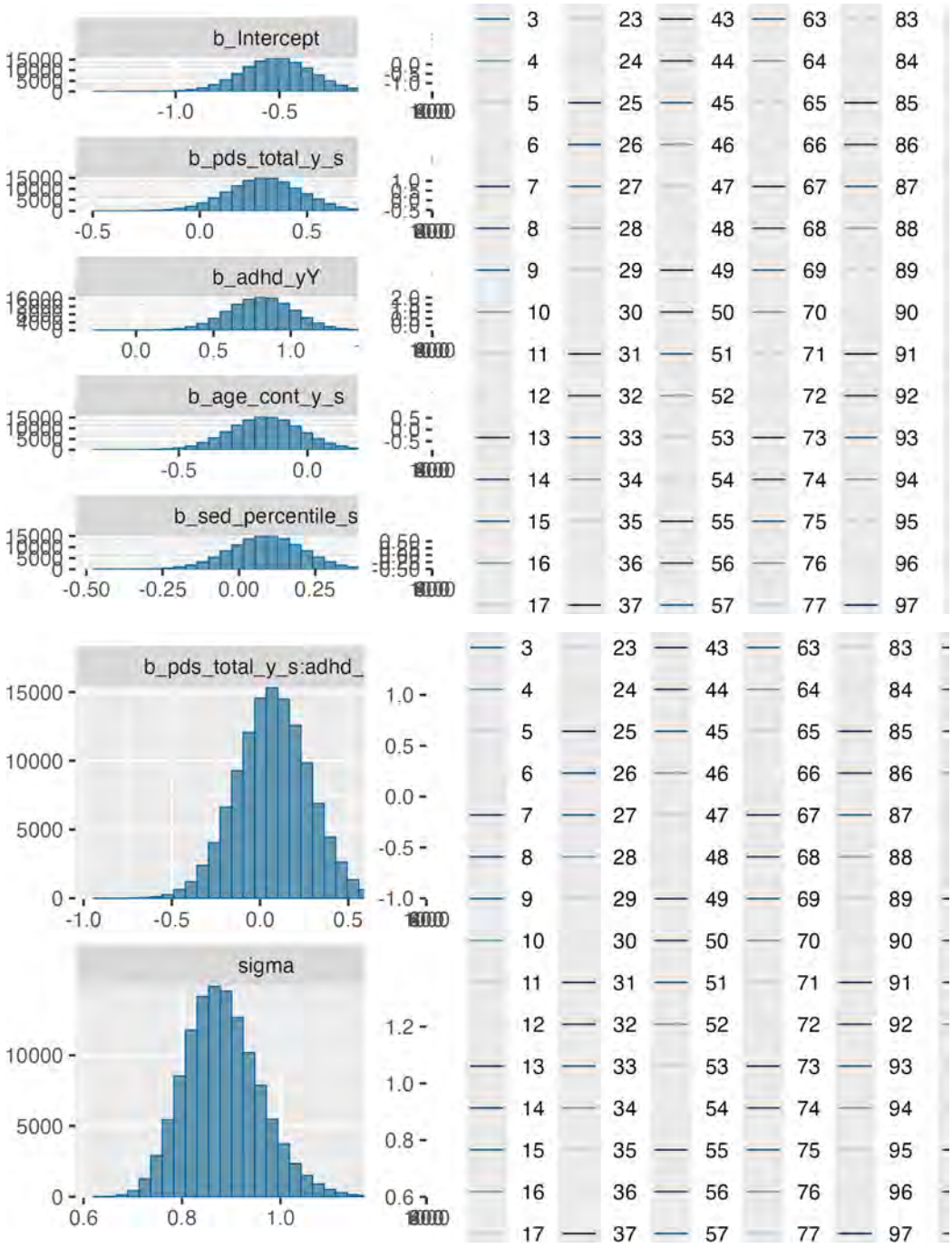
```
=== Prior Summary ===

Priors specified for the model.

          prior     class                  coef group resp dpar nlpar lb ub tag
   normal(0, 1)         b
   normal(0, 1)         b               adhd_yY
   normal(0, 1)         b          age_cont_y_s
   normal(0, 1)         b         pds_total_y_s
   normal(0, 1)         b pds_total_y_s:adhd_yY
   normal(0, 1)         b      sed_percentile_s
   normal(0, 1) Intercept
 exponential(1)     sigma                                                 0
       source
         user
 (vectorized)
 (vectorized)
 (vectorized)
 (vectorized)
 (vectorized)
         user
         user


=== Posterior Predictive Checks ===

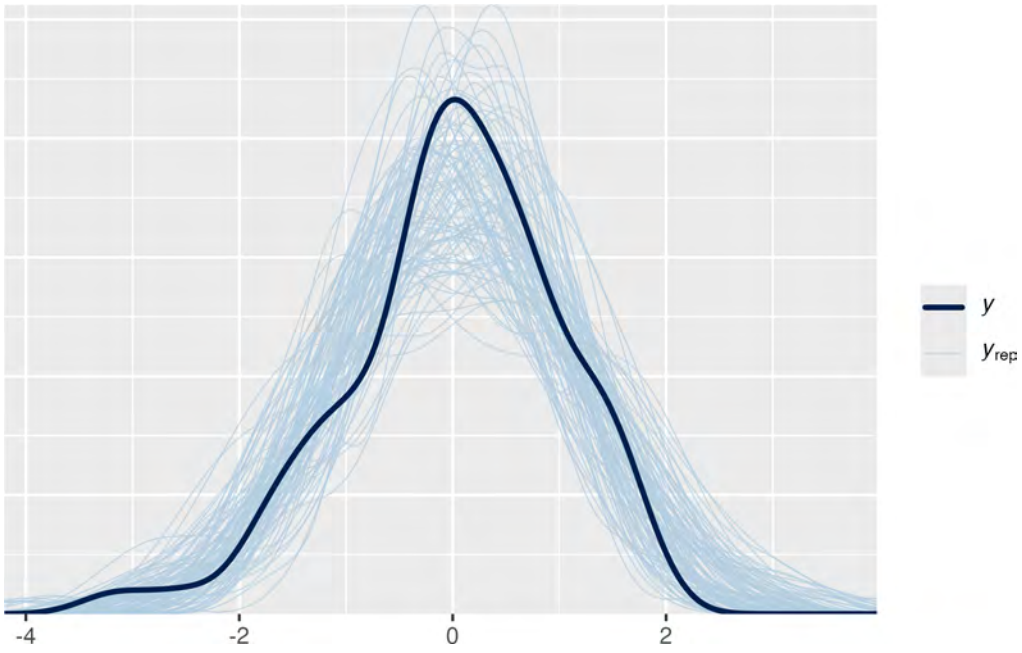
Density plots for each parameter.
```



=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

```
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```



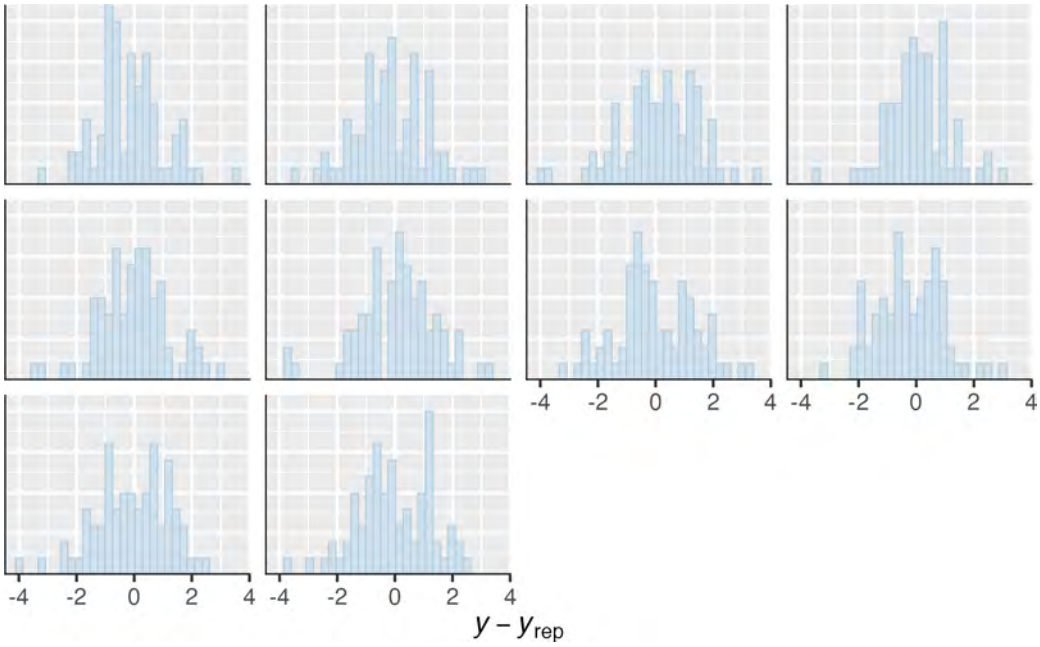

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

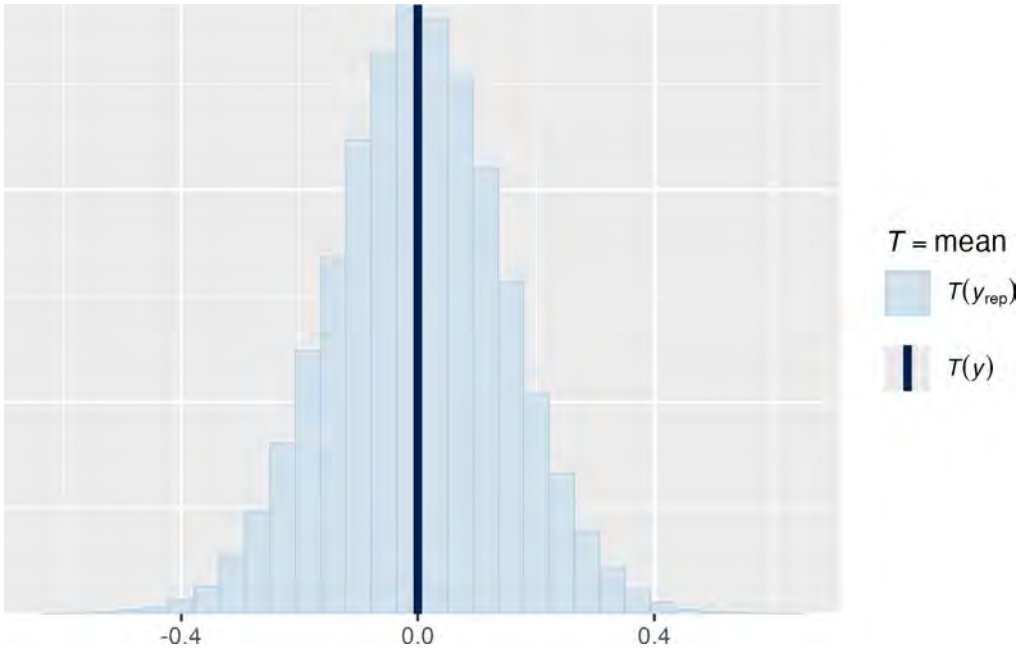
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



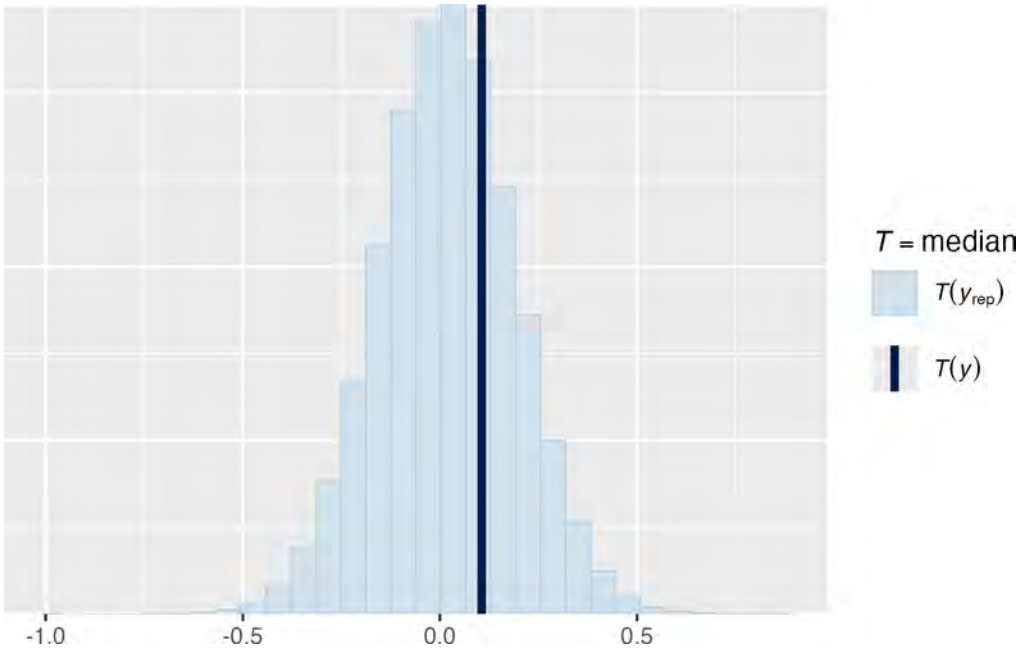

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



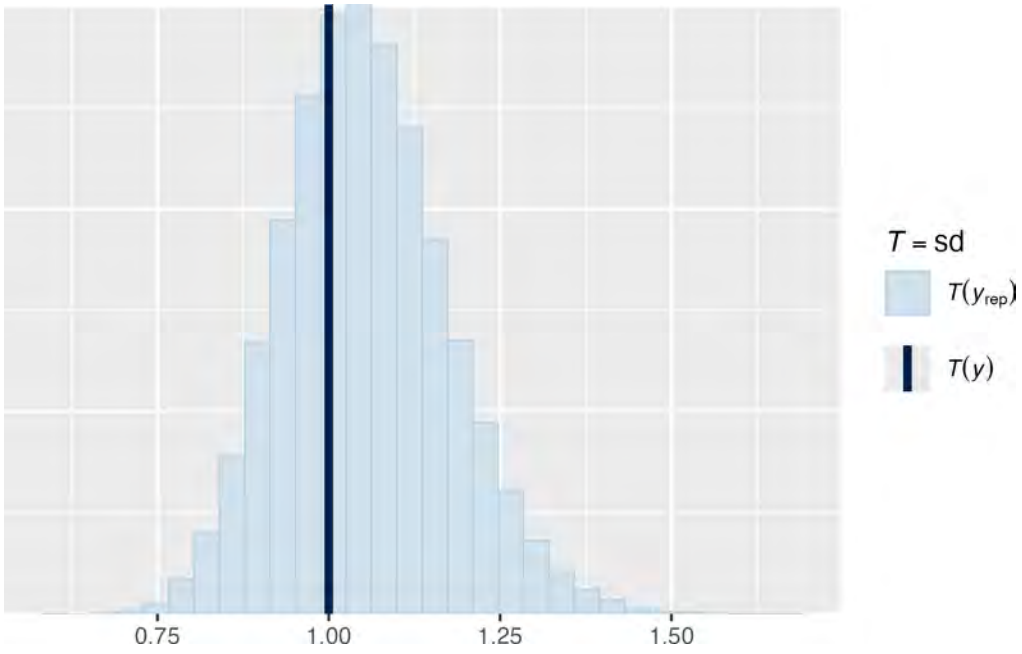

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

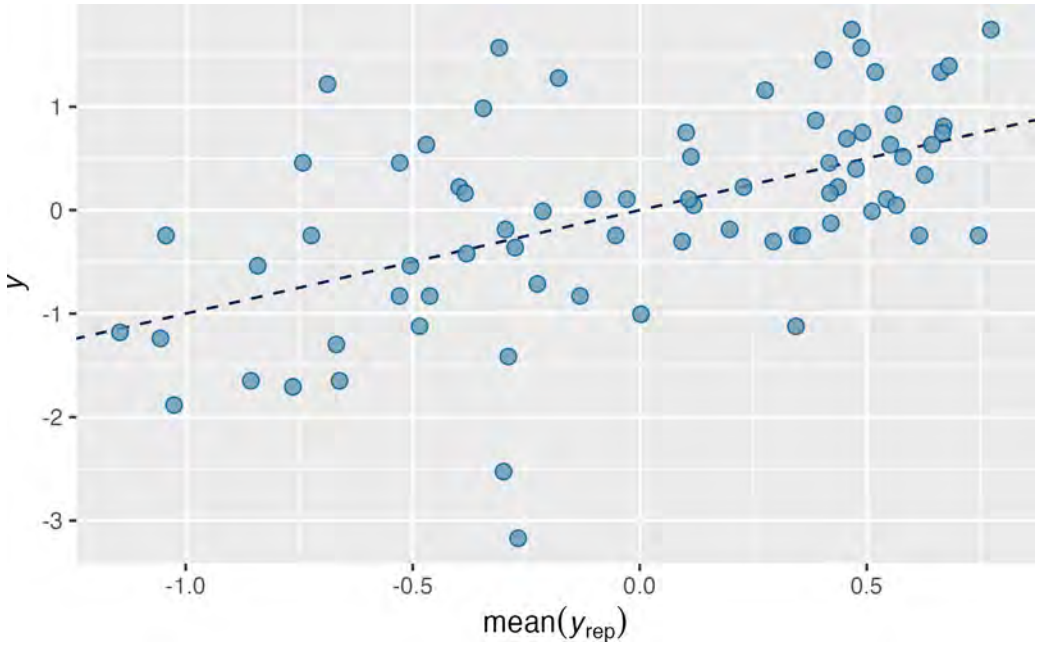



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

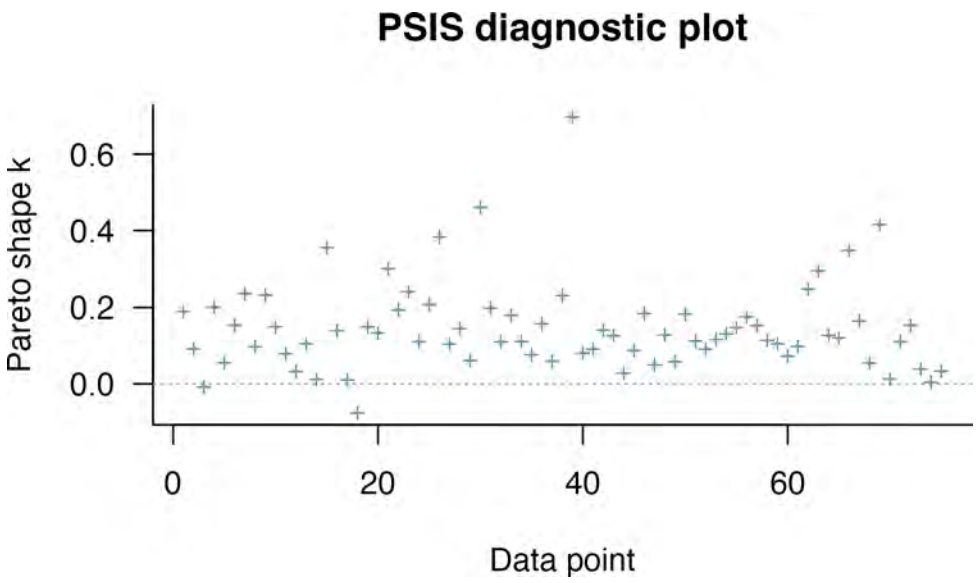
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H1 Young Person Model no interaction output

```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.2956875 0.07365819 0.1442923 0.4288465
```

```
Summary of Posterior Distribution

Parameter          |  Mean | Median |             CI |     pd | % in ROPE | meaningful
-------------------------------------------------------------------------------------
b_Intercept        | -0.51 |  -0.51 | [-0.84, -0.19] | 99.85% |     0.80% |       TRUE
b_pds_total_y_s    |  0.34 |   0.34 | [ 0.07,  0.61] | 99.25% |     4.06% |       TRUE
b_adhd_yY          |  0.81 |   0.81 | [ 0.39,  1.22] | 99.99% |     0.05% |       TRUE
b_age_cont_y_s     | -0.16 |  -0.16 | [-0.43,  0.11] | 87.61% |    29.94% |      FALSE
b_sed_percentile_s |  0.09 |   0.09 | [-0.14,  0.33] | 77.55% |    47.71% |      FALSE
```

```
=== ADHD and pubertal status effects in raw outcome units ===
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

effect	mean	lower	upper
adhd_raw_effect_y	13.83	6.70	20.86
pds_raw_effect_y	2.07	0.41	3.72

H1 Young Person Model no interaction diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

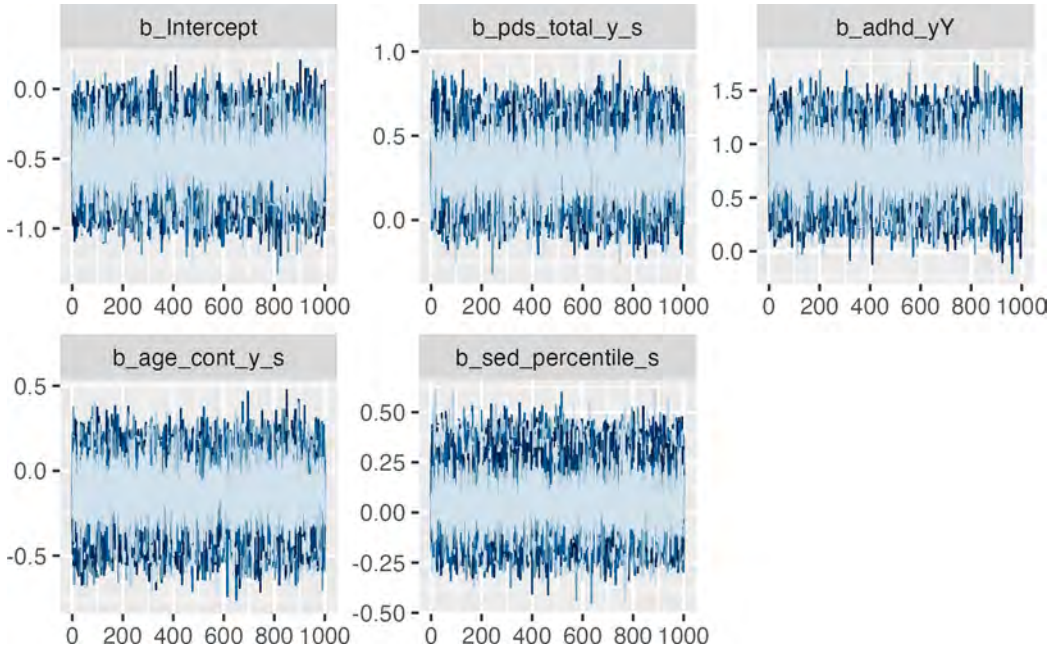
# A tibble: 9 x 4
  variable            rhat ess_bulk ess_tail
  <chr>              <dbl>    <dbl>    <dbl>
1 b_Intercept         1.01   12780.   88231.
2 b_pds_total_y_s     1.01    5102.   57529.
3 b_adhd_yY           1.01    6872.   75410.
4 b_age_cont_y_s      1.02    2872.   16153.
5 b_sed_percentile_s  1.15     496.    2179.
6 sigma               1.01    8896.   89592.
7 Intercept           1.00  165274.   89884.
8 lprior              1.01   14604.   87498.
9 lp__                1.15     514.     750.


=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```



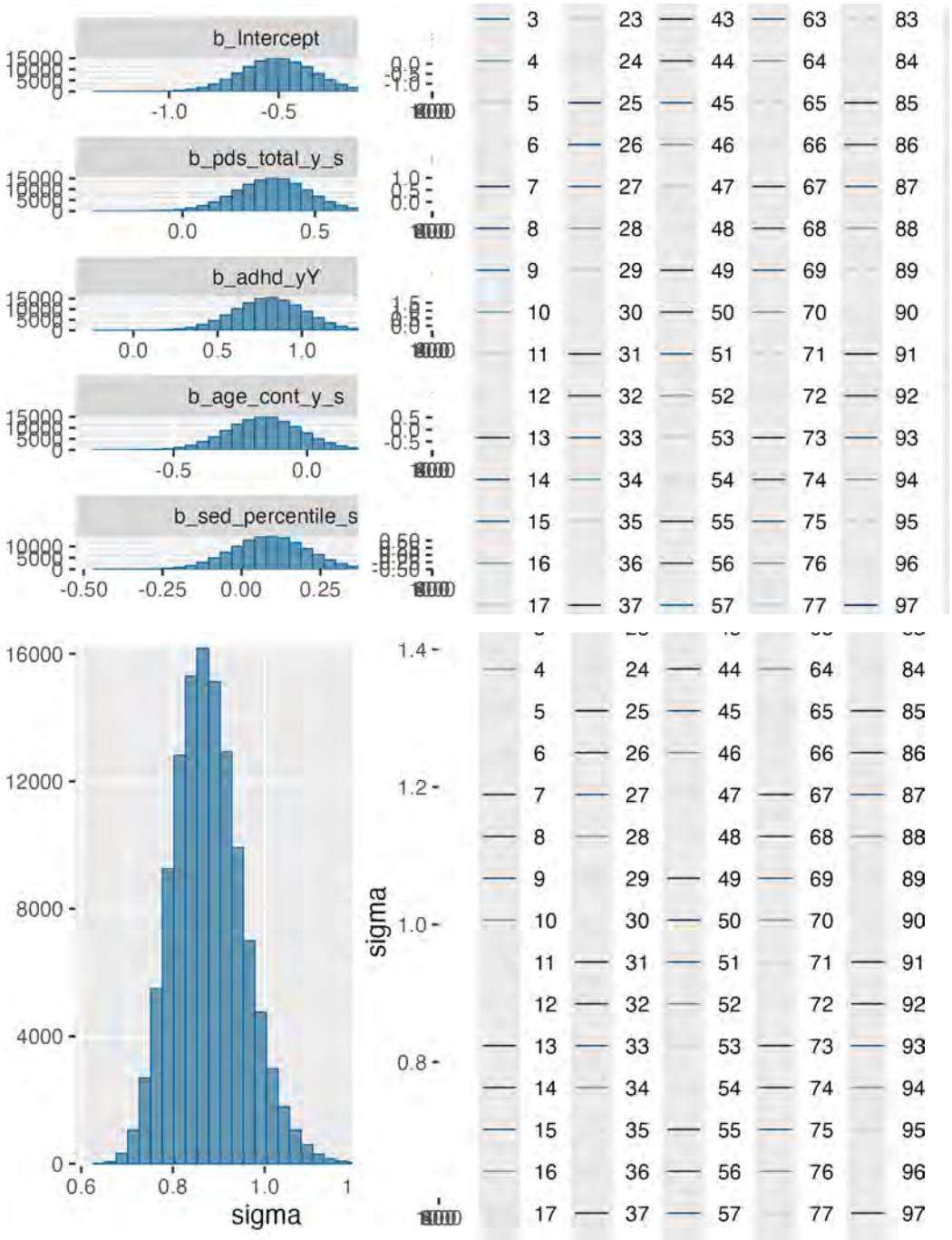

```
=== Prior Summary ===

Priors specified for the model.

          prior     class             coef group resp dpar nlpar lb ub tag
   normal(0, 1)         b
   normal(0, 1)         b          adhd_yY
   normal(0, 1)         b     age_cont_y_s
   normal(0, 1)         b    pds_total_y_s
   normal(0, 1)         b sed_percentile_s
   normal(0, 1) Intercept
 exponential(1)     sigma                                          0
       source
         user
 (vectorized)
 (vectorized)
 (vectorized)
 (vectorized)
         user
         user


=== Posterior Predictive Checks ===
```

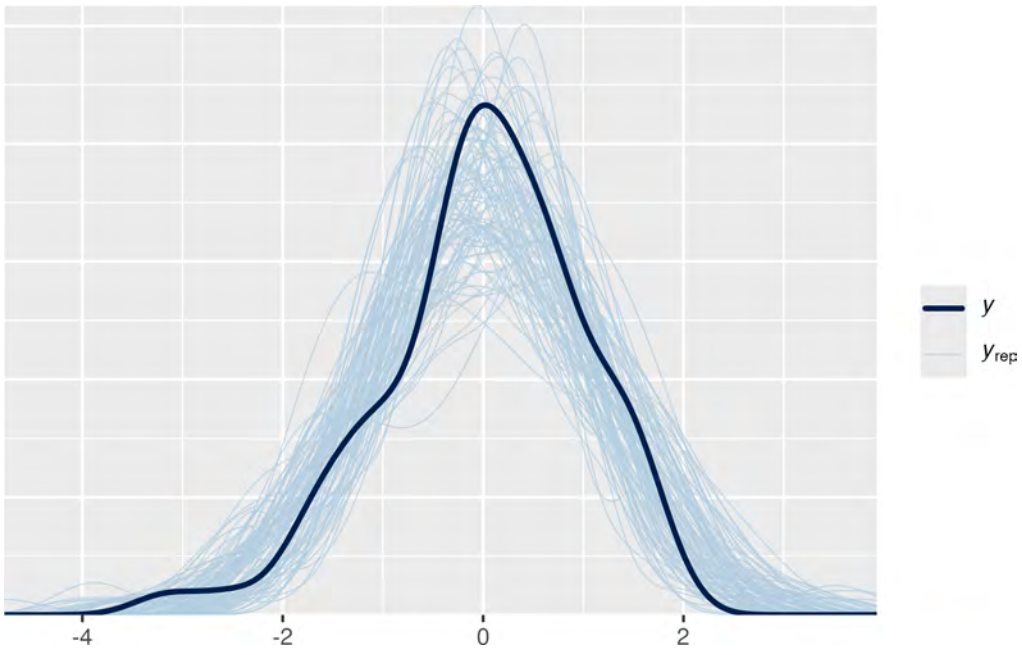
Density plots for each parameter.



```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```



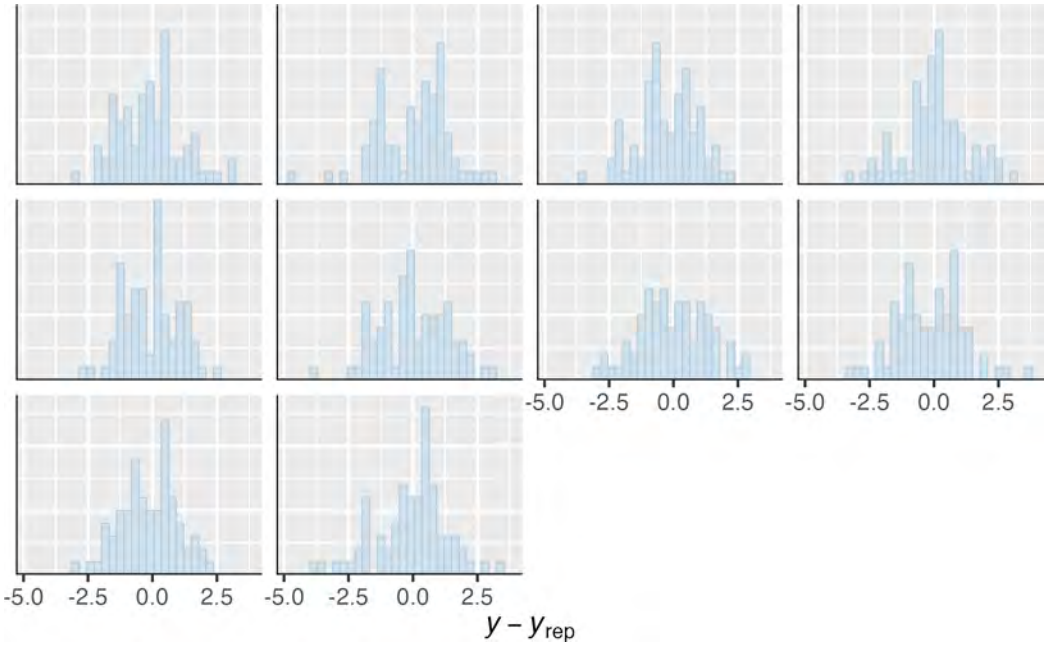

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

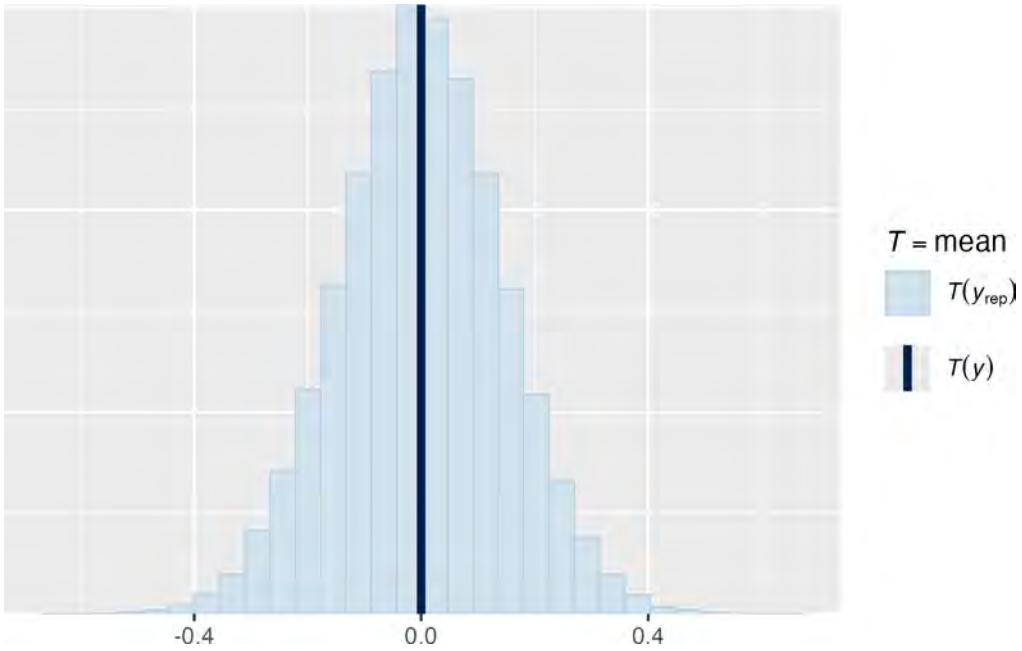
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



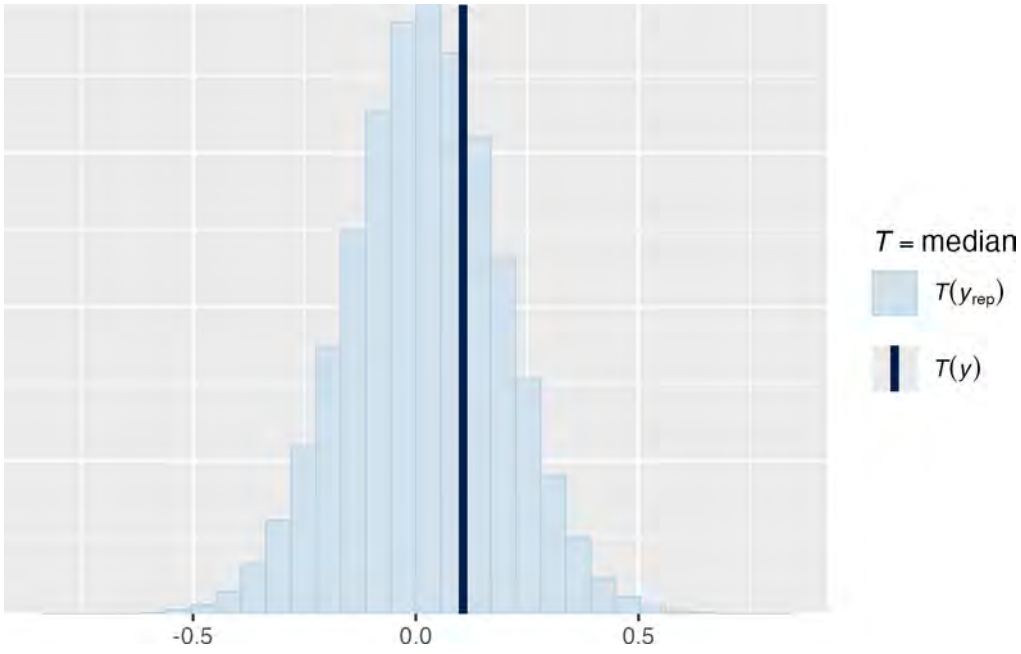

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



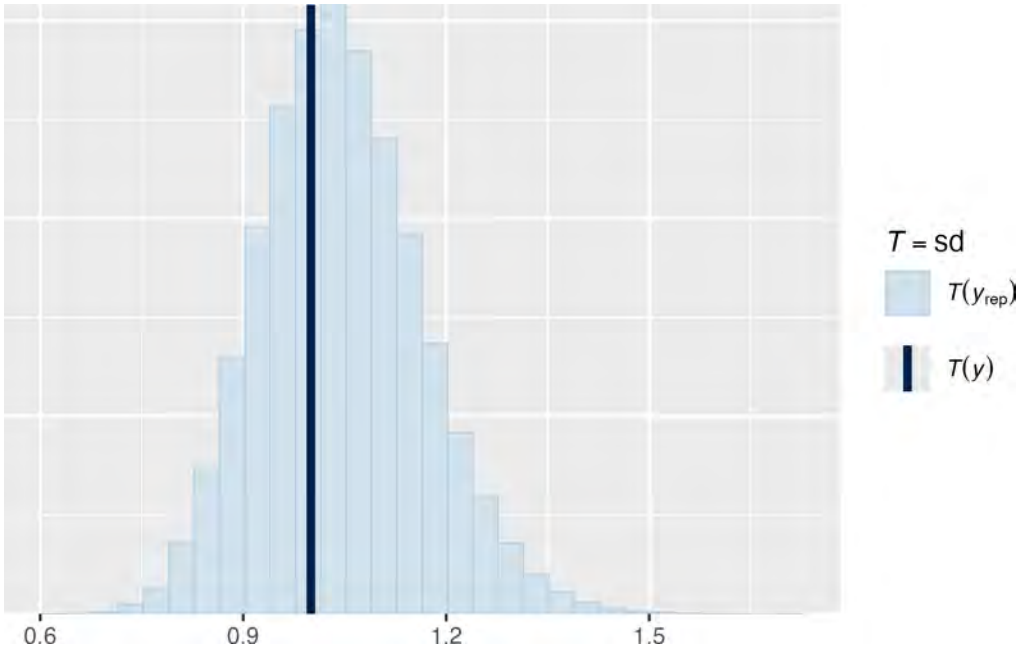

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

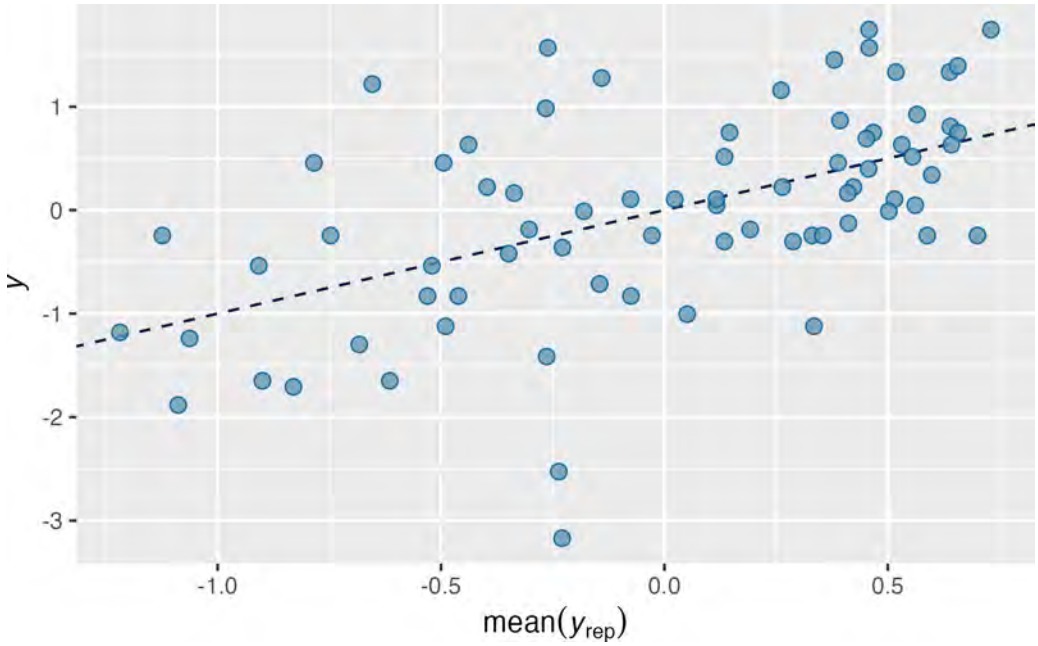



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

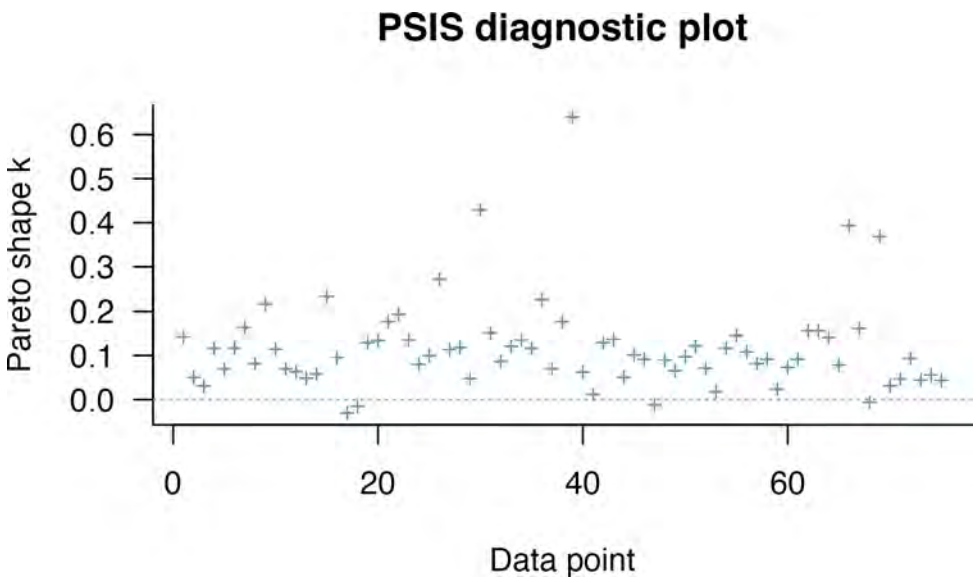
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H2 Young Person Model output

```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.2920913 0.07396603 0.1403149 0.4267302
```

```
Summary of Posterior Distribution

Parameter                     |  Mean | Median |             CI |     pd
------------------------------------------------------------------------
b_Intercept                   | -0.51 |  -0.51 | [-0.84, -0.17] | 99.83%
b_pubertal_timing_y_s         |  0.28 |   0.28 | [-0.02,  0.57] | 96.66%
b_adhd_yY                     |  0.81 |   0.81 | [ 0.39,  1.23] | 99.99%
b_sed_percentile_s            |  0.10 |   0.10 | [-0.14,  0.33] | 79.15%
b_pubertal_timing_y_s:adhd_yY | -0.04 |  -0.04 | [-0.45,  0.37] | 57.62%

Parameter                     | % in ROPE | meaningful
-------------------------------------------------------
b_Intercept                   |        0% |       TRUE
b_pubertal_timing_y_s         |     9.65% |       TRUE
b_adhd_yY                     |        0% |       TRUE
b_sed_percentile_s            |    48.24% |      FALSE
b_pubertal_timing_y_s:adhd_yY |    38.80% |      FALSE
```

H2 Young Person Model diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

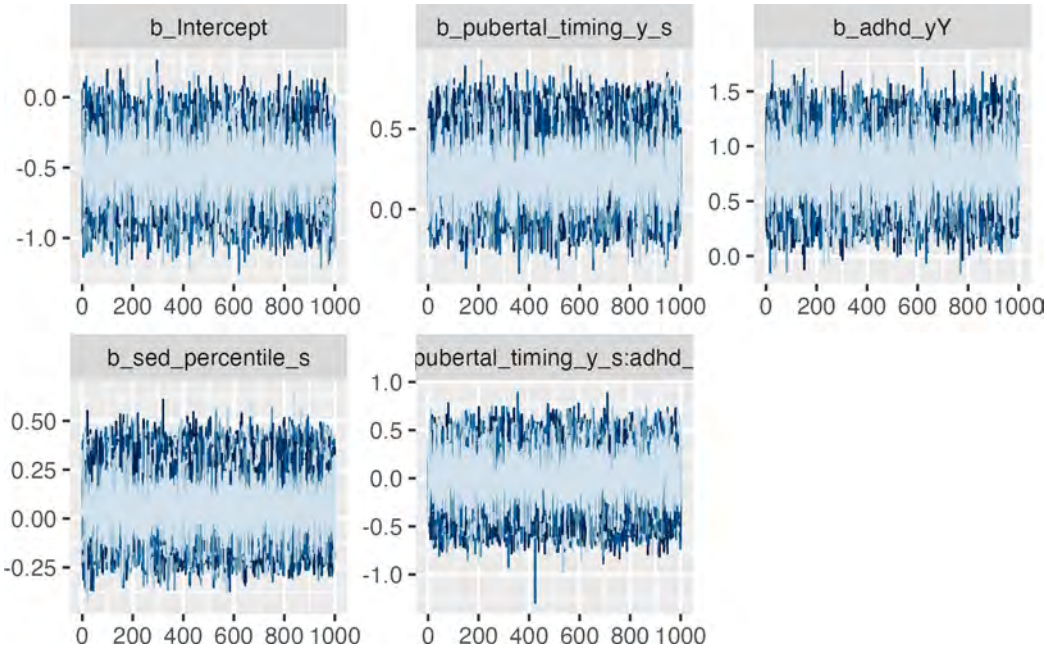
# A tibble: 9 x 4
  variable                      rhat ess_bulk ess_tail
  <chr>                        <dbl>    <dbl>    <dbl>
1 b_Intercept                   1.01   18148.   87680.
2 b_pubertal_timing_y_s         1.02    3714.   28519.
3 b_adhd_yY                     1.01    8505.   86452.
4 b_sed_percentile_s            1.16     487.    2215.
5 b_pubertal_timing_y_s:adhd_yY 1.03    2734.   19214.
6 sigma                         1.01    5698.   54388.
7 Intercept                     1.00  157791.   90730.
8 lprior                        1.01   15925.   85722.
9 lp__                          1.18     440.     633.


=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```



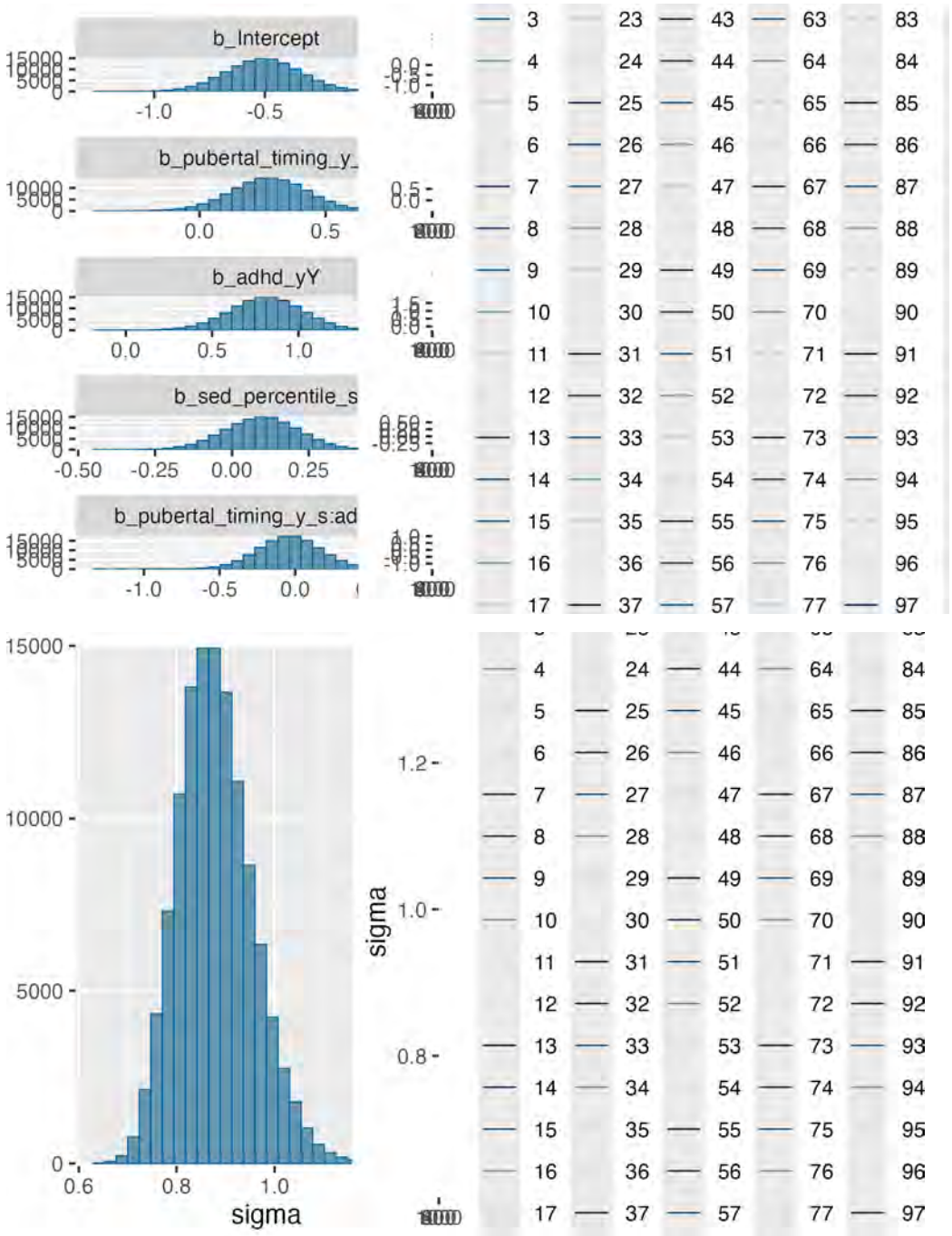

```
=== Prior Summary ===

Priors specified for the model.

          prior     class                         coef group resp dpar nlpar lb
   normal(0, 1)         b
   normal(0, 1)         b                     adhd_yY
   normal(0, 1)         b         pubertal_timing_y_s
   normal(0, 1)         b pubertal_timing_y_s:adhd_yY
   normal(0, 1)         b            sed_percentile_s
   normal(0, 1) Intercept
 exponential(1)     sigma                                                    0
 ub tag       source
               user
       (vectorized)
       (vectorized)
       (vectorized)
       (vectorized)
               user
               user


=== Posterior Predictive Checks ===
```

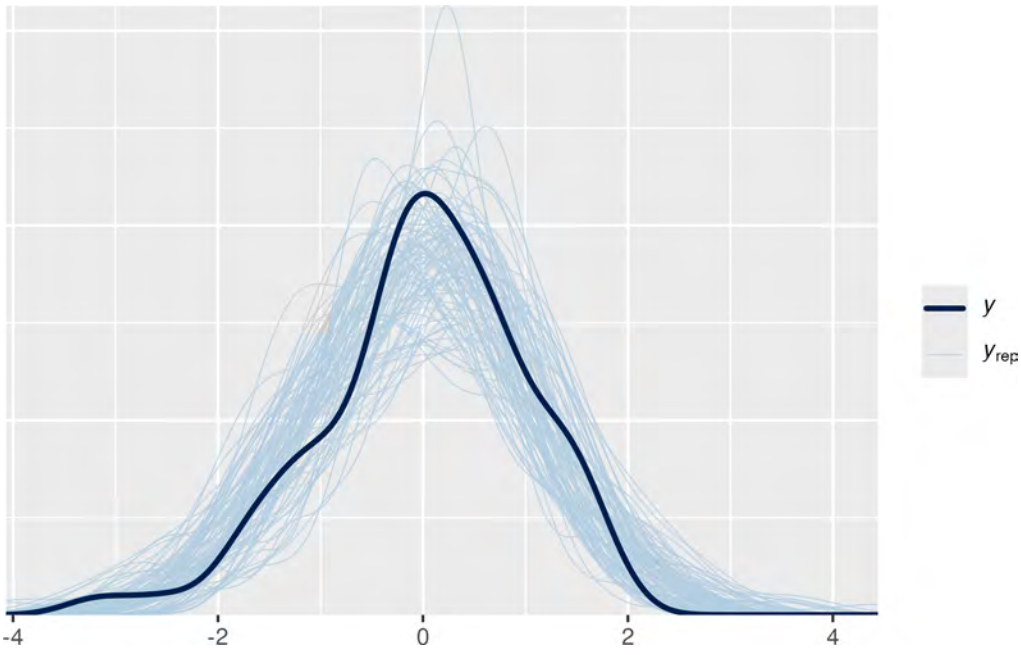
Density plots for each parameter.



```
=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```



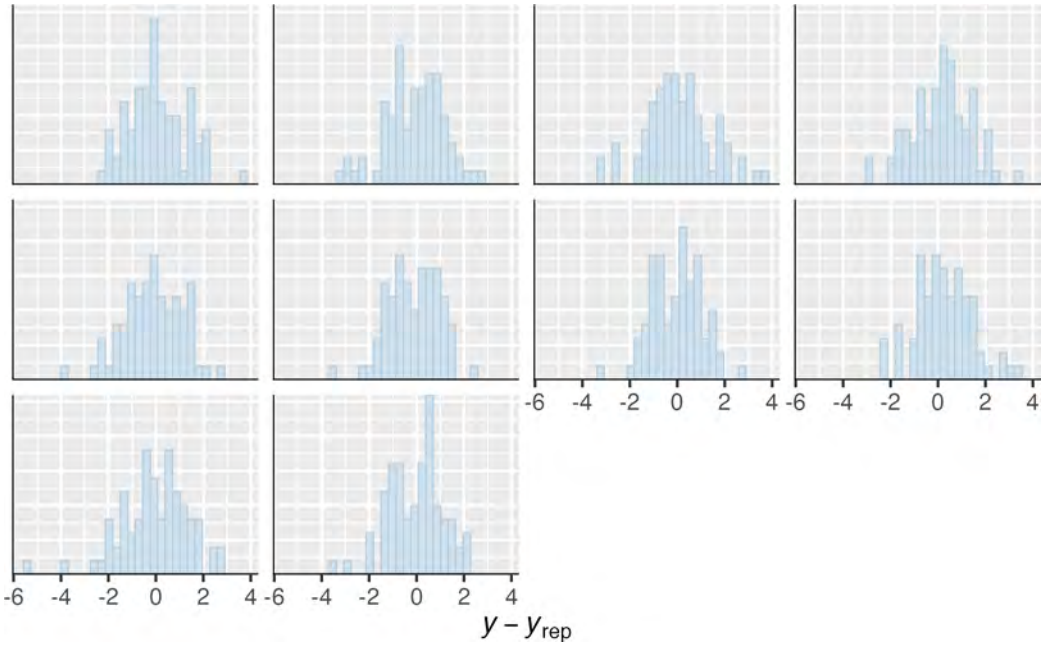

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

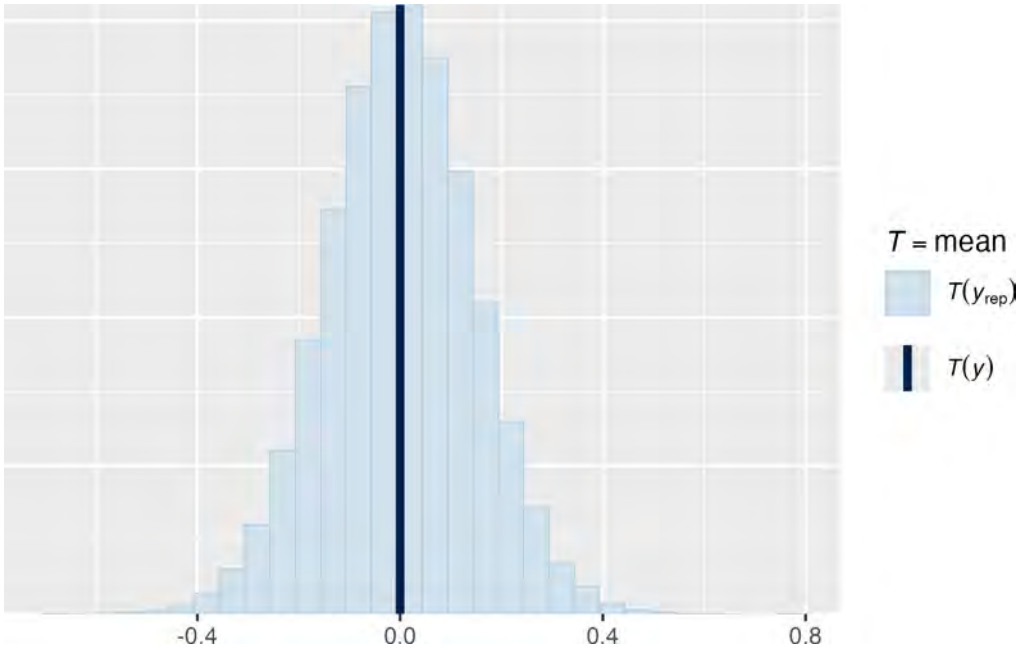
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



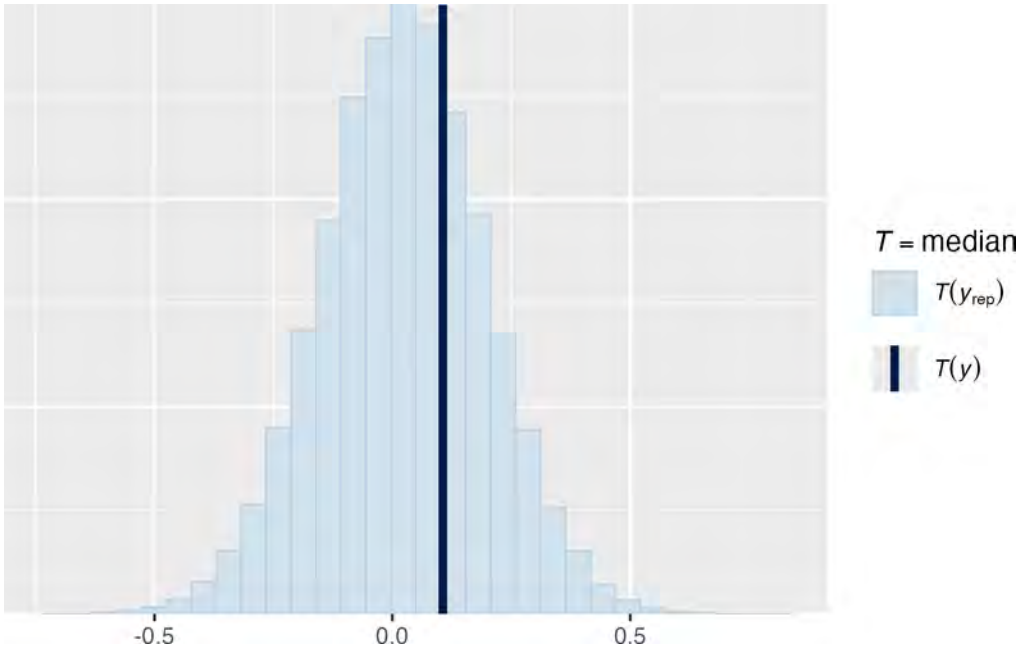

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



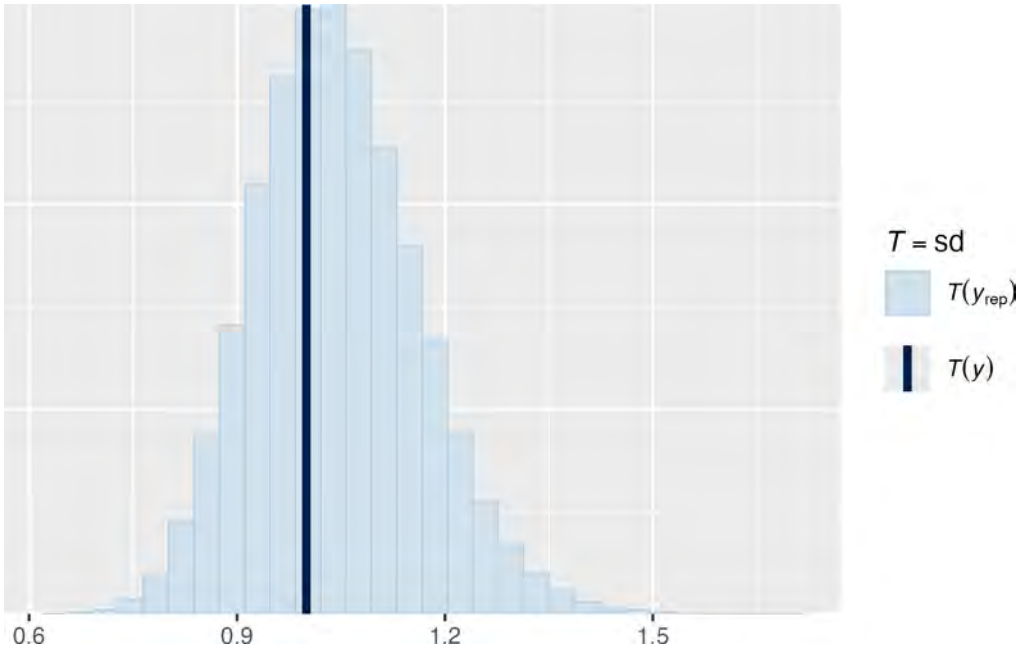

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

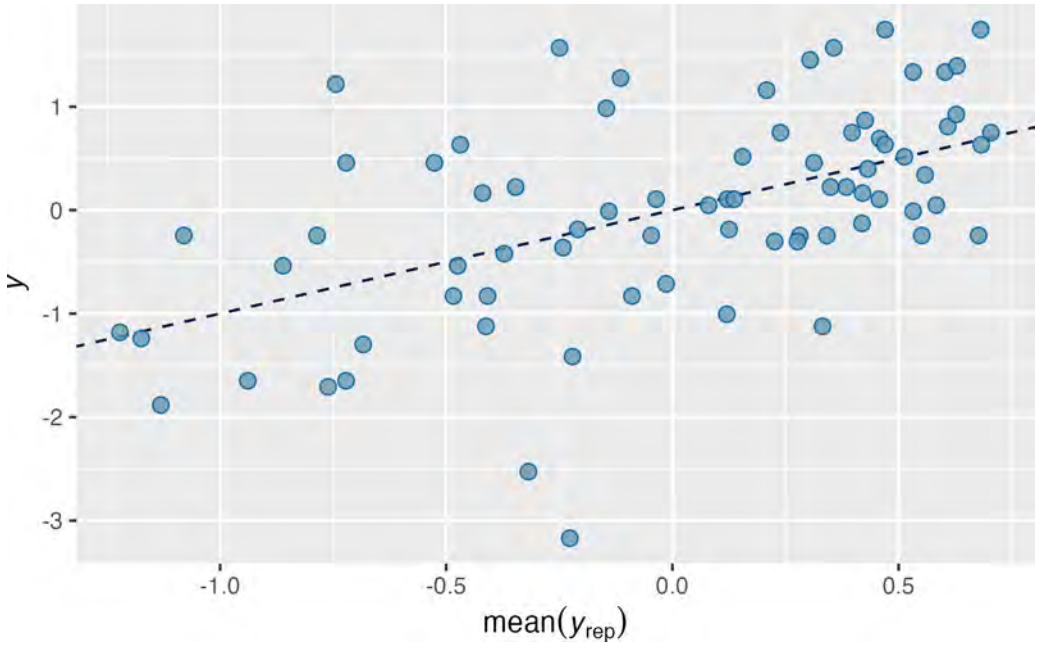



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

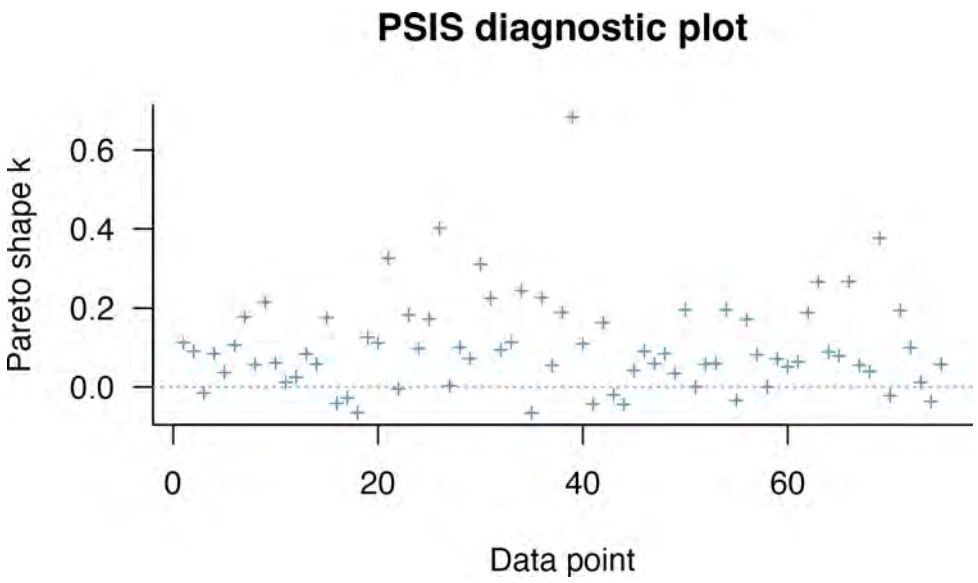
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```



H2 Young Person Model no interaction output

```
    Estimate Est.Error      Q2.5     Q97.5
R2 0.2876003 0.0746618 0.1349938 0.4236757
```

```
Summary of Posterior Distribution

Parameter             |  Mean | Median |             CI |     pd | % in ROPE | meaningful
-------------------------------------------------------------------------------------------
b_Intercept           | -0.51 |  -0.51 | [-0.83, -0.19] | 99.88% |     0.68% |       TRUE
b_pubertal_timing_y_s |  0.26 |   0.26 | [ 0.05,  0.46] | 99.27% |     6.39% |      FALSE
b_adhd_yY             |  0.82 |   0.82 | [ 0.41,  1.23] | 99.99% |     0.05% |       TRUE
b_sed_percentile_s    |  0.10 |   0.10 | [-0.13,  0.32] | 79.25% |    46.58% |      FALSE
```

```
=== ADHD difference in raw outcome units ===
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

adhd_raw_effect_y_mean	adhd_raw_effect_y_lower	adhd_raw_effect_y_upper
13.98	6.95	21.04

H2 Young Person Model no interaction diagnostics and PP checks

```
=== Rhat and ESS ===

Rhat values close to 1 indicate convergence. ESS should ideally be > 400.

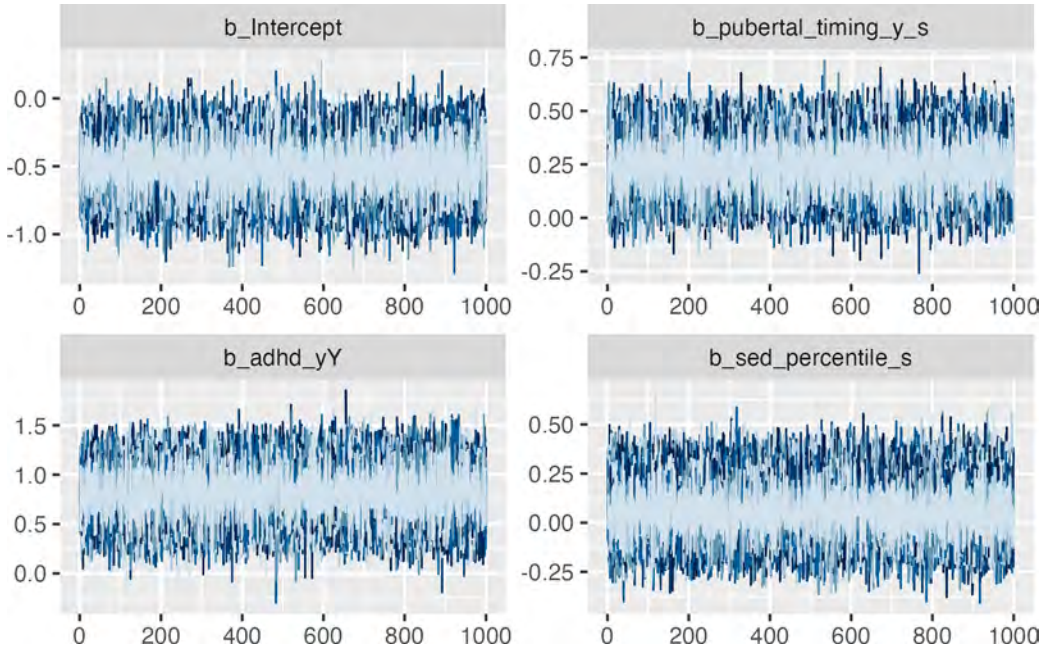
# A tibble: 8 x 4
  variable               rhat ess_bulk ess_tail
  <chr>                 <dbl>    <dbl>    <dbl>
1 b_Intercept            1.01   13835.   92670.
2 b_pubertal_timing_y_s  1.01    4743.   27798.
3 b_adhd_yY              1.01    7551.   85258.
4 b_sed_percentile_s     1.15     490.    2005.
5 sigma                  1.01    6034.   52701.
6 Intercept              1.00  146616.   92410.
7 lprior                 1.01   18676.   89967.
8 lp__                   1.20     394.     584.


=== Divergent Transitions ===

Ideally 0. Large numbers indicate sampler problems.

Number of divergent transitions: 0


=== Trace Plots ===
```



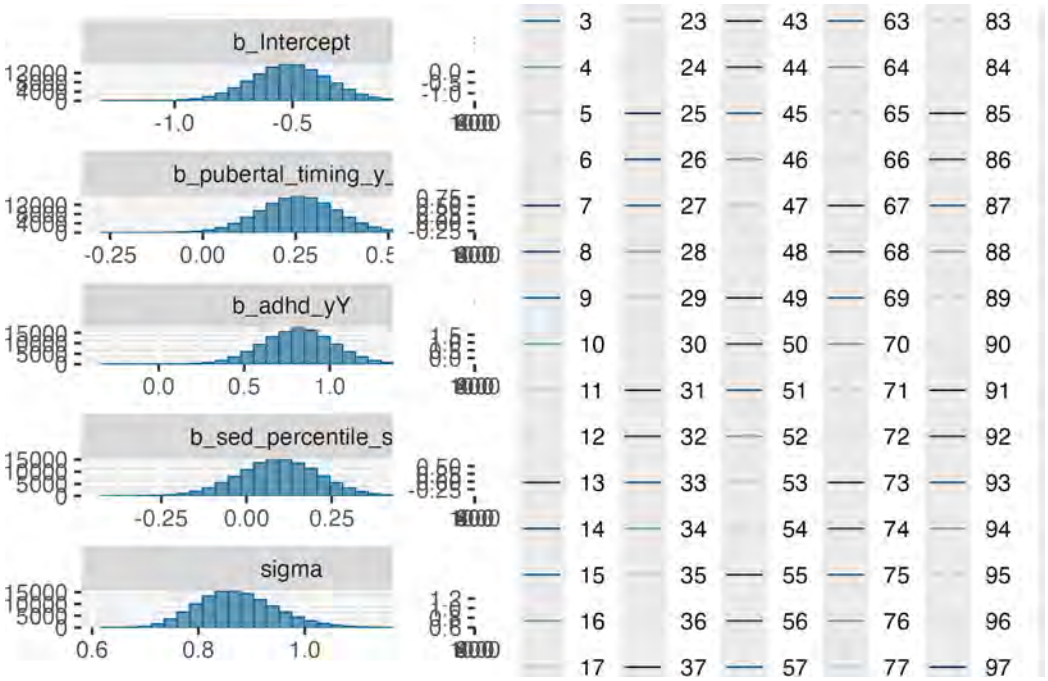

```
=== Prior Summary ===

Priors specified for the model.

          prior     class                coef group resp dpar nlpar lb ub tag
   normal(0, 1)         b
   normal(0, 1)         b             adhd_yY
   normal(0, 1)         b pubertal_timing_y_s
   normal(0, 1)         b    sed_percentile_s
   normal(0, 1) Intercept
 exponential(1)     sigma                                                  0
       source
         user
 (vectorized)
 (vectorized)
 (vectorized)
         user
         user


=== Posterior Predictive Checks ===
```

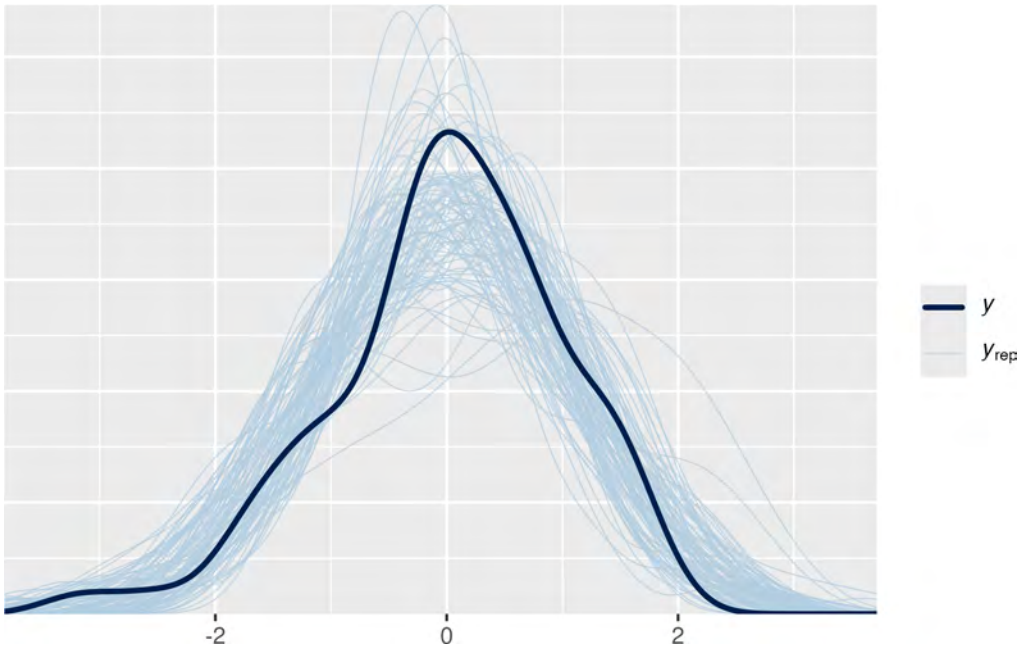
Density plots for each parameter.



=== PPC: Density Overlay ===

y_rep lines should closely follow the observed y distribution.

Warning: Using only the first imputed data set. Please interpret the results with caution until a more principled approach has been implemented.



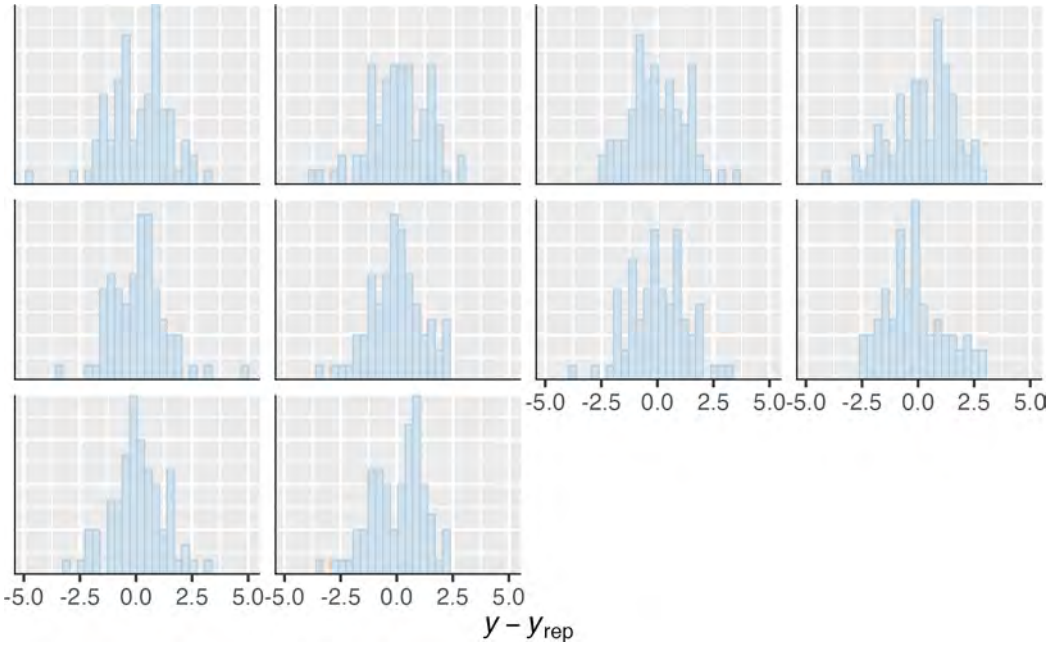

```
=== PPC: Error Histogram ===

Errors should be approximately normally distributed around 0.

Using 10 posterior draws for ppc type 'error_hist' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
=== PPC: Mean ===

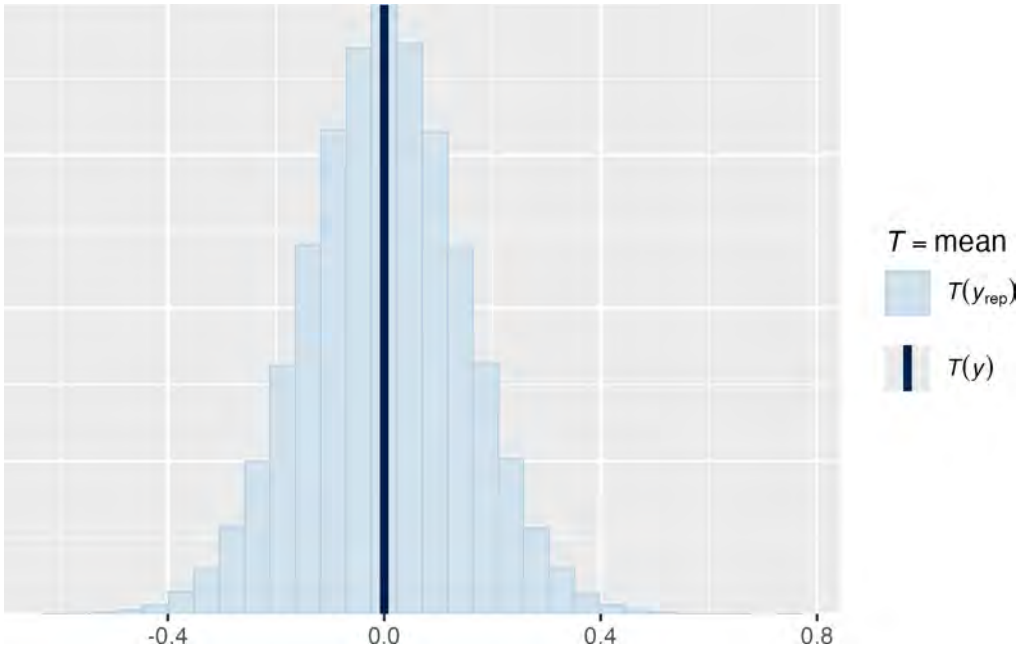
Observed mean (dark line) should fall within the distribution of replicated means.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of inte

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



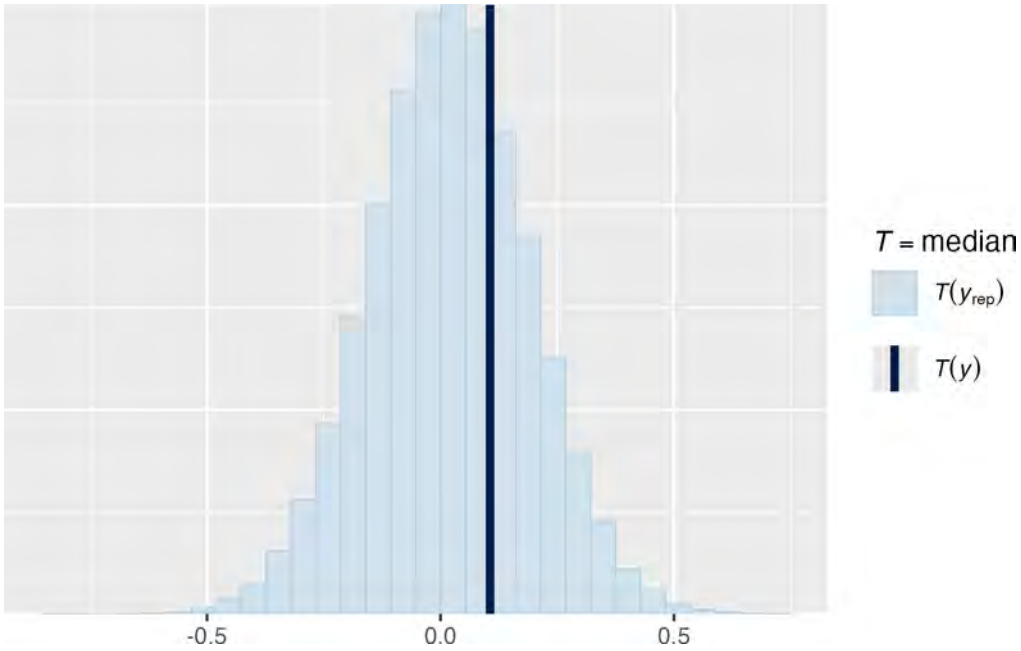

```
=== PPC: Median ===

Observed median (dark line) should fall within the distribution of replicated medians.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



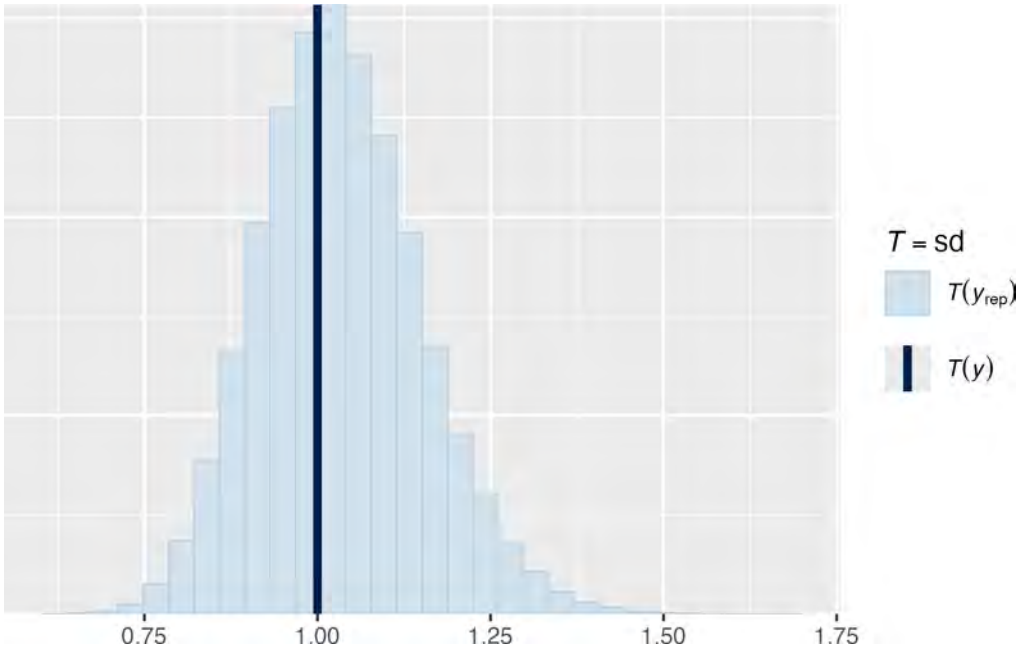

```
=== PPC: SD ===

Observed SD (dark line) should fall within the distribution of replicated SDs.

Using all posterior draws for ppc type 'stat' by default.

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

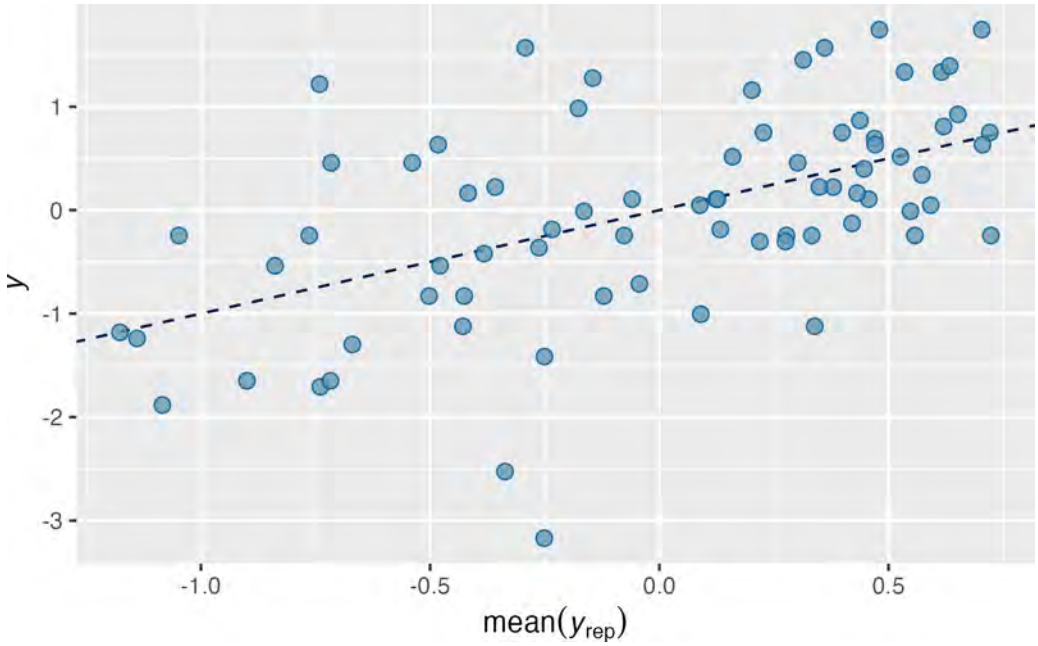



```
=== PPC: Scatter Average ===

Points should cluster around the diagonal. Systematic patterns indicate misfit.

Using all posterior draws for ppc type 'scatter_avg' by default.

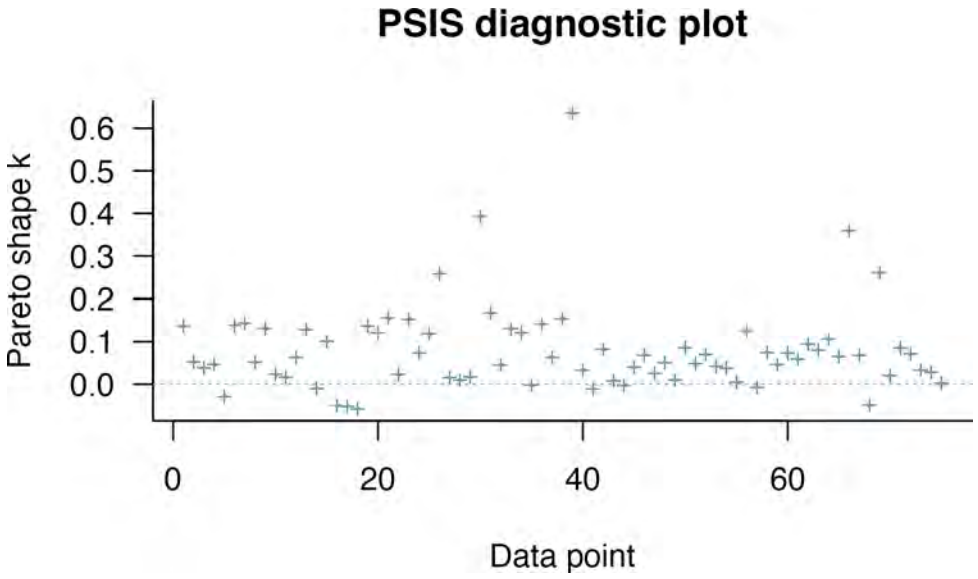
Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.
```




```
=== Pareto-k ===

Warning: Using only the first imputed data set. Please interpret the results
with caution until a more principled approach has been implemented.


All Pareto k estimates are good (k < 0.7).
```


PSIS diagnostic plot
Pareto shape k
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
20
40
60
Data point